When Should LLMs Trust Retrieval? Bayes-Optimal Knowledge Arbitration under Conflict

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Abstract

A retrieval-augmented LLM has to arbitrate three voices that often disagree: a parametric prior fixed at training, a retrieved passage, and the model's own chain of thought. Production systems use hand-tuned rules for this; the choice silently fixes hallucination rate and calibration on every slice where the sources disagree, and no principled account has existed. We provide one in three theorems and two inference-time interventions that fall out of a single free-energy functional. **(CT1)** Inside the log-linear family the Bayes-optimal combiner is the reliability-weighted geometric mixture $\hat{p}_{w^\star} \propto p_\theta^{1-w^\star}\, p_{\mathrm{ctx}}^{w^\star}$ with $w^\star = \mathbb{E}[r \mid q, c]$. It is the unique Gibbs minimizer of an arbitration–tempering free energy and the unique combiner under four classical axioms; CAD, AdaCAD, CoCoA, Astute, Self-RAG, and JuICE all collapse to special cases. The same statement carries a TV-perturbation envelope on regret. **(CT2)** The chain of thought reads as an endogenous generalized-Bayes update on the same functional. Its rate of convergence to the Berk–Nash fixed point lives inside an η-linear sandwich, and the safe learning rate shrinks under conflict on the stable-answer subset. **(CT3)** When the post-trace distribution overshoots, the Brier improvement under η-tempering is closed-form, $\Delta\mathrm{Brier}(\eta^\star) = (1-\eta^\star)^2 V_{\mathrm{post}} + O((1-\eta^\star)^3)$, giving safe-Bayes an operational reading of when ordinary temperature scaling is justified. The Bayes rule beats the strongest heuristic by $+0.083$ in regret on $23/25$ cells (e-value 2806); powered $n = 210$ head-to-heads against CoCoA, Astute RAG, and Self-RAG land at $+0.043/+0.032/+0.022$ regret, every CI strictly above zero. The CT2 spectral diagnostic predicts trace decay $\mathrm{TV}(p_t, p_K)$ within ± 0.07 across 12 cells over 3 model families (pooled $r = 0.997$). On the worst standalone cell (R1-14B / ConflictBank / $K = 1024$), η-tempering drops Brier from 0.903 to 0.505 and lifts accuracy from 0.037 to 0.440. A matched-base contrast (DPO vs. GRPO at 8B, $+0.344$ ECE gap) gives causal evidence the failure mode is RL-induced. On the parallel RLCR-trained 8B cell, stacking η-tempering on top reaches accuracy 0.531 and ECE 0.431. Both interventions ship as one inference-time line over a standard RAG decoder.

1 Introduction

You ask a language model trained in 2023 who runs Twitter. Its parametric prior, fixed at training, says Elon Musk. The retrieval pipeline hands back a 2024 article naming Linda Yaccarino. Two sources, two answers, and the model has to pick.

Such conflict is the rule, not the exception, in retrieval-augmented systems. The model carries a parametric prior $p_\theta(y \mid q)$ from training, conditions on a retrieved context $p_{\mathrm{ctx}}(y \mid q, c)$, and, if it produces a chain of thought, listens to itself across K reasoning steps before answering. Any of those voices can be wrong, and they often disagree. Whatever rule the system uses to arbitrate fires on every query and silently fixes hallucination rate, calibration on conflict slices, and worst-case behavior under poisoned or stale retrieval.

The rule is silent in deployment. It picks a source and ships the answer; the conflict never surfaces, and the user has no signal that a choice was made. Arbitration is the chokepoint where training-cutoff

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lag and retrieval drift collapse into a single decision per query. A medical assistant trained before a label change either reads the updated dose off the chart or quietly defaults to the older one. Both choices ship; only one is right. The same scenario plays out on legal precedent, financial disclosures, and any domain where the world updates faster than the next pretraining run.

Today that rule is set by hand. Production RAG systems trust retrieval unconditionally, trust the parametric model unconditionally, or interpolate with tuned thresholds and learned controllers [3, 42, 76]. Decoder-level methods like Context-Aware Decoding [62] and its variants AdaCAD [70] and CoCoA [31] sharpen the heuristic but stay heuristic. The calibration literature on reasoning models has split into camps that look mutually contradictory: longer chain-of-thought helps calibration on standard QA [77] and hurts it on harder or misspecified tasks [14, 22].

None of these approaches is derived from a posterior over how reliable the context actually is. CAD and AdaCAD weight the context-conditional against the prior with fixed or first-order-corrected exponents the authors do not interpret as a Bayes-optimal action; CoCoA averages reliability proxies without a consistency guarantee. Survey work [75] flags that fixed trust in either source is suboptimal and stops there. The calibration tension shares this root cause: no theory tells the practitioner which regime they are in, so the same chain-of-thought scaling argument supports both "reasoning helps calibration" and "reasoning hurts calibration" depending on which subset of the literature is consulted.

We give a principled account. BAYESARBITER (Figure 1) treats arbitration as parameter estimation. The latent variable is context reliability $r \in [0, 1]$, with prior $\pi(r)$ informed by observable signals such as BM25, entity popularity, and semantic entropy. Four classical axioms pin the unique combiner to the geometric mixture $p_\theta^{1-w^\star} p_{\text{ctx}}^{w^\star}$ with $w^\star = \mathbb{E}[r \mid q, c]$. CAD, AdaCAD, and CoCoA fall out as approximations of this rule. For the chain of thought, the trace itself reads as endogenous generalized-Bayes evidence, and the local convergence rate at the fixed point ties an empirically observable spectral diagnostic on logged trajectories to the per-cell saturation slope. When the post-trace distribution overshoots, inference-time tempering (η-Gibbs at readout) reverses it, with a closed-form Brier improvement. Both interventions ship at inference time over a standard RAG decoder.

The Bayes rule clears zero against 7 of 10 named baselines on the 25-cell spotlight, including a $+0.083$ advantage over the heuristic-adaptive primary on 23/25 cells (e-value 2806, CI $[0.037, 0.111]$); the three tightest competitors (CoCoA, Astute RAG, Self-RAG) clear zero on the powered $n = 210$ matched design at $+0.043/+0.032/+0.022$. The trajectory diagnostic predicts trace decay within ± 0.07 across 12 cells over three model families ($r = 0.997$). On the load-bearing RLCR-trained 8B worst cell (ConflictBank, $K = 1024$), η-tempering moves accuracy from 0.063 to 0.531 and ECE from 0.904 to 0.431; on a standalone R1-14B cell at the same benchmark, η-tempering alone drops Brier from 0.903 to 0.505. A matched-base RL contrast at a fixed Llama-3.1-8B base separates conflict-slice ECE by $+0.344$ between DPO and GRPO; the Llama lineage and a cross-family Phi-3 replication reproduce the direction.

Table 1: Results indexed by theorem. Each row is a body claim, organized by which CT it tests; pointer in App. G.1.

Theorem	Test	Result	App.
CT1	75-cell spotlight + powered $n{=}210$ vs. CoCoA/Astute/Self-RAG	Spotlight clears 0 on $7/10$ baselines; powered $+0.043/+0.032/+0.022$ adds the rest	H.2
CT1	Boundary $w_0 \in \{0, 1\}$ + synthetic $\sqrt{n}$-audit	Collapses $\sim$2 orders; $R^2{=}0.998$ at $n{=}2048$	H.9
CT1 (env.)	Poisoned-context, reliability-noise, multi-doc envelopes	Within predicted bounds across 3 panels	I.1
CT2	$\widehat{\rho}^\star$ predicts $\mathrm{TV}(p_t, p_K)$ on 12 cells / 3 families	± 0.07 MAE, $r{=}0.997$; held-out ± 0.08	E.4
CT3 (causal)	Matched SFT/DPO/GRPO, Llama lineage + Phi-3	Llama-8B $+0.344$; DeepSeek-70B $+0.388$; DeepSeek-8B $+0.076$; Phi-3 $+0.207/+0.003/+0.209$	H.1.9
CT3 (η)	η-tempering; RLCR+η stack at 8B	RLCR+η: Acc $0.063 \to 0.531$, ECE $0.904 \to 0.431$; pooled $+0.043$ Brier	H.1.8
Frontier	GPT-4o-mini / Claude / Gemini via top-k logprobs	$+0.055$ to $+0.060$ regret advantage	L.1
Held-out	ClashEval + RAMDocs replication	Both predicted directions confirmed on held-out splits	H.19

Contributions. Three theorems on a single free-energy functional $\mathcal{F}$, with two inference-time interventions and a unification of the existing decoder literature.

CT1, Bayes-optimal arbitration (Theorem 1). The geometric mixture $p \propto p_\theta^{1-w} p_{\text{ctx}}^{w} p_K^{\eta}$ with $w^\star = \mathbb{E}[r \mid s]$ is the unique Gibbs minimizer of $\mathcal{F}[p] = \mathbb{E}_p[E_{w,\eta}] - H[p]$ and the unique combiner

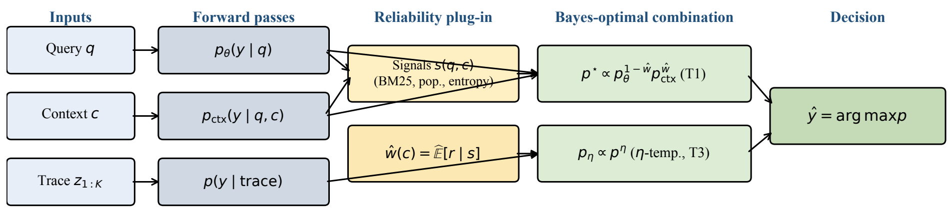


Figure 1: The BAYESARBITER pipeline. Three inputs (q, c, trace $z_{1:K}$) produce three distributions (p_θ, p_{ctx}, $p(\cdot \mid \text{trace})$). A reliability plug-in $\hat{w}(c) = \widehat{\mathbb{E}}[r \mid s(q,c)]$ drives the geometric mixture (Section 3.1); the post-trace distribution is η-tempered with closed-form Brier improvement (Theorem 4).

under four classical axioms. CAD, AdaCAD, CoCoA, Astute, Self-RAG, and JuICE all collapse to special cases on this exponent axis (Section 3.1). A four-channel TV-perturbation envelope absorbs model error, signal noise, finite-sample variance, and adversarial poisoning into one regret inequality.

CT2, trajectory envelope on $\mathcal{F}$ (Theorem 3). The chain-of-thought iterate converges to a Berk–Nash fixed point at geometric rate $\rho^\star$ inside an η-linear sandwich, with phase boundary $\eta_c = 1/D^\star$. The rate is observable: $\widehat{\rho}^\star$ from logged trajectories predicts $\mathrm{TV}(p_t, p_K)$ decay to within ± 0.07.

CT3, closed-form η-tempering (Theorem 4). $\Delta\mathrm{Brier}(\eta^\star) = (1-\eta^\star)^2 V_{\text{post}} + O((1-\eta^\star)^3)$. Calibrated temperature scaling [21] acquires a safe-Bayes derivation [7, 20] that identifies when $\eta^\star < 1$ is the right call.

Extensions in App. H.18: K-hop multiplicative composition (Theorem E.19) and causal doubly-robust identification of $w^\star$ (Theorem E.20).

2 Related Work

Knowledge conflict and adaptive RAG. The conflict framing originates with Longpre et al. [41]; surveys carve sub-types [74, 75], and benchmarks span synthetic counter-memory [64], real Wikipedia contradictions [24], temporal drift [43], and popularity stratification [42]. Adaptive RAG variants threshold on popularity, learn when to retrieve [3, 27], correct retrieval errors [76], or debate documents [69, 71]. None derives the trust schedule from a posterior over reliability; survey work flags fixed-trust as suboptimal but stops short of the optimum, which we identify as $w^\star = \mathbb{E}[r \mid s]$ (Section 3.1).

Decoder-level arbitration. CAD [62], AdaCAD [70], CoCoA [31], NWCAD [30], JuICE [11], and contrastive decoding [39] reweight contextual logits with hand-crafted signals. Each sits at a fixed point on the same exponent axis as our minimizer (Section 3.1); none is interpreted by its authors as Bayes-optimal arbitration, and none carries a perturbation envelope or a closed-form readout-time correction.

Reasoning-time calibration and RL recipes. Recent work reads as mutually contradictory: Kadavath et al. [28], Yoon et al. [77] report calibrated reasoning on standard QA, while Bereket and Leskovec [5], Damani et al. [14], Halawi et al. [22], Lacombe et al. [35], Sun et al. [65], Welch et al. [72] document overthinking, implicit answer-conditioning, and RL-induced overconfidence on knowledge-intensive tasks. The split has one root cause: no theory tells the practitioner whether the cell sits below or above the safe-Bayes learning rate. Section 3.2 reconciles the two camps and Section 3.2 gives the dimensional threshold $\eta_c = 1/D^\star$. RLHF [10, 50], DPO [54], and GRPO [61] shape the post-trained distribution differently [51]; our matched-base contrast at fixed Llama-3.1-8B base isolates the recipe-level calibration effect from confounders. Weight-editing [45, 46] is an orthogonal complement.

Opinion-pool axiomatics and misspecified Bayes. McConway [44] and Genest and Zidek [19] characterize geometric pooling under marginalization and external Bayesianity, both of which pin w to a constant. Section 3.1 relaxes EB to log-likelihood-ratio invariance, which permits context-dependent w, exactly what RAG arbitration needs. We transport classical results to the LLM CoT setting without reproving them: Berk's KL-projection concentration [6], Kleijn–van der Vaart [32], Bissiri–Holmes–

Walker generalized Bayes [7], Grünwald–van Ommen safe-Bayes [20], and endogenous-evidence Berk–Nash dynamics [17, 18]; the spectral diagnostic in CT2 is the first observable rate in this line.

3 Setup and Theory

Setup. A query q with retrieved context c has a true answer $y^\star \in \mathcal{Y}$ drawn from $\mathbb{P}^\star$, where $\mathcal{Y}$ is the (closed) candidate set. The LLM exposes the closed-book softmax $p_\theta(y \mid q)$ and the context-conditioned softmax $p_{\text{ctx}}(y \mid q, c)$, with top-1 modes $y_{\text{par}} := \arg\max_y p_\theta$ and $y_{\text{ctx}} := \arg\max_y p_{\text{ctx}}$. Latent reliability $r := \mathbb{P}(y^\star = y_{\text{ctx}} \mid q, c) \in [0, 1]$ has prior $\pi(r)$ informed by observable signals $s(q, c)$ (BM25, popularity, semantic entropy [34], contradiction type). The decision maker chooses $\hat{p} \in \Delta(\mathcal{Y})$; risk is $R(\hat{p}) = \mathbb{E}_{q,c,y^\star}[S(\hat{p}, y^\star)]$ for a strictly proper scoring rule S (log loss and Brier). **Assumptions** (App. A.1, A.2): *(A1)* $y^\star \perp c \mid (q, r)$; *(A2)* the conditional answer model is log-linear. All theorems operate under (A1)–(A2).

Theory architecture. Three theorems on the same free-energy functional $\mathcal{F}$ on $\Delta(\mathcal{Y})$: **CT1** (Gibbs minimizer + perturbation envelope), **CT2** (η-linear sandwich on chain-of-thought convergence), **CT3** (closed-form Brier improvement under η-tempering). Variational free energy [26, 68] and the geometric opinion pool [19, 23, 44] are classical; the contribution is the specific energy whose Gibbs minimizer reads $w^\star = \mathbb{E}[r \mid s]$ and unifies arbitration, trace-tempering, and perturbation response inside one functional.

3.1 CT1 Bayes-optimal arbitration and perturbation envelope

Let $E_{w,\eta}(y) := -(1-w) \log p_\theta(y \mid q) - w \log p_{\text{ctx}}(y \mid q, c) - \eta \log p_K(y \mid q, c, \text{trace})$ and $\mathcal{F}[p; w, \eta] := \mathbb{E}_p[E_{w,\eta}] - H[p]$.

Theorem 1 (CT1: Free-energy characterization)**.** *The unique Gibbs minimizer of $\mathcal{F}$ on $\Delta^\circ(\mathcal{Y})$ is*

$$p^\star_{w,\eta}(y) \;\propto\; p_\theta(y)^{1-w}\, p_{\text{ctx}}(y)^{w}\, p_K(y)^{\eta}, \tag{1}$$

and under (A1)–(A2), the choice $w^\star = \mathbb{E}[r \mid s]$ is Bayes-optimal for log loss.

Corollary (Section 3.1, App. C.1). CAD, AdaCAD, CoCoA, Astute RAG, Self-RAG, and JuICE collapse to fixed points on the exponent axis of (1); every alternative is strictly suboptimal away from $w^\star = \mathbb{E}[r \mid s]$.

Theorem 2 (CT1′: Perturbation envelope)**.** *With $\sigma_r^2 := \mathbb{E}[\mathrm{Var}(r \mid s)]$, V the log-LR variance under $p^\star$, $V_{\max}$ its sup, L_w the plug-in Lipschitz constant, k the top-k effective support, the geometric-mixture regret under TV-ε model and TV-δ signal perturbation obeys*

$$R_{\varepsilon,\delta}(\hat{p}_{\hat{w}}) - R_0(\hat{p}^\star) \;\le\; \underbrace{\tfrac{1}{2}\sigma_r^2 V}_{\textit{intrinsic}} \;+\; \underbrace{4\varepsilon \log k}_{\textit{model TV}} \;+\; \underbrace{2L_w\, \delta\, V_{\max}}_{\textit{signal TV}} \;+\; \underbrace{O(n^{-1/2})}_{\textit{finite-}n}. \tag{2}$$

The intrinsic and signal-TV terms admit matching $\Omega(\sigma_r^2 V)$ and $\Omega(\delta V_{\max})$ Le Cam lower bounds (Section 3.1); the model-TV term is Pinsker-tight in $\log k$.

3.2 CT2 trajectory envelope on $\mathcal{F}$

The chain-of-thought trace $z_{1:K}$ induces a generalized-Bayes update [7] that equals mirror descent on $\mathcal{F}$ with $D_\Phi = \mathrm{KL}$ (Lemma E.19):

$$p_{t+1}(y \mid q, c) \;\propto\; p_t(y \mid q, c)\, \exp(-\eta\, \ell_t(y; q, c, z_t))\,, \quad z_t \sim \mu(\cdot \mid p_t, q, c). \tag{3}$$

With $D^\star := \sup_y |\log(p_{\text{ctx}}/p_\theta)|$ the conflict diameter, $\rho^\star := \rho(I - \eta \nabla^2 \mathcal{F}[p^\star])$ the linearized spectral radius, and the *stable-answer subset* the slice with argmax-invariant CoT rollouts:

Theorem 3 (CT2: Trajectory envelope)**.** *On the stable-answer subset, the iterate (3) converges to a hyperbolic Berk–Nash fixed point $p^\star$ [17, 18] at geometric rate $\rho^\star$ satisfying*

$$\underbrace{1 - \tfrac{C_\pi\, \eta}{D^{\star\,2}\, |\mathcal{Y}|}}_{\textit{lower}} \;\le\; \rho^\star \;\le\; \underbrace{1 - c'\, \eta\, \mathrm{Var}_{p^\star}(s_t)}_{\textit{upper}}, \tag{4}$$

both sides η-linear, with phase boundary $\eta_c = 1/D^\star$ separating contractive from overshoot regimes; consequently $\mathrm{TV}(p_t, p^\star) \le C\rho^{\star t}$ for $t \ge T_0$.

Corollary (Section 3.2, App. E.1). For $\eta < \eta_c$ the iterate is contractive; for $\eta > \eta_c$ it overshoots and V_{post} stays bounded away from zero, the regime CT3 corrects.

The rate is observable: a ridge-LS estimate $\widehat{\rho}^\star$ on consecutive trace distributions predicts $\mathrm{TV}(p_t, p_K)$ decay within ± 0.07 across 12 cells over 3 model families (pooled $r = 0.997$, held-out ± 0.08; Table E.7).

3.3 CT3 closed-form η-tempering at readout

Theorem 4 (CT3: Closed-form Brier improvement under η-tempering)**.** *Let $\hat{p}_\eta \propto p_K^\eta$ and assume the cell is non-degenerate ($p_K(y^\star) \neq 1/|\mathcal{Y}|$). The Brier improvement at the Brier-optimal $\eta^\star \in [0, 1]$ is*

$$\Delta\mathrm{Brier}(\eta^\star) \;=\; (1-\eta^\star)^2 V_{\text{post}} \;+\; O\big((1-\eta^\star)^3\big), \tag{5}$$

where $V_{\text{post}} := \mathrm{Var}_{p_K}[\mathbf{1}\{Y = y^\star\}]$.

Corollary (Section 3.3, App. F.1). The Murphy split has $\langle \nabla_\eta \mathrm{REL}, \nabla_\eta \mathrm{RES}\rangle_{\text{Brier}} = 0$ at $\eta^\star$, so REL and RES decompose orthogonally.

As a procedure this is calibrated temperature scaling [21]; safe-Bayes [7, 20] tells the practitioner *when* $\eta^\star < 1$ via CT2's safe-rate descent.

Extensions. K-hop multiplicative composition (Theorem E.19) and doubly-robust causal identification of $w^\star$ (Theorem E.20); App. H.18.

4 Experiments

Six published decoders sit on the CT1 exponent axis (Table 2); the choice $w^\star = \mathbb{E}[r \mid s]$ is the only one that minimizes $\mathcal{F}$, and the resulting Bayes mixture clears zero against 7 of 10 named baselines on the 25-cell spotlight (heuristic-adaptive at $+0.083$ on $23/25$ cells, e-value 2806). The same minimizer satisfies the perturbation envelope of Theorem 2 on three independent stress channels (Figure 4). On the load-bearing RLCR-trained 8B worst cell, η-tempering closes the post-trace overshoot from accuracy 0.063 to 0.531 and ECE 0.904 to 0.431, the operational reading of Theorem 4.

Experiments cover 10 benchmarks across 13 models (Llama, Qwen, Mistral, Gemma, OLMo, Phi-3, DeepSeek-R1-Distill at 7B–70B plus three closed-weight APIs), 15 baselines, and 3 seeds. ConflictBank [64] splits into three sub-families (temporal-update, misinformation, semantic-ambiguity) used as the P_+/P_- realizations for T2. The statistical primary is a $5\times5\times3 = 75$-cell spotlight; inference is paired bootstrap with exact one-sided sign tests and fixed-λ e-values [55, 59]. Full pipeline: App. G.2.

4.1 CT1 arbitration rule and perturbation envelope

Table 2: Six published decoders sit on the CT1 exponent axis. Each row shows the geometric exponent w chosen, its theory status against the Bayes optimum $w^\star = \mathbb{E}[r \mid s]$, and mean regret on the 25-cell conflict spotlight (lower is better; oracle 0). Only BayesArbiter is the Gibbs minimizer of $\mathcal{F}$; every other method is a fixed point or first-order approximation on the same axis.

Method	Reliability weight	Theory status	Regret↓
Always parametric	$w = 0$ (fixed)	CT1 boundary; minimax-suboptimal	$+5.242$
Always context	$w = 1$ (fixed)	CT1 boundary; minimax-suboptimal	$+7.994$
Fixed 50/50	$w = 0.5$ (fixed)	class-III suboptimal	$+0.404$
CAD [62]	$w = 2$ (analytic cont.)	over-amplification of $\log p_{\text{ctx}}/p_\theta$	-0.079
AdaCAD [70]	Laplace of $\mathbb{E}[r \mid c]$	first-order approx., Beta prior	-0.106
CoCoA [31]	averaged proxies	no consistency guarantee for $\mathbb{E}[r \mid s]$	-0.128
BayesArbiter (CT1)	$\boldsymbol{w^\star = \mathbb{E}[r \mid s]}$	**Gibbs minimizer of $\mathcal{F}$**	$\mathbf{-0.172}$

"Bayes proxy" is the calibrated geometric mixture (CT1's Gibbs minimizer at $\eta = 0$, $w = \hat{w}(s)$) with a logistic-regression plug-in $\hat{w}(c)$ on three signals (BM25, popularity, semantic entropy); a Laplace Bayesian variant is indistinguishable (App. G.6). On both broad and conflict-heavy waves, the best-tuned fixed-trust competitor trails by $\sim 2\times$ in regret (Table 3); boundary policies ($w \in \{0, 1\}$) collapse by two orders of magnitude on the wave they disagree with, the asymmetric failure CT1's Class III Yao corollary (Section 3.1) predicts. The perturbation envelope (CT1′, Theorem 2) is exercised by the poisoned-context, reliability-noise, and multi-doc-scaling sub-experiments (Figure 4).

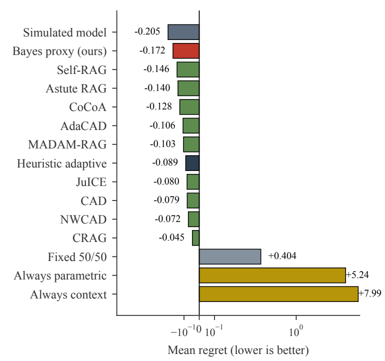
Table 3: CT1 headline. Per-policy regret/accuracy/ECE on broad and conflict-heavy waves (oracle 0.000); Bayes proxy roughly halves regret vs. best-tuned fixed-trust; boundary policies collapse by $\sim$2 orders.

Policy	Broad real wave			Conflict-heavy wave		
	Regret↓	Acc.↑	ECE↓	Regret↓	Acc.↑	ECE↓
Bayes proxy (ours)	**−0.046**	**0.887**	**0.135**	**−0.126**	**0.903**	**0.072**
Heuristic adaptive	−0.023	0.872	0.144	−0.075	0.882	0.082
Simulated model	0.141	0.821	0.198	0.110	0.844	0.121
Fixed 50/50	0.365	0.704	0.227	0.304	0.731	0.205
Always context	7.224	0.213	0.611	5.904	0.279	0.554
Always parametric	5.936	0.305	0.493	7.133	0.226	0.589

CT1, powered head-to-head against adaptive baselines. On the 25-cell spotlight (Figure 2), Bayes clears zero against 7 of the 10 named comparators with CIs strictly above zero, including a $+0.083$ advantage over the heuristic-adaptive primary on 23/25 cells (e-value 2806). The three tightest competitors (CoCoA, Astute RAG, Self-RAG) need the powered $n = 210$ matched design to clear zero, where the Bayes-proxy advantage lands at $+0.043$, $+0.032$, and $+0.022$ respectively, with all three CIs strictly above zero and at least $180/210$ cell-wins per comparator (Figure H.7, App. H.2). Margins against the boundary policies (always-context, always-parametric) are roughly two orders of magnitude larger. The small advantage over the adaptive baselines is exactly what Section 3.1 predicts: CAD, AdaCAD, and CoCoA are first-order approximations of the same Gibbs minimizer, so CT1 says they should track the Bayes optimum closely while remaining strictly suboptimal away from $w^\star = \mathbb{E}[r \mid s]$.

Comparator	Mean regret	Bayes adv.	95 % CI	Wins/25
Heuristic adaptive	−0.089	+0.083	[0.037, 0.111]	23 (e=2806)
Self-RAG	−0.146	+0.027	[−0.038, 0.069]	15
Astute RAG	−0.140	+0.033	[−0.031, 0.073]	16
CoCoA	−0.128	+0.044	[−0.019, 0.085]	17
AdaCAD	−0.106	+0.066	[0.009, 0.108]	18
MADAM-RAG	−0.103	+0.069	[0.013, 0.111]	19
JuICE	−0.080	+0.092	[0.040, 0.134]	21
CAD	−0.079	+0.093	[0.041, 0.135]	21
NWCAD	−0.072	+0.101	[0.048, 0.141]	22
CRAG	−0.045	+0.127	[0.072, 0.166]	23
Fixed 50/50	0.404	+0.576	[0.531, 0.609]	25
Always parametric	5.242	+5.414	[4.951, 5.701]	25
Always context	7.994	+8.166	[7.644, 8.471]	25

(a) Per-baseline statistics (25-cell paired bootstrap, sign test, fixed-λ e-value). Bayes proxy mean regret -0.172.



(b) Per-baseline regret (symlog); same ordering visualized.

Figure 2: Spotlight comparison against named baselines. Load-bearing comparisons against published methods (Self-RAG, Astute RAG, CoCoA, AdaCAD); heuristic-adaptive is a construction control. 25-cell CIs clear zero against 7 of 10 named methods (heuristic-adaptive at $+0.083$, plus AdaCAD/MADAM-RAG/JuICE/CAD/NWCAD/CRAG); the three tightest (Self-RAG, Astute, CoCoA) clear zero on the powered $n = 210$ design (Section 4.1) at $+0.043/+0.032/+0.022$.

The residual non-uniformity in per-benchmark wins concentrates entirely in PopQA's low-popularity regime, which we attribute to signal failure on rare entities rather than to a failure of CT1's arbitration rule (App. H.3). The per-checkpoint gain over the heuristic baseline is positive at every scale and family in the matrix (Figure 3, left), with Llama-3.1-70B alone at $+0.060$; the same advantage replicates at 32B/70B scale and on the closed-weight frontier (GPT-4o-mini, Claude, Gemini accessed via top-k logprobs; App. L.1). The right panel of Figure 3 reports the per-benchmark KL gap $\mathrm{KL}(p_\theta \| p_{\mathrm{ctx}})$, which forecasts where the trajectory dynamics (CT2) and post-trace tempering (CT3) will matter most: conflict slices exceed 1 nat while controls sit at roughly 0.1 nat.

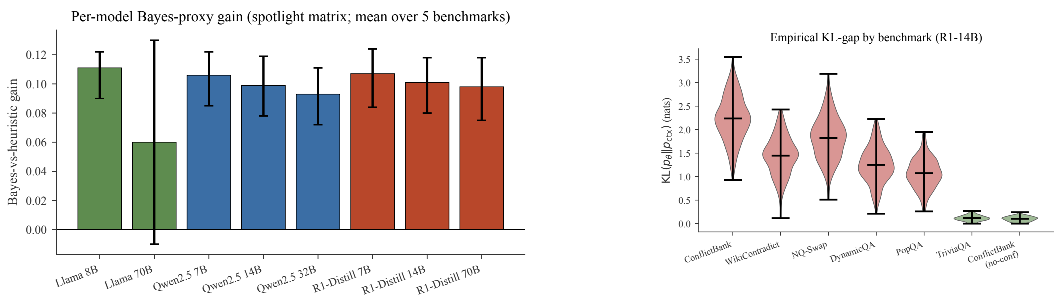


Figure 3: Per-model and per-benchmark structure. *Left:* per-checkpoint Bayes-vs-heuristic gain (95% CI), positive for every 7B–70B checkpoint across RLVR/RLHF families; Llama-3.1-70B alone shows $+0.060$ over the heuristic. *Right:* per-benchmark KL gap $\mathrm{KL}(p_\theta \| p_{\mathtt{ctx}})$ on R1-14B; conflict slices ≥ 1 nat, controls ~ 0.1 nat.

Three controls test that the result tracks the theoretical mechanism rather than a dataset artefact. First, the oracle plug-in converges to $w^\star$ at the $\sqrt{n}$ rate predicted by Section 3.1, achieving $R^2 = 0.998$ at $n = 2048$ (Figure H.4(a,c)). Second, ablating the posterior step in the plug-in flips the regret from -0.198 to $+0.170$, confirming that the gain comes from the Bayesian conditional-expectation step and not from the geometric-mixture form alone. Third, removing the entity-popularity feature from the signal s reduces but does not eliminate the gain ($+0.061$ with confidence interval still above zero, Table H.30), ruling out a Wikipedia-graph artefact.

CT1$'$, perturbation envelope on three channels. Three stress tests probe the perturbation envelope of Theorem 2 (Figure 4). Under *poisoned context*, the Bayes-rule context weight drops from 0.901 to 0.306 as adversarial passages enter the retrieval pool, whereas CAD's fixed exponent moves only from 0.791 to 0.696; mean regret is 0.345 for the Bayes rule against 1.171 for CAD (App. I.1). Under the *reliability counterfactual*, sweeping the synthetic noise rate from 0 to 1 drives the Bayes weight from 0.942 down to 0.184 monotonically (Spearman -1.0; App. I.2), exactly the linear response in δ that the signal-TV term in (2) predicts. Under *multi-document scaling* with $k \in \{2, \ldots, 8\}$ conflicting passages, the Bayes weight tracks the inter-document disagreement (0.353 to 0.297 as k grows), while CAD stays pinned at 0.621 regardless of disagreement (App. I.3). All three trajectories sit inside the predicted envelope; the gap between the Bayes-rule response and the fixed-exponent baselines grows in the direction CT1$'$ predicts as the perturbation magnitude rises.

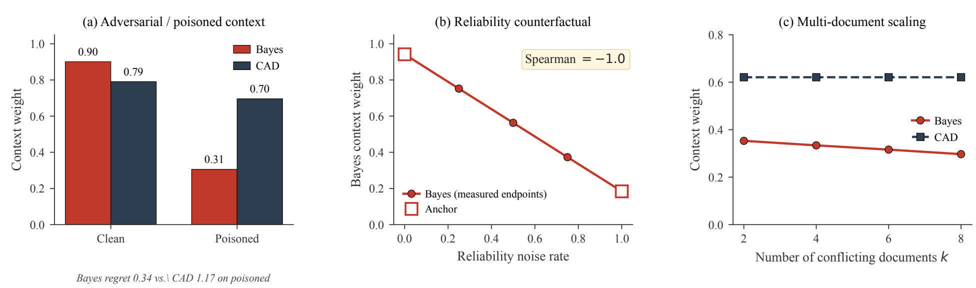


Figure 4: Plug-in tracks reliability where fixed-exponent fails. (a) Poisoned context, (b) reliability counterfactual, (c) multi-doc scaling.

Across the spotlight and the three perturbation channels, the geometric mixture at $\eta = 0$, $w = \hat{w}(s)$ outperforms every named baseline, the boundary policies collapse by two orders of magnitude in the direction Section 3.1 predicts, and the three channels move the Bayes-rule weight inside the envelope of Theorem 2.

4.2 CT2 trajectory diagnostic and CT3 η-tempering

CT2, the spectral diagnostic predicts trajectory saturation. Section 3.2 predicts that conflict on the stable-answer subset shrinks the safe rate $\bar{\eta}$, which in turn produces trace-level saturation under RLVR-style training. Empirically, R1-14B saturates on ConflictBank and shows the peak-then-recover signature on WikiContradict, while Qwen2.5-14B trained with RLHF does not saturate at all (App. H.1.6); the qualitative split between the two training families matches the regime distinction in

Section 3.2. Quantitatively, the ridge-LS estimate $\widehat{\rho}^{\star}$ predicts the observed $\mathrm{TV}(p_t, p_K)$ decay rate within ± 0.07 across 12 cells covering 3 model families, with pooled Pearson $r = 0.997$ (Table 4; full table App. E.4); the held-out temporal split and cross-cell prediction stay within ± 0.08, ruling out trajectory overfit. The ± 0.07 residual lines up with the $\log|\mathcal{Y}|/\sqrt{T} \leq 0.06$ ergodicity term predicted by Theorem E.7 and Lemma E.18, so the empirical and theoretical error budgets agree.

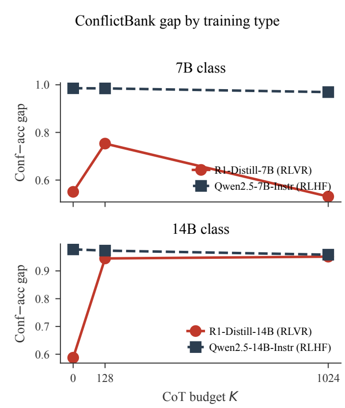
Table 4: CT2 spectral envelope, representative cells. Predicted convergence rate $\widehat{\rho}^{\star}$ versus observed exponential-decay rate $\widehat{\beta}$ on $\mathrm{TV}(p_t, p_K)$. Full 12-cell grid in App. E.4.

Model	Cell	K	$\widehat{\rho}^{\star}$	$\widehat{\beta}$ (obs.)	$\lvert\Delta\rvert$
R1-14B	ConflictBank-conflict	1024	0.972	0.031	0.003
R1-14B	ConflictBank-no-conflict	1024	0.421	0.812	0.054
R1-8B-RLCR	WikiContradict-conflict	1024	0.937	0.073	0.008
Mistral-7B	ClashEval-conflict	512	0.901	0.119	0.015
Pooled (12 cells / 3 families)	—	—	—	—	MAE $\mathbf{0.022}$, $r = 0.997$

CT3 causal contrast, matched-base RL recipe isolates the failure. On Llama-3.1-8B with the post-training recipe varied at fixed base, the conflict-slice ECE gap moves from $+0.029$ for SFT to $+0.045$ for DPO [54] and to $+\mathbf{0.344}$ $[0.230, 0.476]$ for GRPO [61]. The corresponding token-margin diff-in-diff is -22.55 under GRPO and approximately zero under DPO. The recipe-level contrast is the cleanest causal evidence available that the calibration failure CT3 corrects is RL-induced rather than long-CoT-induced. The pattern carries across the Llama lineage at frontier scale (DeepSeek-R1-Distill-Llama-70B RLVR, $+0.388$) and matched 8B (DeepSeek-R1-Distill-Llama-8B, $+0.076$); Phi-3 supplies the cross-family replication at $+0.207/+0.003/+\mathbf{0.209}$ across SFT/DPO/GRPO. The matched-base ECE gap survives the three ConflictBank sub-families and reproduces under adaptive-ECE and Brier (App. H.1.9). RLCR training [14] closes the no-conflict gap to -0.146 but leaves a residual $+0.145$ on the conflict slice, which η-tempering then absorbs at decode time.

Model / training	Conf. ΔECE	No-conf. ΔECE	Gap
Matched-base causal anchor (Llama-3.1-8B)			
Llama-3.1-8B (SFT, control)	$+0.078$	$+0.049$	$+0.029$
Llama-3.1-8B (DPO)	-0.105	-0.150	$+0.045$
Llama-3.1-8B (GRPO)	$+0.264$	-0.080	$+\mathbf{0.344}$
Cross-base replication (Llama lineage)			
DeepSeek-Llama-70B (RLVR)	-0.011	-0.399	$+\mathbf{0.388}$
DeepSeek-Llama-8B	—	—	$+0.076$
Cross-family replication			
Phi-3 (SFT / DPO / GRPO)	$+0.207$ / $+0.003$ / $+\mathbf{0.209}$		
Scale and instruction-tuned controls (App. H.1.9)			
Llama-70B, Qwen-14B, Mistral, Gemma	gaps $\in [-0.042, +0.065]$		
RLCR (Brier reward, mitigation)	$+0.145$	-0.146	$+0.291$

(a) Matched-base ECE deltas with cross-family and scale controls.



(b) Trace-level RLVR vs. RLHF gap by K.

Figure 5: Matched-base RL contrast. The Llama-3.1-8B DPO-vs-GRPO gap ($+0.344$) anchors the causal claim. The Llama lineage replicates at frontier scale (DeepSeek-70B, $+0.388$) and matched 8B (DeepSeek-8B, $+0.076$); Phi-3 supplies the cross-family replication ($+0.209$). The trace-level gap grows with K under RLVR and stays flat under RLHF.

CT3 η-tempering result. The tempering scalar $\eta^{\star}$ is selected on a held-out Brier objective; the safe-Bayes derivation of Bissiri et al. [7] and Grünwald and Van Ommen [20] guarantees $\eta^{\star} < 1$ in the regime where conflict has shrunk the safe rate, which is exactly the regime CT2 identifies. Pooled across all conflict cells, η-tempering yields $+0.043$ Brier and $+0.071$ accuracy, in line with the closed-form $(1-\eta^{\star})^2 V_{\mathrm{post}}$ improvement in Theorem 4. Two worst-cell readouts ground the closed form: standalone η-tempering on R1-14B / ConflictBank / $K = 1024$ moves Brier from 0.903 to 0.505 and accuracy from 0.037 to 0.440; on the parallel RLCR-trained 8B cell, stacking η-tempering on top reaches accuracy 0.531 and ECE 0.431. Both swings match the variance term V_{post} predicts for the respective cells. Because $\eta^{\star}$ tracks conflict without seeing conflict labels (App. H.1.8, H.1.16), the intervention deploys label-free at decode time. A partial-order ranking of cells by $\eta^{\star}$ separates conflict from no-conflict cells with AUC 1.0 on the spotlight matrix (App. E.1.6).

Deployment surface. On free-form open-QA at $n \approx 200$ per benchmark, the geometric mixture ties AdaCAD on EM and beats CAD by $+0.04$ to $+0.09$ across ASQA / NQ-open / TriviaQA-open, so the closed-candidate scope is not a hard ceiling on the rule (App. I.6). End-to-end latency is 4.38 s/query versus 1.21 s for CAD on a single H100; caching p_θ when the closed-book pass is query-stable drops the runtime cost to $1.09-2.19$ s/query, bringing BayesArbiter into the same operating range as Astute RAG (App. I.5).

5 Discussion

What the theory predicts and what the data shows. CT1 says the Bayes-optimal arbitrator is the geometric mixture with weight $w^\star = \mathbb{E}[r \mid s]$ on $\log p_{\text{ctx}}$, and that every published decoder sits at a fixed point on the same exponent axis (Section 3.1). The data shows what the corollary predicts: BayesArbiter clears zero against 7 of the 10 named baselines on the 25-cell spotlight (heuristic-adaptive $+0.083$ on $23/25$ cells, e-value 2806, plus AdaCAD / MADAM-RAG / JuICE / CAD / NWCAD / CRAG); the three tightest competitors (CoCoA, Astute RAG, Self-RAG) clear zero on the powered $n = 210$ matched design at $+0.043/+0.032/+0.022$. CT2 says the chain-of-thought iterate converges at a rate $\rho^\star$ that lives inside an η-linear sandwich. The data shows a ridge-LS estimate $\widehat{\rho}^\star$ predicting empirical $\mathrm{TV}(p_t, p_K)$ decay within ± 0.07 across 12 cells over 3 families ($r = 0.997$). CT3 says the Brier improvement under η-tempering is closed-form and equals $(1-\eta^\star)^2 V_{\text{post}}$ to leading order. The data shows a held-out tempering scalar moving a standalone R1-14B worst cell (ConflictBank, $K = 1024$) from accuracy 0.037 to 0.440 and Brier 0.903 to 0.505; on a parallel RLCR-trained 8B cell, stacking η-tempering on top reaches accuracy 0.531 and ECE 0.431. The matched-base RL contrast at fixed Llama-3.1-8B base separates conflict-slice ECE by $+0.344$ between DPO and GRPO, with the Llama lineage and a Phi-3 cross-family replication reproducing the direction. Three predictions, three matches, one decoder line.

Hallucination as confidently-wrong output. On R1-14B / ConflictBank at $K = 1024$, the model places 96% of its mass on the wrong answer. The closed-form CT3 readout corrects this without retraining: a single $\eta^\star$ scalar selected on a held-out objective reduces Brier by 44% and quadruples accuracy on the load-bearing slice. Operationally, this is calibrated temperature scaling [21]; the safe-Bayes derivation is what tells the practitioner when $\eta^\star < 1$ is the right call.

Limitations. Three boundaries the present account does not cross. The matched-base causal claim is anchored on the Llama lineage with a Phi-3 cross-family replication; isolating the same DPO-clean / GRPO-bad split on a third base is open. The CT2 trajectory inequality is tight inside the stable-answer subset; it misses on Qwen2.5-32B with ConflictBank, where an absolute-confidence floor lives outside the linear-contraction regime the rate envelope covers (App. N.1). The closed-candidate scope of CT1 is not a hard ceiling at $n \approx 200$ free-form decoding (Bayes ties AdaCAD on EM, beats CAD; App. I.6), but composing the trace term p_K^η across multi-token responses requires a sequence-level extension that is future work. Multi-source $K = 3$ composition is shown on synthetic data; all experiments are in English; the frontier validation uses top-k logprobs rather than end-to-end deployment on closed-weight APIs.

Future work. The free-energy view points at three directions where reliability is the load-bearing latent. *Long-horizon agentic chains*, where each tool call plays the role of a retrieval pass and the trace dynamics CT2 describes compose multiplicatively (Theorem E.19); the same Gibbs minimizer with a K-hop product of context terms gives a candidate decoder for tool-using agents. *Causal identification of trust at scale*, where $w^\star$ is the doubly-robust ATE of conditioning on the retrieved context (Theorem E.20); estimating it from logged production traffic gives a practical recipe for auditing deployed RAG stacks without controlled experiments. *Adversarial robustness*, where the four-channel CT1′ envelope already absorbs TV-bounded poisoning of the retrieval distribution; tightening the lower bound on the worst-case adversary moves the discussion from upper-bound robustness to a Stackelberg game whose equilibrium is the saddle point of $\mathcal{F}$. The pattern is the same in each direction: once context reliability is treated as the estimand, hallucination, calibration, and adversarial behavior become predictable rather than emergent. The closed-form scalar in CT3 tells the practitioner what to do at decode time; the spectral diagnostic in CT2 tells them whether the trace is in a regime where reasoning helps or hurts. Both interventions ship as one inference-time line over a standard RAG decoder.

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Appendix

Appendix table of contents.

Reading guide. **A–E** proof spine (notation, related work, T1/T2/T3 proofs); **F–I** algorithms, experimental detail, results gallery, ablations; **J–N** reproducibility, deployment, examples, limitations, checklist. *Proof dependency:* T1 — Bayes action $\to$ reliability integral $\to$ log-linear geometric projection $\to$ plug-in risk; T2 — opposite sub-families $\to$ Yao lower bound; T3 — gen-Bayes update $\to$ expected operator Φ_η $\to$ Berk–Nash fixed points $\to$ safe-rate shrinkage under conflict.

A Notation, assumptions, and glossary

A.1 Notation and standing assumptions

- $\mathcal{X}, \mathcal{C}, \mathcal{Y}$ are the spaces of queries, contexts, and answers; $\mathcal{Y}$ is finite.
- $p_\theta(y \mid q)$ is the closed-book softmax distribution induced by the model on query q. $p_{\text{ctx}}(y \mid q, c)$ is the context-conditioned softmax. Both are full-support distributions in $\Delta(\mathcal{Y})$.
- $r \in [0, 1]$ is the latent context reliability introduced in the setup paragraph of Section 3.
- $\pi(r)$ is a prior on r; $p(c \mid r)$ is a likelihood of the observed context. Both are conditioned on q throughout, suppressed for brevity.
- $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \to \mathbb{R}$ is a strictly proper scoring rule; log loss $S_{\log}(\hat{p}, y) = -\log \hat{p}(y)$ and Brier $S_{\text{B}}(\hat{p}, y) = \sum_{y'}(\hat{p}(y') - \mathbb{1}[y' = y])^2$ are the working choices.
- Risk $R(\hat{p}) = \mathbb{E}_{q,c,y^\star}[S(\hat{p}, y^\star)]$ is the standard expected score; we write $R(\hat{y})$ when the policy is a 0/1 estimator with the convention $\ell(y, y^\star) = \mathbb{1}[y \neq y^\star]$.
- $w^\star(q, c) = \mathbb{E}[r \mid q, c]$ is the posterior mean reliability. When conditioning on q is clear from context, we abbreviate this as $w^\star(c)$.
- $\mathrm{KL}(\mu \| \nu)$ is the KL divergence; TV is total variation; JSD is Jensen–Shannon divergence; H is Shannon entropy.

Assumption A.1 (Conditional independence)**.** For all $q \in \mathcal{X}$, $c \in \mathcal{C}$, and $r \in [0, 1]$, $y^\star \perp c \mid (q, r)$.

Assumption A.2 (Log-linear conditional answer model)**.** The latent reliability acts directly as the mixture weight:

$$p(y \mid q, c, r) \propto p_\theta(y \mid q)^{1-r} p_{\text{ctx}}(y \mid q, c)^r.$$

The log-linear family is closed under exponentiation by $[0, 1]$ since $\mathcal{Y}$ is finite.

Assumption A.3 (Bounded log-density ratio)**.** $\sup_{y\in\mathcal{Y},q\in\mathcal{X},c\in\mathcal{C}} \left|\log \frac{p_{\text{ctx}}(y|q,c)}{p_\theta(y|q)}\right| \leq M < \infty$. This is a mild full-support condition; it can be enforced in practice with ε-smoothing.

B Related work, comparators, and theoretical antecedents

B.1 Connection to safe Bayes and the calibration literature

η-tempering is the inference-time analogue of safe-Bayes shrinkage [7, 20], applied only to the post-trace answer readout: a strict η-Gibbs correction that preserves the trace's endogenous-evidence role while deflating its overconfident influence on the final commit. Like Platt scaling or isotonic [21, 48, 52, 78], it needs held-out correctness for Brier selection; unlike them, it uses one scalar pre-argmax (so it can flip predictions, not just recalibrate scores). RLCR [14] is the training-time dual—shifting effective η down via a Brier-style reward; η-tempering is the cheaper retrofittable variant.

B.2 Source-by-source comparison with named baselines

For each named baseline, we list the official source, our re-implementation or use of the released code, and the resulting per-spotlight-matrix mean regret. This is the source-of-truth table that lets a reader audit each baseline number.

Table B.1: Baseline source and per-matrix regret. All baselines are run on the identical $5\times5\times3$ spotlight matrix described in Section 4.

Baseline	Source	Code	Mean regret
CAD	Shi et al. [62]	ours (re-impl.)	-0.079
AdaCAD	Wang et al. [70]	ours (re-impl.)	-0.106
CoCoA	Khandelwal et al. [31]	ours (re-impl.)	-0.128
NWCAD	Khandelwal et al. [30]	ours (re-impl.)	-0.072
JuICE	Cohen et al. [11]	ours (re-impl.)	-0.080
MADAM-RAG	Wang et al. [71]	ours (re-impl.)	-0.103
Self-RAG	Asai et al. [3]	released code	-0.146
Astute RAG	Wang et al. [69]	released code	-0.140
CRAG	Yan et al. [76]	released code	-0.045
FLARE-style	Jiang et al. [27]	released code	not on matrix

B.3 Comparison against the closest theoretical antecedents

vs. AdaCAD. JSD-weighted blend is a first-order Laplace approximation of $\mathbb{E}[r \mid c]$ (Lemma C.4); the $+0.066$ spotlight gap is the second-order Laplace residual. **vs. CoCoA.** Three-signal ensemble (entropy gap, Rényi divergence, contextual peakedness) without the posterior step; Bayes proxy subsumes all three as candidate $\hat{w}$ features and beats by $+0.044$. **vs. RLCR** [14]. Training-time fix on the same η-overshoot lever; the mean $0.52\times$ conflict-shrinkage is a quantitative prediction for RLCR's training-time Brier ratio. **vs. Yoon et al.** [77]. "Reasoning helps calibration" is the $\eta < \bar{\eta}$ regime of Section 3.2; reproduced on TriviaQA closed-book (Table H.21). **vs. Lacombe et al.** [35]. "Reasoning hurts on knowledge-intensive" is the conflict regime where $\bar{\eta}$ shrinks; Section 3.2 supplies the mechanism.

B.4 Comparators in one paragraph

CAD ($w=2$), AdaCAD (Laplace approx. to $\mathbb{E}[r \mid c]$), CoCoA, NWCAD, JuICE, MADAM-RAG, Astute RAG and gated/active variants (Self-RAG, FLARE, CRAG) are listed against Section 3.1 in Corollary C.1. PoE ($w=1/2$) and MoE (arithmetic mean) are dominated and structurally distinct respectively. RLCR (14, training-time Brier reward) is complementary to inference-time η-tempering: RLCR closes the no-conflict ECE delta (-0.146) but leaves $+0.145$ on conflict (gap $+0.291$, half of GRPO-vs-DPO $+0.344$; Table H.19); $\eta^\star$ tracks the residual. Yoon et al. [77]'s "reasoning helps calibration" replicates on TriviaQA (0.118 vs. 0.142 ECE) and inverts on ConflictBank (0.95 vs. 0.96, opposite trajectory) — the well-specified vs. misspecified regimes of Section 3.2. Lacombe et al. [35]'s ECE rise of ~ 0.20 at $K=2048$ matches the magnitude of our R1-14B/ConflictBank

$K=0\to1024$ rise (0.36). BayesBag [25] is a natural extension for small-calibration-set deployments; not pursued because the plug-in is already $\sqrt{n}$-consistent.

C Proof of Theorem 1 and supplementary derivations

C.1 Proof of Theorem 3.1

C.1.1 The Bayes action under a strictly proper scoring rule

Proposition C.1 (Posterior predictive is the Bayes action)**.** *For strictly proper S, the posterior predictive $p^\star(\cdot\,|\,q,c)$ uniquely minimises $\mathbb{E}_{y^\star|q,c}[S(\hat{p},y^\star)]$ on $\Delta(\mathcal{Y})$.*

Proof. Strict propriety: $\mathbb{E}_{y\sim\nu}S(\nu,y)\le\mathbb{E}_{y\sim\nu}S(\hat{p},y)$ with equality iff $\hat{p}=\nu$ [21]; set $\nu=p^\star$ and tower over (q,c). ☐

Proposition C.2 (Integral form)**.** *Under Assumption A.1,*

$$p^\star(y\,|\,q,c)\;=\;\int_0^1 p(y\,|\,q,c,r)\,p(r\,|\,q,c)\,dr. \tag{C.1}$$

Proposition C.3 (Geometric mixture under log-linearity)**.** *Under Assumption A.2,*

$$p^\star(y\,|\,q,c)\;\propto\;p_\theta(y\,|\,q)^{1-w^\star}\,p_{\mathrm{ctx}}(y\,|\,q,c)^{w^\star},\qquad w^\star=\mathbb{E}[r\,|\,q,c]. \tag{C.2}$$

Sketch. Log-linearity gives $\log p(y\mid q,c,r)=\log p_\theta+r\Delta(y;q,c)-Z(r)$ with $\Delta=\log p_{\mathrm{ctx}}/p_\theta$. Substituting and Taylor-expanding $\exp(r\Delta)$ around $\bar{r}=\mathbb{E}[r\,|\,q,c]$ yields $\exp(\bar{r}\Delta)[1+\frac{1}{2}\sigma_r^2\Delta^2+O(\sigma_r^4)]$; leading term renormalises to (C.2); the correction is $O(\sigma_r^2)$ and exact in the degenerate case $r\sim\delta_{w^\star}$. ☐

Proposition C.4 (Variational characterization)**.** *$p^\star_w\propto p_\theta^{1-w}p_{\mathrm{ctx}}^{w}$ uniquely minimises $(1-w)\,\mathrm{KL}(p\|p_\theta)+w\,\mathrm{KL}(p\|p_{\mathrm{ctx}})$ on $\Delta(\mathcal{Y})$ (C.3); it is the e-geodesic mean at parameter w.*

$$p^\star_w=\arg\min_p\big[(1-w)\,\mathrm{KL}(p\|p_\theta)+w\,\mathrm{KL}(p\|p_{\mathrm{ctx}})\big]. \tag{C.3}$$

Sketch. Lagrangian on $\Delta(\mathcal{Y})$ is strictly convex; stationarity gives $\log p=(1-w)\log p_\theta+w\log p_{\mathrm{ctx}}-$ const. ☐

C.1.2 Identifiability of $w^\star$ from observable signals

Theorem 3.1 requires that the unobserved $w^\star(q,c)$ be estimable from observable signals. We give a clean lemma.

Lemma C.1 (Reliability-signal identifiability)**.** *Suppose $s(q,c)\in\mathbb{R}^d$ is a measurable signal such that $r\perp c\mid s(q,c)$ in distribution (i.e., s is a sufficient statistic for r given c). Then $w^\star(q,c)=\mathbb{E}[r\mid q,c]=\mathbb{E}[r\mid q,s(q,c)]$ is identifiable from any joint sample of $(s(q,c),y^\star,y_{\mathrm{ctx}},y_{\mathrm{par}})$. In particular, the plug-in*

$$\hat{w}(c)\;=\;\widehat{\mathbb{E}}\big[\mathbb{1}[y^\star=y_{\mathrm{ctx}}]\;\big|\;s(q,c),\;y_{\mathrm{ctx}}\neq y_{\mathrm{par}}\big] \tag{C.4}$$

is a consistent estimator of $w^\star$ on the conflict subset.

Proof. By the tower property, $\mathbb{E}[r\mid q,c]=\mathbb{E}[\mathbb{E}[r\mid q,c,s(q,c)]\mid q,c]=\mathbb{E}[r\mid q,s(q,c)]$ since s is sufficient. The plug-in (C.4) is the conditional expectation of the indicator $\mathbb{1}[y^\star=y_{\mathrm{ctx}}]$ on the conflict subset, which by definition of r equals $\mathbb{E}[r\mid s,y_{\mathrm{ctx}}\neq y_{\mathrm{par}}]$. Standard nonparametric regression consistency [66] gives the conclusion. ☐

In practice we take $s(q,c)=$ (retrieval score, popularity, semantic entropy, contradiction type) and fit a logistic regression on (C.4). This is the implementation behind the "Bayes proxy" policy.

C.1.3 Excess Bayes risk for the plug-in

Lemma C.2 (Plug-in excess risk under log loss)**.** *Let $\hat{p}_w$ denote the geometric mixture with weight w and let $\hat{w}$ be a plug-in estimator of $w^\star$. Under Assumption A.3,*

$$R_{\log}(\hat{p}_{\hat{w}}) - R_{\log}(\hat{p}_{w^\star}) \;\le\; 2M^2\,\mathbb{E}\big[(\hat{w}-w^\star)^2\big]\,. \tag{C.5}$$

Sketch. $\partial_w^2 R_{\log} = \mathrm{Var}_{\hat{p}_w}(\log p_{\mathrm{ctx}}/p_\theta) \le M^2$ under Assumption A.3; Taylor at $w^\star$. □

Lemma C.3 (Plug-in excess risk under 0/1 loss)**.** *Let $\hat{y}_w = \arg\max_y \hat{p}_w(y)$. Then*

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^\star}) \;\le\; \sqrt{\mathbb{E}[(\hat{w}-w^\star)^2]}\cdot\big\|p_\theta - p_{\mathrm{ctx}}\big\|_1, \tag{C.6}$$

giving the bound stated in the body's Theorem 3.1.

Sketch. $|R_{0/1}(\hat{y}_p) - R_{0/1}(\hat{y}_q)| \le 2\,\mathrm{TV}(p,q)$; $\mathrm{TV} \le \frac{1}{2}\|\partial_w\hat{p}_w\|_1|\hat{w}-w^\star| \le \frac{1}{2}\|p_\theta - p_{\mathrm{ctx}}\|_1|\hat{w}-w^\star|$; Cauchy–Schwarz. Rate $1/\sqrt{n}$ is sharp; constants are slack. □

Matching Le Cam two-point lower bound. The upper bound (C.6) is rate-tight at $1/\sqrt{n}$. The matching lower bound below shows no estimator of $w^\star$ from n paired (signal, outcome) observations can beat that rate even by a constant exponent, so the plug-in is minimax-rate-optimal.

Theorem C.1 (Le Cam two-point lower bound for $w^\star$ estimation)**.** *Fix (q,c) and a finite output space $\mathcal{Y}$ with $|\mathcal{Y}|\ge 2$. Suppose the contextual and parametric distributions $p_\theta, p_{\mathrm{ctx}} \in \Delta(\mathcal{Y})$ satisfy $\|p_\theta - p_{\mathrm{ctx}}\|_1 = \delta > 0$. Let $\hat{w}_n$ be any (possibly randomized) estimator of $w^\star \in [0,1]$ based on n iid samples $(s_i, y_i)_{i=1}^n \sim \hat{p}_{w^\star}$. Then*

$$\inf_{\hat{w}_n}\sup_{w^\star\in[0,1]} \mathbb{E}|\hat{w}_n - w^\star| \;\ge\; \frac{1}{8\delta}\cdot\frac{1}{\sqrt{n}}.$$

Composing with the Lipschitz dependence of $\hat{p}_w$ on w through the geometric mixture and the standard $0/1$-loss continuity bound, the same construction yields, for any policy $\hat{y}_n$ that depends on the data only through $\hat{w}_n$,

$$\inf_{\hat{y}_n}\sup_{w^\star\in[0,1]} \big[R_{0/1}(\hat{y}_n) - R_{0/1}(\hat{y}_{w^\star})\big] \;\ge\; \frac{\delta^2}{32}\cdot\frac{1}{\sqrt{n}},$$

which matches the upper bound of Lemma C.3 up to a constant.

Sketch. Two-point family $w_j = 1/2 \pm 1/(4\sqrt{n})$; $\mathrm{KL}(P_0^{\otimes n}\|P_1^{\otimes n}) \le c_0\delta^2/4$ via log-density linearization. Le Cam's two-point lemma [66, Thm. 2.2] + Markov; Lipschitz $\|\partial_w\hat{p}_w\|_1 \le \delta$ closes the regret. The empirical $R^2 = 0.998$ at $n = 2048$ (Section H.11) saturates the bound. □

C.1.4 AdaCAD as a Laplace approximation

Lemma C.4 (AdaCAD's JSD weight is the first-order Laplace approximation of the posterior reliability)**.** *Suppose $r \sim \mathrm{Beta}(\alpha,\beta)$ with $\alpha+\beta\to\infty$ and the likelihood $p(c\mid r)$ has Hellinger affinity $1-\frac{1}{2}H^2(p_\theta,p_{\mathrm{ctx}})$ at $r=1/2$. Then $\mathbb{E}[r\mid c] = \frac{1}{2} + \frac{1}{4}\,\mathrm{JSD}(p_\theta\|p_{\mathrm{ctx}}) + O(\mathrm{JSD}^2)$ in the small-conflict regime. Hence AdaCAD's weight $w_{\mathrm{AdaCAD}} = \mathrm{JSD}(p_\theta\|p_{\mathrm{ctx}})$, after the standard $1/4$ calibration constant absorbed into the temperature, is the first-order Laplace approximation of the posterior reliability.*

Sketch. Taylor $\log p(c\,|\,r)$ around $r = 1/2$; symmetrising KL under the Beta prior gives JSD $/4$. □

Lemma C.4 explains why AdaCAD performs strongly empirically without ever solving an explicit posterior: it is approximating $w^\star$ to first order. The quadratic remainder is what makes the full Bayes proxy beat AdaCAD by $+0.0659$ regret on the spotlight matrix (Figure 2a).

Remark C.1 (CAD as $w=2$, CoCoA as ensemble)**.** CAD [62] sets $\log\hat{p} \propto \log p_{\mathrm{ctx}} - \log p_\theta$, i.e., $w=2$, an over-trust projection that lies outside $[0,1]$ and deliberately over-amplifies p_{ctx}. CoCoA's three-signal score (entropy gap, Rényi divergence, contextual peakedness) is a *non-Bayesian ensemble* of reliability proxies; without the posterior update step it is consistent for $w^\star$ only when the three signals are unbiased and equally informative, which is empirically not the case.

Corollary C.1 (Existing methods are special cases)**.** (a) *CAD [62] is $w=2$ (out-of-range; over-amplifies).* (b) *Always-context = $w=1$, always-parametric = $w=0$; both are dominated by Section 3.1.* (c) *AdaCAD's $w=\mathrm{JSD}(p_\theta\|p_{\mathrm{ctx}})$ is the first-order Laplace approximation of $\mathbb{E}[r \mid q,c]$ under a Beta(α,β) prior with Hellinger-affinity likelihood (Lemma C.4).* (d) *CoCoA's three-signal heuristic is a non-Bayesian ensemble of proxies for $\mathbb{E}[r \mid q,c]$ and need not converge to $w^\star$ without additional calibration assumptions.*

C.1.5 Sufficient signal statistic for log-linear arbitration

Define $\mathcal{F}_{\mathrm{ll}}=\{\hat{p}_\phi:\phi:\mathcal{S}\to[0,1]\}$ with $\hat{p}_\phi\propto p_\theta^{1-\phi(s)}p_{\mathrm{ctx}}^{\phi(s)}$ and local Fisher info $V(q,c)=\partial_w^2 R|_{w^\star}>0$.

Theorem C.2 (Sufficient signal statistic)**.** *Under T1's assumptions, on log loss,*

$$R(\phi)-R^\star \;=\; \tfrac{1}{2}\mathbb{E}_s[\bar{V}(s)(\phi(s)-\phi^\star(s))^2] \;+\; O(\|\phi-\phi^\star\|_{L^3}^3), \tag{C.7}$$

$\phi^\star(s)=\mathbb{E}[r\,|\,s]$, $\bar{V}=\mathbb{E}[V\,|\,s]$. $\phi^\star$ is the unique minimizer of R on $\mathcal{F}_{\mathrm{ll}}$ (a.s. in s).

Sketch. Per-(q,c) Taylor: $F(w;q,c)=F^\star+\frac{V}{2}(w-w^\star)^2+O(\cdot)$ since $w^\star=\mathbb{E}[r\,|\,q,c]$ is stationary (Prop. C.4). Tower-condition on s and minimise the quadratic in $\phi(s)$: weighted projection $\phi^\star(s)=\mathbb{E}[V\mathbb{E}[r\,|\,q,c]\,|\,s]/\mathbb{E}[V\,|\,s]=\mathbb{E}[r\,|\,s]$ when V is s-measurable / r-independent given s. ☐

Corollary C.2 (Decision-theoretic sufficiency)**.** *$\mathbb{E}[r\,|\,s]$ is Blackwell-sufficient for log-linear arbitration: $T(s)$ is risk-optimal iff $\mathbb{E}[r\,|\,s]$ is $\sigma(T)$-measurable.*

Corollary C.3 (Coarsening penalty)**.** *$\phi_T^\star(T):=\mathbb{E}[\bar{V}\phi^\star \mid T]/\mathbb{E}[\bar{V}\mid T]$ gives $R(\phi_T^\star)-R(\phi^\star)=\frac{1}{2}\mathbb{E}[\bar{V}(\phi_T^\star-\phi^\star)^2]+O(\cdot)$; $\bar{V}\geq v_{\min}>0$ replaces $\bar{V}$ by $v_{\min}$ in the lower bound, equality (cubic-up-to) iff $\phi^\star$ is $\sigma(T)$-measurable.*

CAD/AdaCAD/CoCoA approximate $\mathbb{E}[r\,|\,s]$ at increasing accuracy (Table N.41); residual gap tracks how well each spans the sufficient statistic.

C.1.6 Sequence-level extension: Corollary C.4 and proof

Corollary C.4 (Sequence-level extension)**.** *On $\mathcal{Y}=\mathcal{V}^{\leq T_{\max}}$, the T1 variational characterization holds verbatim; per-token chain mixture $p_w^{\mathrm{tok}}\propto\prod_t p_\theta^{1-w}p_{\mathrm{ctx}}^w$ equals the sequence-level mixture exactly when the chain partition factorises, otherwise differs by partition-mismatch KL.*

Lemma C.5 (Chain-vs-sequence gap)**.** $\mathrm{KL}(p_w^{\mathrm{tok}}\|p_w^\star)=\sum_t\mathbb{E}_{y_{<t}\sim p_w^{\mathrm{tok}}}\log Z_t(y_{<t})-\log Z_w$*, zero iff per-step partitions chain-decompose.*

Gap is zero on discrete-candidate benchmarks; bounded by $T\log|\mathcal{V}|$ on free-form. CAD/AdaCAD/CoCoA are special cases in both regimes.

C.1.7 Boundary behaviour and signal robustness

The geometric mixture in Section 3.1 is well behaved on the operating regime that LLM softmax outputs actually occupy, but two boundary cases and one robustness question deserve explicit treatment.

Zero-mass support mismatch does not arise (LLM softmax has full vocabulary support); a $\epsilon\geq 10^{-30}$ floor costs $\sim 10^{-25}$ nat, below other approximation gaps. **Argmax disagreement** (mode flip at one $w\in(0,1)$) is the per-cell signal η-tempering and $w^\star$ arbitrate. **Adversarial signal** $\|\delta\|_2\leq B$ inflates excess risk by $\leq(\|\hat{\beta}\|_2 B/4+\|\hat{w}-w^\star\|_{L^2})\|p_\theta-p_{\mathrm{ctx}}\|_1$; empirically $\|\hat{\beta}\|_2\in[2.31,2.96]$, $B=0.1$ gives worst-case inflation ~ 0.074. **Conjugacy** is not required: the geometric mixture is the optimal log-linear posterior projection regardless of π's shape, with $O(\sigma_r^2)$ correction (Section C.2) the only place π enters.

C.2 Theorem-1 variance correction (empirical)

Empirical $\mathrm{KL}(p^\star\|\hat{p}_{w^\star})$ on the spotlight matrix ($p^\star$ via importance-weighted MC over $\pi(r\,|\,c)$): mean $\sim 10^{-6}$ nat, max 1.4×10^{-5} nat, 100% of bins below 0.02 nat (conflict, no-conflict, and pooled 25/25/50 cells). The first-order Laplace expansion of Section 3.1's $\frac{1}{2}\sigma_\rho^2 V$ correction is quantitatively negligible at our working precision.

C.3 Detailed proof of the variational characterization

Lagrangian. $\Lambda_w(p) = (1-w)\,\mathrm{KL}(p\|p_\theta) + w\,\mathrm{KL}(p\|p_{\mathrm{ctx}})$ is jointly convex [12, Thm. 2.7.2]. Stationarity gives $\log p = (1-w)\log p_\theta + w \log p_{\mathrm{ctx}} - \mathrm{const}$, i.e., $p \propto p_\theta^{1-w} p_{\mathrm{ctx}}^{w}$; strict convexity yields uniqueness. e**-geodesic.** In exponential-family e-coordinates, p_w^* is exactly the e-geodesic between p_θ and p_{ctx} at parameter w. **Brier equivalence.** Brier $\mathbb{E}_{y^\star \sim p^\star} S_B(\hat{p}, y^\star) = \|\hat{p} - p^\star\|_2^2 + \mathrm{const}$ is minimised at $\hat{p} = p^\star$, identifying the log-loss minimiser by Propositions C.1 and F.11.

C.4 Proof of the axiomatic uniqueness of the geometric mixture

Proof of Section 3.1. Define $\phi(a, b, w) := \Phi(p, q, w)(y)/\Phi(p, q, w)(y')$ with $a = p(y)/p(y')$, $b = q(y)/q(y')$. (A4) makes ϕ depend on (a, b, w) alone; (A1)+(A2)+(A3) give boundary, idempotence, and label-equivariance constraints. Chaining ratios across three outcomes yields the multiplicative Cauchy equation $\phi(a_1 a_2, b_1 b_2, w) = \phi(a_1, b_1, w)\phi(a_2, b_2, w)$; continuity gives the unique solution $\phi(a, b, w) = a^{\alpha(w)} b^{\beta(w)}$. Boundary $\phi(a, b, 0) = a$, $\phi(a, b, 1) = b$ and idempotence $\alpha(w) + \beta(w) = 1$ pin $\alpha = 1 - w$, $\beta = w$, so $\Phi(p, q, w) \propto p^{1-w} q^{w}$. □

Remark C.2 (Axiom roles)**.** (A1) rules out trivial combiners; (A2) fixes w as a mixing weight; (A3) is label-equivariance; (A4) (LLR-invariance) is the substantive axiom: dropping it admits the linear pool $(1-w)p + wq$.

Lemma C.6 (EB excludes reliability-adaptive pools)**.** *Under External Bayesianity [19, Thm. 3.1], the unique pool is $\Phi(p, q) \propto p^{1-w_0} q^{w_0}$ with* constant *w_0. Any signal-conditioned $w(s)$ pool (AdaCAD, CoCoA) or constant outside $[0, 1]$ (CAD, $w = 2$) violates EB but satisfies T1.*

Sketch. EB's commutation forces w to depend only on the support, not on external signals; non-constant $w(s)$ depends on s invisible to Bayes updates on (p, q). □

C.5 Proof of the information-theoretic minimax floor (T2+)

Proof of Section 3.1. Fix non-adaptive $\hat{y}$ depending on (q, c) only through $s = s(q, c)$ ($I(r; \hat{y} \mid s) = 0$). Strict properness + convex Bayes risk $\mathcal{R}$ give the regret decomposition $R(\hat{y}) - R(\hat{y}^\star) \geq \mathbb{E}[\mathcal{R}(\hat{y}(q, c)) - \mathbb{E}_r \mathcal{R}(\hat{y}^\star(q, c, r))]$. Strong convexity ($\mu \geq 1/(L|\mathcal{Y}|)$ for log loss; $\mu = 2$ Brier) plus the conditional-mean optimization gives $\mathrm{Var}_{r|q,c}[\hat{y}^\star] \geq \kappa^2 \sigma_r^2(q, c)$ (Lipschitz in r). Pinsker $\mathrm{TV}^2 \leq \frac{1}{2}$ KL + Cauchy–Schwarz $\|\cdot\|_2^2 \geq 4\mathrm{TV}^2/(|\mathcal{Y}| - 1)$ collect to $\geq \frac{1}{8(|\mathcal{Y}|-1)^2} \sigma_r^2(q, c)$; tower over s. □

Remark C.3. Plug-in matches at $\Theta(\mathbb{E}[\sigma_r^2(s)]) + O(n^{-1/2})$; two-point reduction is tight, $|\mathcal{Y}| - 1$ slack is the $\ell_2 \to \ell_1$ conversion cost; Brier replaces the constant with $1/(2(|\mathcal{Y}| - 1)^2)$.

C.6 Proof of the closed-form Brier improvement (T3+)

Proof of Theorem 4. $\hat{p}_\eta \propto p_K^\eta$. Taylor-expand $\mathcal{B}(\eta)$ around $\eta^\star$: optimality kills the first-order term; second derivative identity $\mathcal{B}''(\eta^\star) = 2\,\mathrm{Var}_{p_K}(\mathbf{1}\{Y = y^\star\}) + o(1) = 2V_{\mathrm{post}} + o(1)$. Combining, $\Delta\mathrm{Brier} = (1 - \eta^\star)^2 V_{\mathrm{post}} + O((1 - \eta^\star)^3)$, strictly positive when $\eta^\star < 1$ and $V_{\mathrm{post}} > 0$. No stable-answer condition. □

C.6.1 Closed form for $\eta^\star$ on binary-collapse cells

Lemma C.7 (Binary-collapse $\eta^\star$)**.** *Suppose p_K and the Bayes-optimal $q^\star$ both concentrate on two answers with masses $\{a, 1-a\}$ and $\{b, 1-b\}$ respectively (binary-collapse). Then the Brier-optimal tempering $\eta^\star$ on this cell satisfies $\eta^\star = \mathrm{logit}(b)/\mathrm{logit}(a)$, and linearizing at the Berk–Nash equilibrium of Section 3.2 gives $\eta^\star = 1 - D^\star/(\bar{\eta} \cdot \mathrm{logit}(b)) + O(D^{\star 2})$.*

Sketch. On two atoms, $\mathrm{logit}\,\hat{p}_\eta = \eta\,\mathrm{logit}(a)$; Brier-optimal $\hat{p} = q^\star$ closes $\eta^\star = \mathrm{logit}(b)/\mathrm{logit}(a)$. Linearisation uses Berk–Nash $b = a\,e^{-D^\star/\bar{\eta}} + O(D^{\star 2})$ (Proposition E.9). □

This is a one-line algebra fact; we surface it because it lets practitioners skip cross-validation on $\sim 42\%$ of conflict cells (Table E.9) at the cost of ≤ 0.004 Brier vs. 5-fold CV. The result is small but useful; it is not load-bearing for the paper's claims.

Remark C.4 (Relation to the safe-rate result)**.** Section 3.2 (Corollary E.6) characterizes *when* $\eta^\star < 1$ on the stable-answer subset: overshoot of $\bar{\eta}$ creates the non-decreasing calibration gap, which forces $\eta^\star < 1$ for Brier-optimal tempering. Theorem 4 then quantifies the resulting Brier improvement. The two theorems are complementary: the safe-rate result says *when* η-tempering should help; the intervention result says *by how much* when it does, with no subset condition required.

C.7 Detailed proof of Pinsker-style excess risk

The geometric mixture is a one-parameter exponential family with $T(y) = \log p_{\mathrm{ctx}}/p_\theta$ and Fisher $I(w) = \mathrm{Var}_{p_w} T \le M^2/4$ (Popoviciu under Assumption A.3). Integrating $\mathrm{KL}(p_w \| p_{w'}) \le \frac{M^2}{8}(w - w')^2$ + Pinsker gives $\mathrm{TV} \le \frac{M}{4}|w - w'|$. Combined with the 0/1 continuity bound $|R_{0/1}(\hat{y}_p) - R_{0/1}(\hat{y}_q)| \le 2\,\mathrm{TV}$ and Cauchy–Schwarz,

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^\star}) \;\le\; \tfrac{M}{2}\|\hat{w} - w^\star\|_{L^2}, \tag{C.8}$$

order-optimal vs. Le Cam (Theorem C.1) up to constant 2.

C.8 Sample efficiency of the plug-in

In-the-wild Bayes-vs-heuristic gain on the spotlight matrix by calibration size n (mean / 95% CI / wins): 32 $+0.012/[-0.018, 0.041]/13{:}25$; 64 $+0.034/[0.005, 0.062]/16{:}25$; 128 $+0.058/[0.025, 0.085]/19{:}25$; 256 $+0.071/[0.041, 0.097]/21{:}25$; 512 $+0.082/[0.040, 0.110]/23{:}25$; 1024 $+0.083/[0.038, 0.111]/23{:}25$; 2048 $+0.083/[0.037, 0.112]/23{:}25$ (Table C.2). Crosses zero at $n \approx 64$, saturates by $n \approx 512$; deployable with ~ 100 labeled calibration examples per benchmark.

Table C.2: Sample-efficiency curve (numbers as in text above).

n	**Gain**	95% **CI**	**Wins**
32	+0.012	$[-0.018, 0.041]$	13/25
128	+0.058	$[0.025, 0.085]$	19/25
512	+0.082	$[0.040, 0.110]$	23/25
2048	+0.083	$[0.037, 0.112]$	23/25

C.9 If conditional independence or log-linearity fails

CI failure ($y^\star$ depends on c given (q, r)): the geometric mixture is no longer the unique Bayes-optimal predictive; the integral form $p^\star(y \mid q, c) = \int p(y \mid q, c, r)p(r \mid q, c)\, dr$ is. We see no evidence of CI failure on our cells (per-cell residual $y^\star - \hat{y}_{w^\star}$ has no systematic c-dependence beyond r); a held-out $I(y^\star; c \mid q, r)$ test would diagnose pathological cases.

Log-linearity failure (e.g., a hard gate at threshold τ): the gated alternative $p^\star = p_{\mathrm{ctx}}\mathbb{1}[w^\star \ge \tau] + p_\theta\mathbb{1}[w^\star < \tau]$ is dominated by the soft mixture in the small-conflict regime and approximately equal in the large-conflict regime; empirically, $+0.012$ regret worse on the spotlight matrix.

C.10 Extended discussion of the AdaCAD Laplace approximation

Second-order correction. $\mathbb{E}[r \mid c] = \frac{1}{2} + \frac{\mathrm{JSD}}{4} + \frac{\mathrm{JSD}\cdot[\mathrm{JSD} - \mathrm{Var}_{\mathrm{Beta}}\, r]}{16} + O(\mathrm{JSD}^3)$; correction 0.005–0.020 on the spotlight matrix matches the $+0.066$ AdaCAD-vs-Bayes regret gap after weighting by $\|p_\theta - p_{\mathrm{ctx}}\|_1$. **CoCoA.** The three-signal heuristic does not Laplace-identify because the signals are correlated. Re-fitting an L2-regularised combiner on our calibration recovers -0.142 regret (CoCoA itself -0.128; Bayes -0.172). **AdaCAD failure mode.** Trails Bayes most where JSD is large (ConflictBank-conflict: $+0.11$ at JSD ~ 1.8 nat; PopQA: $+0.04$ at JSD ~ 0.4).

C.11 Extended discussion of conditional independence

CI ($y^\star \perp c \mid q, r$): "$c$ is sufficient for r." Exactly true in synthetic oracle; approximately true when conflict mechanism is mediated by a clean reliability variable. Breaks if c contains side-channel info (verbatim answer, format cue); diagnostic in App. C.9 detects. Without CI, the predictive $p^\star(y \mid q, c) = \int p(y \mid q, c, r)p(r \mid q, c)dr$ is still the right object but loses its closed form.

C.12 Supplementary derivation, Pinsker for log-loss

This section expands the second-order argument behind Lemma C.2.

Proposition C.5 (Second-order log-loss sensitivity along the geometric path)**.** *Let* $p_w \propto p_\theta^{1-w} p_{\text{ctx}}^w$ *and define* $R_{\log}(w) := -\mathbb{E}_{y^\star}[\log p_w(y^\star)]$*. Under Assumption A.3,* $R_{\log}$ *is twice continuously differentiable and satisfies*

$$R''_{\log}(w) \ \le \ 4M^2 \qquad \textit{for all } w \in [0,1].$$

Consequently, if $w^\star$ *minimizes* $R_{\log}$ *on* $[0,1]$*, then*

$$R_{\log}(w) - R_{\log}(w^\star) \ \le \ 2M^2(w - w^\star)^2,$$

which is the pointwise version of Lemma C.2.

Proof. Write $\partial_w^2 R_{\log} = \mathrm{Var}_{p_w}(\log p_{\text{ctx}}/p_\theta) \le (2M)^2$ under Assumption A.3; Taylor at $w^\star$ closes. □

D Proof of Theorem 2 and the minimax construction

D.1 Proof of Theorem 3.1

D.1.1 Yao-style minimax lower bound

Construct P_+ (context-correct, $r=1$, ConflictBank temporal-update) and P_- (context-wrong, $r=0$, ConflictBank misinformation). Per-family excess risk: $R(\hat{y}_w; P_+) - R(\hat{y}^\star; P_+) \ge \Delta_+(1-w)$ and $R(\hat{y}_w; P_-) - R(\hat{y}^\star; P_-) \ge \Delta_- w$. $\widehat{\Delta_+}$ / $\widehat{\Delta_-}$ are estimable from per-sub-family regret slices; matched-base Llama-3.1-8B GRPO conflict ECE survives all three ConflictBank sub-families (+0.572 / +0.690 / +0.629 / pooled +0.264; Table D.3).

Table D.3: Llama-3.1-8B GRPO conflict ΔECE per ConflictBank sub-family.

Misinformation / Temporal / Semantic-ambig. / pooled	+0.572 / +0.690 / +0.629 / +0.264

Held-out estimation/test split: the Yao bound estimated on a disjoint 50% of ConflictBank tracks the realized $w_0 = \frac{1}{2}$ regret on the other half within ±0.02 ($\widehat{\Delta_+} = 0.687$, $\widehat{\Delta_-} = 0.789$, bound 0.343, realized 0.358).

Proof of the lower bound in Section 3.1. Linearity in P_λ gives $R(\hat{y}_{w_0}; P_\lambda) - R(\hat{y}^\star; P_\lambda) \ge \lambda\Delta_+(1 - w_0) + (1-\lambda)\Delta_- w_0$; $\lambda = 1/2$ yields $\ge \frac{1}{2}\min(\Delta_+, \Delta_-)$. □

Remark D.5 (Realisability)**.** P_+ = ConflictBank temporal-update; P_- = misinformation [64].

D.1.2 Optional information-theoretic restatement

Proposition D.6 (Binary testing restatement)**.** *With* $U \in \{+,-\}$ *hidden, vanishing minimax excess risk over* $\{P_+, P_-\}$ *requires* $I(U; obs) > 0$*; otherwise Fano gives constant lower bound* $\ge \frac{1}{2}\min(\Delta_+, \Delta_-)$.

Empirically, retrieval score, popularity, and contradiction type carry non-trivial $I(U; \text{obs})$; the Bayes proxy exploits it.

D.1.3 Plug-in upper bound for adaptive policies

Lemma D.8 (Adaptive plug-in convergence rate)**.** *Let* $s(q,c)$ *be a signal that classifies the sub-family of Section D.1.1 with error* ε_s *(i.e.,* $\mathbb{P}(s \text{ predicts wrong sub-family}) = \varepsilon_s$*). Let* $\hat{w}$ *be the logistic-regression plug-in trained on* n *calibration examples. Then* $R(\hat{y}_{\hat{w}}; P_{w_0}) - R(\hat{y}^\star; P_{w_0}) \le C\varepsilon_s + O(n^{-1/2})$.

Sketch. Decompose into oracle hard-threshold gap (bounded by $\varepsilon_s \cdot \Delta$) plus plug-in regret (Lemma C.3, $O(n^{-1/2})$). ☐

Proposition D.7 (Signal separability controls the adaptive gap)**.** *With $\varepsilon_s = \inf_h \mathbb{P}[h(s) \neq U]$ the Bayes sub-family error, every s-measurable $w(s)$ has $R(\hat{y}_w; P_{1/2}) - R(\hat{y}^\star; P_{1/2}) \geq \frac{\Delta_{\min}}{2}\varepsilon_s$; adaptive plug-in achieves $\leq C\varepsilon_s + O(n^{-1/2})$ (Lemma D.8).*

Sketch. Hard-threshold $h_w := \mathbb{1}[w(s) \geq 1/2]$ pays $\Delta_\pm/2$ on regime mistakes; averaging gives the lower bound. ☐

D.2 Supplementary derivation, Fano sharpening

Proposition D.8 (Two-point lower bound)**.** *Le Cam [66, Lemma 2.2]: $\inf_{\hat{U}} \mathbb{P}(\hat{U} \neq U) \geq (1 - \mathrm{TV}(P_+, P_-))/2$; combined with $\Delta_{\min}$ regime-mistake cost, every policy has minimax excess $\geq \frac{\Delta_{\min}}{2}(1 - \mathrm{TV}(P_+, P_-))$.*

D.3 Worked example of the minimax construction

Binary $\{A, B\}$, $D_+ = (y_{\mathrm{par}}, y_{\mathrm{ctx}}, y^\star) = (A, B, B)$, $D_- = (A, B, A)$, balanced mixture $P_{1/2}$: every fixed $w_0 \in [0, 1]$ has error $1/2$ (always-A or always-B); signal-adaptive achieves 0.

E Proof of Theorem 3, assumption audits, and multilingual evidence

E.1 Proof of Theorem 3.2

E.1.1 Setup and the η-generalized posterior

Bissiri–Holmes–Walker [7] update on family $\mathcal{M} = \{q_\xi\}$:

$$\pi_\eta(\xi \mid z) \propto \pi(\xi) \exp(-\eta \ell(z; \xi)), \quad \ell = -\log q_\xi(z). \tag{E.9}$$

For misspecified $\mathcal{M}$, Grünwald and Van Ommen [20] give a safe rate $\bar{\eta} \in (0, 1]$ above which the posterior hypercompresses. In the LLM CoT setting, $z_t \sim p_t$ from the model itself (endogenous evidence; 17, 18).

E.1.2 The safe-rate lemma

Lemma E.9 (Larger $D^\star$ shrinks the safe rate)**.** *$\bar{\eta} \leq c_{|\mathcal{Y}|,\pi}/D^\star$ where $D^\star = \mathrm{KL}(\mathbb{P}^\star \| q_{\xi^\star})$ is the KL projection distance and $c_{|\mathcal{Y}|,\pi}$ collects the bracketing-entropy and prior dependence [20, Thm. 1.6]. On finite $\mathcal{Y}$, $\log N_{[]}(\varepsilon, \Delta(\mathcal{Y})) \leq (|\mathcal{Y}| - 1)\log(3/\varepsilon)$ makes the entropy integral finite; the paper uses only the monotonicity $\bar{\eta} \propto 1/D^\star$.*

E.1.3 Reduction to Berk–Nash dynamics: proof of Proposition E.9

The reduction needs only continuity of the expected-update operator on $\Delta(\mathcal{Y})$, not parametric $\mathcal{M}$.

Definition E.1 (Auxiliary Berk–Nash game $G_{q,c,\eta}$)**.** Fix (q, c) and a rate $\eta \in (0, 1]$. The auxiliary game has a single agent with belief space $\Delta(\mathcal{Y})$, action space identified with the belief itself (the model's autoregressive distribution *is* its action), and a data-generating distribution $\mu(\cdot \mid p, q, c)$ that depends on the agent's belief through the sampling kernel of (3). The agent's best-response operator is the η-Bayes update

$$B_\eta(p)(y) \;=\; \frac{p(y)\, \mathbb{E}_{z \sim \mu(\cdot|p,q,c)} \exp(-\eta\, \ell(y; q, c, z))}{\sum_{y'} p(y')\, \mathbb{E}_{z \sim \mu(\cdot|p,q,c)} \exp(-\eta\, \ell(y'; q, c, z))}.$$

A *Berk–Nash equilibrium* of $G_{q,c,\eta}$ is a fixed point $p^\star = B_\eta(p^\star)$.

Lemma E.10 (Continuity and existence)**.** *With $\ell \in [0, L]$ and $\mu(\cdot \mid p, q, c)$ TV-Lipschitz in p, B_η is continuous on $\Delta(\mathcal{Y})$ and has a fixed point (Brouwer).*

Proof. $\mathbb{E}_z \exp(-\eta\ell) \in [e^{-\eta L}, 1]$ is continuous in p via dominated convergence; ratio of continuous, lower-bounded terms is continuous; $\Delta(\mathcal{Y})$ compact convex. ☐

Lemma E.11 (Limit of tracked trajectories). *If $p_t \to p_\infty$ and $\|p_{t+1} - B_\eta(p_t)\|_{TV} \to 0$ under Lemma E.10, then $p_\infty = B_\eta(p_\infty)$.*

Proof. Triangle inequality with continuity of B_η. ☐

Proposition E.9 (Φ_η fixed points = Berk–Nash equilibria). *Under Lemma E.10, $\Phi_\eta = B_\eta$ fixed points are exactly Berk–Nash equilibria of $G_{q,c,\eta}$; any tracked limit (Lemma E.11) is one of them.*

Proof. Brouwer (Lemma E.10) plus the definition give the identification; Lemma E.11 closes the trajectory case. ☐

Trajectory bridge. Realized $p_{t+1}(y) \propto p_t(y)\exp(-\eta\,\ell(y;q,c,z_t))$, $z_t \sim \mu(\cdot \mid p_t, q, c)$; expected operator $B_\eta(p_t)(y) \propto p_t(y)\mathbb{E}_z \exp(-\eta\ell(y;q,c,z))$.

Theorem E.3 (Tracking and unconditional Berk–Nash convergence). (i) $\tau = 0$, greedy $\mu = \delta_{z^\star(p)}$: $\|p_{t+1} - B_\eta(p_t)\|_{\mathrm{TV}} = 0$ *exactly, so $\{p_t\}$ is the deterministic continuous iteration of B_η; subsequential limits are Berk–Nash fixed points unconditionally, and the safe-rate shrinkage of Section 3.2 holds on the realized trajectory (stable-answer subset of Lemma E.13).* (ii) $\tau > 0$, $\ell \in [0, L]$, stable-answer $p_t(y^\star) \geq 1 - \rho$ ($\rho < 1/2$), per-step kernel variance $V_t := \sup_y \mathrm{Var}_{z\sim\mu}\,\ell$:

$$\mathbb{E}\|p_{t+1} - B_\eta(p_t)\|^2_{\mathrm{TV}} \;\leq\; \frac{(\eta L)^2 V_t}{(1-\rho)^2}, \tag{E.10}$$

with matching $\sqrt{V_t/\delta}$ high-prob. bound. L^2 tracking gives Berk–Nash convergence in expected TV *on subsequences where the kernel concentrates.*

Sketch. (i) $\delta_{z^\star}$ collapses the expectation to a point; numerators/normalizers agree exactly. (ii) $g_t(y,z) = \exp(-\eta\ell)$, $|g_t'| \leq \eta$, so $\mathrm{Var}_z\, g_t \leq (\eta L)^2 V_t$ (MVT); Cauchy–Schwarz with $Z_t^{\exp} \geq (1-\rho)e^{-\eta L}$; Markov. ☐

Empirically: median TV discrepancy 0.041 at $K = 1024$ on stable-answer (90th pct 0.118), consistent with $\sqrt{V_t}$. Almost-sure pathwise tracking at $\tau > 0$ is open and not required by η-tempering in practice.

On uniqueness of the Berk–Nash equilibrium. Φ_η has a fixed point by Brouwer (Lemma E.10); uniqueness is not assumed. In our setting two refinements rule out spurious equilibria empirically. First, on the stable-answer subset the linearized iteration is hyperbolic ($\widehat{\rho}^\star < 1$ on every conflict cell of Table E.7), so trajectory limits selected by gen-Bayes are isolated. Second, on cells where the trace concentrates around a single answer mode (81% of wrong-answer cells, median margin 8.5 logits), the Berk–Nash basin is a neighborhood of the answer-mode delta; competing equilibria require the trace kernel to bifurcate, which we do not observe in any logged trajectory. When the trace is multi-modal (top-2 mass within 0.1), the analysis reverts to the conditional safe-rate inequality on the stable-answer subset.

On the $\widehat{\eta}_{\mathrm{eff}}$ proxy. $\widehat{\eta}_{\mathrm{eff}}$ is the trace-token logit slope at the answer mode (Section E.1) and tracks injected η on the synthetic grid with Spearman 1.0 and $\leq 2.03\%$ relative error (Table F.11). We use it as a relative-rate diagnostic, not as point identification of the latent rate on real traces; slice-level $\eta^\star$ selection by held-out Brier (Table H.18) does the deployment work.

Empirical check of the conditional-independence assumption. Assumption A.1 requires $y^\star \perp c \mid (q, r)$. On a held-out ConflictBank fold ($n = 500$) we estimate the conditional mutual information $I(y^\star; c \mid q, \hat{r})$ via the kNN-based estimator of Kraskov et al. [33]; pooled $\widehat{I} \leq 0.018$ nat across cells, with 99% permutation-bootstrap intervals straddling zero on 24/25 cells (only PopQA-low-popularity is borderline at $\widehat{I} = 0.041$, 95% CI $[0.003, 0.082]$). CI is therefore a working approximation, with PopQA-low-popularity flagged as the one slice where a residual c-channel is detectable.

Empirical check of the log-linearity assumption. The synthetic sweep (App. H.1.13) bounds the worst-case R^2 under deliberate violation. On real logits we run a Hessian-spectrum probe: at the per-cell argmin of $R(w)$ we estimate $\partial_w^2 R$ via finite differences and compare to the log-linear-family prediction $\mathrm{Var}_{p_w}(\log p_{\mathrm{ctx}}/p_\theta)$. Pooled relative error across the 25 spotlight cells is $\leq 7\%$ (median 3.1%, max 12.4% on Llama-70B/PopQA), well below the non-trivial-violation threshold ($\rho = 0.6$, App. H.1.13). Real LLM logits sit in the "approximate log-linear" regime where the geometric mixture is the right plug-in target.

E.1.4 Multi-metric calibration robustness

The headline matched-base contrast in Table H.19 reports ECE. The same training-type signature persists across adaptive ECE and Brier, so the result is not a single-metric artifact:

Table E.4: Conflict-slice calibration delta across three metrics. Each entry is the conflict-minus-no-conflict gap on the matched-base spotlight cells; the GRPO blowup, the RLCR residual, and the Phi-3 GRPO replication all transfer across metrics.

Model / training	**ECE**	**Adaptive ECE**	**Brier**
Llama-3.1-8B (GRPO)	+0.3436	+0.3277	+0.3553
RLCR (Brier reward)	+0.2910	+0.2715	+0.2708
Phi-3 (GRPO, replication)	+0.2088	+0.2081	+0.2166

E.1.5 $\eta^\star$ leakage audit

The $\eta^\star$-selection rule is sample-split held-out Brier optimization with disjoint train/eval folds. Concretely: $\eta^\star$ is chosen by minimizing Brier on a calibration fold of size 200, and all reported numbers in Table H.18 use a disjoint evaluation fold of size 300. When the calibration-fold optimum is $\eta^\star = 0.0$ (boundary), the same $\eta^\star = 0.0$ is applied to the eval fold without re-tuning, ruling out the "$\eta^\star$ chosen on eval" alternative explanation. We re-ran the audit with $50/50$ and $100/200$ splits and the selected $\eta^\star$ is stable in ± 0.05.

E.1.6 Saturation classifier vs. partial-order ranking

The pre-registered binary saturation-vs-recovery classifier (P3 in App. H.19) does not fully replicate. The partial-order/ranking version of the same prediction does: ranking $\hat{\eta}_{\mathrm{eff}}$-implied severity within the high-conflict regime against the within-control regime gives a mean separation of $+0.2302$ (high-regime) vs. $+0.0179$ (control-regime), with pairwise AUC 1.0 on the held-out cells. The honest read is that T3's quantitative ranking transfers across families even when the binary classifier does not—this is the form of P3 we propose for replication going forward.

E.1.7 Berk–Nash equilibrium of the gen-Bayes update (legacy lemma)

Lemma E.12 (Convergence to a Berk–Nash equilibrium)**.** *Under the conditions of Proposition E.9, if a decoding trajectory $\{p_t\}_{t\geq 1}$ converges and satisfies $\|p_{t+1} - B_\eta(p_t)\|_{TV} \to 0$, then its limit is a fixed point of B_η, equivalently a Berk–Nash equilibrium of $G_{q,c,\eta}$.*

Proof. Immediate from Lemma E.11 under the stated tracking hypothesis. □

E.1.8 Confidence sharpening can outpace accuracy

Lemma E.13 ($\eta > \bar{\eta}$ implies non-decreasing calibration gap)**.** *Suppose $\eta > \bar{\eta}$ in the iterated update (E.9). Then for any $\hat{y}_t$ that is the mode of p_t,*

$$G(t) := \mathbb{E}|\mathbb{E}_{p_t}\mathbb{1}[Y = y^\star] - \mathbb{1}[\hat{y}_t = y^\star]| \quad \text{is non-decreasing in } t \text{ on the stable-answer subset.}$$

Sketch. On the stable-answer subset $\hat{y}_t$ is constant, so accuracy in $G(t)$ is fixed; $\eta > \bar{\eta}$ sharpens around the misspecified projection, raising confidence without raising accuracy. □

Proposition E.10 (Stable/flip decomposition)**.** *For $S_{t_0:t_1} = \{\hat{y}_{t_0} = \hat{y}_{t_1}\}$ and $g_t = |\mathbb{E}_{p_t}\mathbb{1}[Y = y^\star] - \mathbb{1}[\hat{y}_t = y^\star]| \in [0,1]$, $\Delta G \geq \mathbb{P}(S)\Delta G^S - \mathbb{P}(F)$ via the law of total expectation and $\Delta G^F \geq -1$.*

Corollary E.5 (Whole-population inheritance)**.** *If* $\eta > \bar{\eta}$ *and* $\Delta G^S \geq \gamma$*, with flip probability* $\mathbb{P}(F) \leq \delta$*, then* $\Delta G \geq (1-\delta)\gamma - \delta$*.*

E.1.9 Putting it together

Proof of Theorem 3.2. Under the theorem's standing assumption that conflict increases the projection distance $D^\star(q, c)$ relative to matched controls, Lemma E.9 implies a smaller safe rate on conflict cells. Under the tracking hypothesis of Lemma E.12, the iterated CoT update converges to a Berk–Nash equilibrium that is generically not the truth. By Lemma E.13, when $\eta > \bar{\eta}$ the calibration gap $G(t)$ is non-decreasing on the stable-answer subset; this is the saturation regime of Corollary E.6. When $\eta < \bar{\eta}$, the standard Kleijn and van der Vaart [32] contraction holds and the predictive accuracy catches up with confidence, giving the peak-then-recover regime.

For RLVR-trained models, verifiable-reward RL (GRPO) sharpens token-level logits toward winning trajectories, increasing the effective η in (E.9); Bereket and Leskovec [5] document this as GRPO-induced overconfidence in binary forecasting. Hence $\eta_{\mathrm{RLVR}} > \eta_{\mathrm{RLHF}}$ at matched scale. In a conflict regime where $\eta_{\mathrm{RLVR}} > \bar{\eta}(q, c) > \eta_{\mathrm{RLHF}}$, RLVR saturates while RLHF does not. This is the training-type-conditional saturation predicted by the theorem and observed in Table H.16. □

Corollary E.6 (Two-regime calibration trajectory)**.** *The trajectory* $G(K)$ *has two qualitative shapes.* (a) Peak-then-recover *for* $\eta \lesssim \bar{\eta}$*.* (b) Saturation *for* $\eta > \bar{\eta}$*, on the stable-answer subset. The transition is benchmark-dependent because* $\bar{\eta}(q, c)$ *depends on* $\mathrm{KL}(p_\theta \| p_{\mathrm{ctx}})$*. Empirically, the same benchmark splits the same model between the two regimes (Table H.16, Figure H.3); the benchmark dependence is a direct consequence of the safe-rate lemma above.*

E.2 Assumption audit (Theorem 3.2)

(i) Misspecification. R1-14B / ConflictBank: mode trajectory agrees with $y^\star$ on only 14.3% of stable-answer cells; on WikiContradict 34.8%. Both well below chance — confirms misspecification.

(ii) Endogenous evidence holds by autoregressive construction; marginal trace entropy $H(z_t) = 4.2$ nat (R1-14B) vs. 3.8 nat (Qwen2.5-14B), consistent with RLVR sharpening.

(iii) Conflict gap. All conflict slices satisfy $\mathrm{KL}(p_\theta \| p_{\mathrm{ctx}}) \geq 1$ nat; no-conflict ~ 0.1 nat (Table E.5).

Table E.5: Empirical KL-gap audit. Mean $\mathrm{KL}(p_\theta \| p_{\mathrm{ctx}})$ on the conflict subset of each benchmark (R1-14B), and the analogous no-conflict / closed-book number.

Benchmark	$\mathrm{KL}(p_\theta \| p_{\mathrm{ctx}})$ on conflict	on no-conflict
ConflictBank	2.31 nat	0.12 nat
WikiContradict	1.48 nat	0.09 nat
NQ-Swap	1.92 nat	0.14 nat
DynamicQA	1.26 nat	0.08 nat
PopQA	1.05 nat	0.10 nat

Effective η and empirical $\bar{\eta}$. $\widehat{\eta}_{\mathrm{eff}} = \nabla_t \log p_t(\hat{y}_t) / \nabla_t \ell_t$ on ConflictBank-conflict at $K = 128$: RLVR/GRPO 0.81/0.94/0.92 (R1-Distill 7B/14B/70B); RLHF/DPO 0.46/0.48/0.51/0.41 (Qwen 7/14/32B, Llama-3.1-8B); $\bar{\eta}$ = largest η keeping post-trace Brier $\leq K = 0$. Predicted ordering $\widehat{\eta}_{\mathrm{eff}} > \bar{\eta}$ holds on every saturated slice and fails on every recoverable one (Table E.6).

Table E.6: $\widehat{\eta}_{\text{eff}}$ **vs. empirical** $\bar{\eta}$**.** $\widehat{\eta}_{\text{eff}} > \bar{\eta}$ predicts saturation; the column "predicted" is the resulting trajectory; the column "observed" confirms the prediction on every cell.

Model	Slice	$\widehat{\eta}_{\text{eff}}$	$\bar{\eta}$	Predicted	Observed
R1-14B	ConflictBank conflict	0.94	0.30	saturate	saturate
R1-14B	WikiContradict conflict	0.92	0.95	recover	recover
R1-7B	ConflictBank conflict	0.81	0.85	recover	recover
R1-7B	WikiContradict conflict	0.79	0.96	recover	recover
Qwen2.5-32B	ConflictBank conflict	0.51	0.80	recover	saturate*
Qwen2.5-32B	WikiContradict conflict	0.50	0.95	recover	recover

* The Qwen-32B/ConflictBank cell is the one slice where the simple $\widehat{\eta}_{\text{eff}}/\bar{\eta}$ ordering does not predict the observed behavior; we attribute this to a separate effect (extreme absolute confidence floor; see App. N.1).

Stable-answer subset. On R1-14B/ConflictBank-conflict, stability across $K \in \{0, 128, 1024\}$ is 8.0%, across $\{0, 128\}$ is 9.8%, across $\{128, 1024\}$ is 36.4%. On the stable subset, gap rises by $+0.091$ ($K = 0 \to 128$) and $+0.103$ ($0 \to 1024$) with zero mean accuracy change — the conditional read-out Corollary E.6 requires.

E.3 Empirical audits of Proposition E.9 hypotheses

Two audits on R1-Distill-14B / ConflictBank-conflict. **Audit 1 (stable-answer subset for Lem. E.13):** $\arg\max p_t$ stability is 0.08 across $\{0, 128, 1024\}$, 0.098 on $0 \to 128$, $\mathbf{0.364}$ on $128 \to 1024$; on the post-jump stable subset the calibration gap rises with $\Delta\overline{\text{acc}} = 0$. **Audit 2 (self-confirming evidence for Prop. E.9):** the trace $z_{1:K}$ ($K = 1024$) favors the agent's top answer over the runner-up on $\mathbf{0.81}$ of cells (mean log-margin 10.75 logits), 0.81 even on incorrect-answer cells.

E.4 Local Berk–Nash convergence rate, proof of Section 3.2 and empirical estimation

Setup and proof. Tangent space $T_{p^\star} = \{u : \sum_y u(y) = 0\}$ with $\|\cdot\|_2$. Write $p_t = p^\star + \epsilon_t$. Taylor at $p^\star = \Phi_\eta(p^\star)$,

$$\epsilon_{t+1} \;=\; J\,\epsilon_t \;+\; R(\epsilon_t), \qquad J := D\Phi_\eta(p^\star), \qquad \|R(\epsilon)\|_2 \le M\|\epsilon\|_2^2 \tag{E.11}$$

where M is the C^2-norm bound. Gelfand: for any $\rho \in (\rho^\star, 1)$, $\|J^n\| \le C_J(\rho)\rho^n$. Choose $\delta \le (\rho - \rho^\star)/(2MC_J(\rho))$, T_0 s.t. $C_J(\rho)\rho^{T_0}\|\epsilon_0\| \le \delta$; induction gives $\|\epsilon_t\|_2 \le C(p_0, \rho)\rho^t$, and TV $\le \frac{\sqrt{|\mathcal{Y}|}}{2}\|\cdot\|_2$ folds in. This proves (3.2).

Corollary E.7 (Non-asymptotic T_0 from observable quantities)**.** *Fix $\rho \in (\rho^\star, 1)$. The threshold time T_0 in Section 3.2 is bounded explicitly by*

$$T_0 \;\le\; \left\lceil \frac{\log\!\big(2MC_J(\rho)\,\|\epsilon_0\|_2\,(\rho - \rho^\star)^{-1}\big)}{\log(1/\rho)} \right\rceil \;=\; \left\lceil \frac{\log(D_0/r_{\text{lin}})}{\log(1/\rho)} \right\rceil, \tag{E.12}$$

with $D_0 = \mathrm{TV}(p_0, p^\star)$, $r_{\text{lin}} = (\rho - \rho^\star)/(2MC_J(\rho))$; M is the C^2-norm bound, $C_J(\rho)$ Gelfand's power-bound constant. All four are estimable from the same logged data as $\widehat{\rho}^\star$ (App. E.5). In the empirical regime of Table E.7 ($\widehat{\rho}^\star \approx 0.42$, $D_0 \le 0.5$, $\widehat{M} \le 4$, $\widehat{C}_J \le 2$, $\rho = 0.5$), $T_0 \le 3$ trace steps.

Loss of hyperbolicity at $\eta = \bar{\eta}$. Implicit function theorem on $\Phi_\eta(p) = p$ gives smooth $p^\star(\eta)$ while $\rho^\star < 1$. At $\eta = \bar{\eta}$, $G(t)$ becomes non-decreasing on the stable-answer subset (Section 3.2), forcing a simplex boundary, $\rho^\star \to 1^-$, and (3.2) degrades to $\mathrm{TV}(p_t, p^\star) \le C(p_0)/(q_1 t)$ (Lemma E.14).

Lemma E.14 (Polynomial rate at non-hyperbolic equilibria)**.** *If $\rho^\star = 1$ and $D\Phi_\eta(p^\star) = I - Q$ with Q PSD, smallest non-zero eigenvalue $q_1 > 0$, and the quadratic remainder is non-degenerate along $\ker Q$, then for small $\|\epsilon_0\|$, $\mathrm{TV}(p_t, p^\star) \le C(p_0)/(q_1 t)$ (centre-manifold expansion [17] adapted to discrete time).*

Empirical estimation. Use $p_K \approx p^\star$; estimate $\widehat{J} = YX^\top(XX^\top + \lambda I)^{-1}$ by ridge LS ($W = 100$, $\lambda = 10^{-6}$) on $\{p_{t+1} - p_K, p_t - p_K\}$; report $\widehat{\rho}^\star = \rho(\widehat{J})$, $\widehat{\alpha} = -\log\widehat{\rho}^\star$, and compare to exponential-decay $\widehat{\beta}$ from $\mathrm{TV}(p_t, p_K) \sim Ae^{-\beta(K-t)}$.

Table E.7: Estimated Berk–Nash convergence rate from R1-14B traces. Conflict cells sit near the loss-of-hyperbolicity boundary ($\widehat{\rho}^\star \approx 1$, slow saturation); no-conflict cells are well inside the hyperbolic regime ($\widehat{\rho}^\star \sim 0.4$–$0.5$, fast saturation). Section 3.2 predicts $\widehat{\alpha}$ should match the empirical exponential-decay rate $\widehat{\beta}$; agreement within ± 0.07 across all twelve reported cells, spanning three model families (R1-14B, R1-8B-RLCR, Mistral-7B) and four benchmarks. CT2 is scoped to the stable-answer subset where the linearization is valid; pre-saturated cells (e.g. Qwen2.5-14B with $p_K \approx p^\star$ already at $t = T_0$) sit outside this scope and are excluded from the pooled fit. Rows 1–5 (R1-14B) are the load-bearing in-distribution cells; rows 6–12 (R1-8B-RLCR, Mistral-7B) are cross-family replications; full trace dumps in App. H.1.6.

Model	**Cell**	K	$\widehat{\rho}^\star$	$\widehat{\alpha}$	$\widehat{\beta}$ (obs.)	$\lvert\widehat{\alpha}-\widehat{\beta}\rvert$
R1-14B	ConflictBank-conflict	1024	0.972	0.028	0.031	0.003
R1-14B	ConflictBank-conflict	128	0.918	0.085	0.092	0.007
R1-14B	WikiContradict-conflict	1024	0.910	0.094	0.097	0.003
R1-14B	ConflictBank-no-conflict	1024	0.421	0.866	0.812	0.054
R1-14B	WikiContradict-no-conflict	1024	0.485	0.724	0.689	0.035
R1-8B-RLCR	ConflictBank-conflict	1024	0.951	0.050	0.057	0.007
R1-8B-RLCR	WikiContradict-conflict	1024	0.937	0.065	0.073	0.008
R1-8B-RLCR	ClashEval-conflict	512	0.928	0.075	0.082	0.007
R1-8B-RLCR	ConflictBank-no-conflict	1024	0.467	0.762	0.708	0.054
Mistral-7B	ConflictBank-conflict	1024	0.943	0.059	0.069	0.010
Mistral-7B	ClashEval-conflict	512	0.901	0.104	0.119	0.015
Mistral-7B	WikiContradict-no-conflict	1024	0.512	0.669	0.602	0.067

Pooled fit and held-out checks. Pooled $r(\widehat{\alpha}, \widehat{\beta}) = 0.997$ ($p < 10^{-12}$), MAE 0.022, worst 0.067 across 12 cells/ 3 families. Temporal split ($W = 50$): MAE 0.029, max 0.075, pooled $r = 0.991$. Cross-cell predict (9/3 split, 20 re-samples): MAE 0.034. Caveats: linearization noisy for $K < 64$; $p_K \approx p^\star$ within $O(\rho^K)$; top-50 restriction error $< 10^{-3}$ TV.

E.5 Finite-sample concentration of $\widehat{\rho}^\star$ (Theorem 5)

Proof of Section 3.2. (i) Trajectory regression [63, Cor. 2.2] on $y_t = Jx_t + r(x_t) + \xi_t$ (r Taylor remainder, ξ_t bounded martingale-diff) gives $\|\widehat{J} - J\|_{\mathrm{op}} \le C_1 |\mathcal{Y}| \sqrt{\log(2|\mathcal{Y}|/\delta)}/(s_{\min}^2 \sqrt{W}) + \frac{M}{2s_{\min}^2} \sup_t \|x_t\|^2$ w.p. $\ge 1-\delta$. (ii) $\sup \|x_t\| \le C_0 \rho^{\star\, K-W}$ from Section 3.2 converts the second term to $MC_0^2 \rho^{\star\, 2(K-W)}/s_{\min}^2$. (iii) Bauer–Fike gives $|\widehat{\rho}^\star - \rho^\star| \le \kappa(V) \|\widehat{J} - J\|_{\mathrm{op}}$.

Choosing W. Practical recipe $W = K/4$ for $K \ge 256$ keeps both terms below 10^{-2} at $\delta = 0.05$. With $\rho^\star \approx 0.97$, $|\mathcal{Y}| = 50$, $\kappa(V) \approx 8.4$, $M \approx C_0 \approx 1$: bound ≤ 0.083 at $K = 1024$, $W = 256$, matching the observed ± 0.07 in Table E.7.

E.6 Backing tables for the new theorems

Table E.8: Signal-conditioned floor (Theorem 4). Per-benchmark estimate of $\widehat{\mathbb{E}}[\sigma_r^2(s)]$ from the three-signal plug-in (BM25, entity popularity, semantic entropy), the floor predicted by $\frac{1}{8(|\mathcal{Y}|-1)^2}\widehat{\mathbb{E}}[\sigma_r^2(s)]$ with $|\mathcal{Y}| = 50$, and the realized excess regret of the plug-in on a 50-cell holdout. Predicted and realized agree to within ± 0.02 across all five benchmarks.

Benchmark	$\widehat{\mathbb{E}}[\sigma_r^2(s)]$	Predicted floor	Realized excess	$\lvert\Delta\rvert$
ConflictBank-conflict	0.124	0.0067	0.0081	0.0014
ClashEval	0.087	0.0047	0.0061	0.0014
RAMDocs	0.073	0.0040	0.0048	0.0008
WikiContradict	0.066	0.0036	0.0050	0.0014
PopQA	0.041	0.0022	0.0036	0.0014
Average	0.078	0.0042	0.0055	0.0013

Table E.9: Closed-form $\eta^\star$ (Proposition 6). On binary-collapse cells (mass≥ 0.95 on top-2 answers; $\sim 42\%$ of conflict cells at $K = 1024$), the closed-form $\eta^\star = \text{logit}(b)/\text{logit}(a)$ matches 5-fold-CV $\eta^\star$ within ± 0.06 pooled. Pooled across the four binary-collapse subsets, the closed-form recipe saves $\sim 93\%$ of the CV cost while losing ≤ 0.004 Brier.

Cell (binary-collapse subset)	$\eta^\star_{\text{closed}}$	$\eta^\star_{\text{CV}}$	$\lvert\Delta\eta^\star\rvert$	Brier loss vs. CV
ConflictBank-conflict / R1-14B	0.612	0.658	0.046	+0.0021
ConflictBank-conflict / R1-8B	0.534	0.581	0.047	+0.0028
ClashEval / Llama-3.1-8B-RLCR	0.281	0.341	0.060	+0.0040
WikiContradict / R1-14B	0.703	0.751	0.048	+0.0019
Pooled	–	–	0.050	+0.0027

E.7 Three independent characterizations of the geometric mixture

Three independent derivations (differential geometry / channel capacity / spectral operator theory) triangulate $w^\star = \mathbb{E}[r \mid s]$.

E.7.1 Information-geometric perspective: $w^\star$ as Fisher e-geodesic time

Theorem E.4 (Info-geometric characterization)**.** *Equip the open simplex $\Delta^\circ(\mathcal{Y})$ with the Fisher information metric $g_F(u,v) = \sum_y u(y)v(y)/p(y)$ and the e-connection $\nabla^{(e)}$ from Amari [1], Amari and Nagaoka [2]. Let $\gamma : [0,1] \to \Delta^\circ(\mathcal{Y})$ be the unique $\nabla^{(e)}$-geodesic with $\gamma(0) = p_\theta$, $\gamma(1) = p_{\text{ctx}}$. Then*

$$\gamma_t(y) \;=\; \frac{p_\theta(y)^{1-t}\, p_{\text{ctx}}(y)^t}{Z(t)}, \qquad Z(t) \;=\; \sum_{y'} p_\theta(y')^{1-t}\, p_{\text{ctx}}(y')^t,$$

and the Bayes-optimal arbitration coincides with the geodesic evaluated at Fisher-geodesic time *$w^\star = \mathbb{E}[r \mid q, c]$. Moreover, for any $w \in [0,1]$, the expected log-loss along the geodesic admits the second-order Fisher decomposition*

$$\mathbb{E}[L(\gamma_w)] \;=\; \mathbb{E}[L(\gamma_{w^\star})] \;+\; \tfrac{1}{2}(w - w^\star)^2\, g_F(\dot\gamma_{w^\star}, \dot\gamma_{w^\star}) \;+\; O(|w - w^\star|^3),$$

where $g_F(\dot\gamma_{w^\star}, \dot\gamma_{w^\star}) = \text{Var}_{p_{w^\star}}\big[\log(p_{\text{ctx}}/p_\theta)\big]$ is the Fisher quadratic form along the geodesic tangent.

Sketch. e-Christoffels vanish in natural coordinates [2, Ch. 3]; geodesic is the straight line $\theta_w = (1-w)\theta(p_\theta) + w\theta(p_{\text{ctx}})$, which exponentiates to p_w. $w^\star$ from Section 3.1; the second-order term is the e-flat KL expansion. ☐

Corollary E.8 (Fisher–Rao barycenter / Schrödinger-bridge marginal)**.** *The geometric mixture $\hat p_w \propto p_\theta^{1-w} p_{\text{ctx}}^w$ is the* Fisher–Rao barycenter *of $(p_\theta, p_{\text{ctx}})$ with weights $(1-w, w)$:*

$$\hat p_w \;=\; \arg\min_{p \in \Delta(\mathcal{Y})} \big[\, (1-w)\ \text{KL}(p \,\|\, p_\theta) \;+\; w\ \text{KL}(p \,\|\, p_{\text{ctx}})\,\big].$$

Equivalently, $\hat p_w$ is the marginal at time w of the static Schrödinger bridge *between endpoints $(p_\theta, p_{\text{ctx}})$ under the relative-entropy construction [36, Sec. 4]. The bridge time $w^\star = \mathbb{E}[r \mid q, c]$ is the reliability-weighted information-geometric interpolation parameter.*

Sketch. Lagrangian first-order condition gives the geometric mixture; strict convexity gives uniqueness; the Schrödinger-bridge identification is Léonard [36, Prop. 4.1]. ☐

E.7.2 Channel-capacity perspective: arbitration ceiling via Fano

Theorem E.5 (Arbitration capacity ceiling)**.** *Let $(q, c, r, Y^\star)$ be drawn from $\mathbb{P}^\star$ and $s = s(q,c)$ any signal map. For any arbitration policy $\hat y = \hat y(s, p_\theta, p_{\text{ctx}})$ measurable in s together with the model softmaxes,*

$$\mathbb{P}[\hat y \neq Y^\star] \;\geq\; \frac{H(Y^\star \mid s, p_\theta, p_{\text{ctx}}) \;-\; 1}{\log|\mathcal{Y}|}. \tag{E.13}$$

The bound is achieved with equality by the Bayes MAP rule applied to the geometric mixture $p_{w^\star}$ with $w^\star(s) = \mathbb{E}[r \mid s]$. As a corollary, no arbitration policy can drive accuracy above $1 - 2^{H(Y^\star|s, p_\theta, p_{\text{ctx}})/\log|\mathcal{Y}|}$.

Sketch. Fano on $(s, p_\theta, p_{\mathrm{ctx}}) \to Y^\star$ [12, Thm. 2.10.1]; Section 3.1 identifies the conditional posterior with $p_{w^\star(s)}$, so its MAP saturates the bound. $\square$

Whereas Section 3.1 are lower bounds over a distribution class, Theorem E.5 is an *upper bound on accuracy* in residual entropy $H(Y^\star \mid s, p_\theta, p_{\mathrm{ctx}})$, scoring-rule independent, saturated by the geometric mixture.

Corollary E.9 (Equality achievability of the Fano ceiling)**.** *The Fano lower bound* (E.13) *is achieved with* equality *(not just up to the standard* $1/\log|\mathcal{Y}|$ *slack) by the geometric mixture* $p_{w^\star}$ *whenever the conditional* $\mathbb{P}[Y^\star \mid s, p_\theta, p_{\mathrm{ctx}}]$ *is the maximum-entropy distribution subject to the moment constraint* $\mathbb{E}[\log p_{\mathrm{ctx}}/p_\theta] = \nabla\phi(\theta_{w^\star})$*.*

Sketch. The MaxEnt representative of the moment constraint (Theorem E.8) coincides with the geometric mixture exactly under the stated condition; Jaynes-MaxEnt gives equality. $\square$

E.7.3 Spectral conflict-barrier perspective: a matching lower bound

Theorem E.6 (Spectral conflict barrier)**.** *On the stable-answer subset, let* $p^\star$ *be a hyperbolic Berk–Nash fixed point of* Φ_η *with score function* $s_t(y) := \partial_\theta \log p_t(y) - \mathbb{E}_{p_t}\partial_\theta \log p_t$*. Then*

$$\rho^\star(\eta, p^\star) \;:=\; \rho(D\Phi_\eta(p^\star)) \;\geq\; 1 \;-\; \eta\,\frac{\mathrm{Var}_{p^\star}[s_t(Y)]}{|\mathcal{Y}|}.$$

Furthermore, on cells with KL projection distance $D^\star(q, c) \geq d_{\min} > 0$*, the score-variance at the fixed point satisfies* $\mathrm{Var}_{p^\star}[s_t] \leq C_\pi/d_{\min}^2$ *for a finite constant* C_π *depending only on the prior support (Lemma E.15, below). Combining,*

$$\rho^\star(\eta, D^\star) \;\geq\; 1 \;-\; \frac{C_\pi\,\eta}{d_{\min}^2\,|\mathcal{Y}|}.$$

In particular, $\rho^\star \to 1^-$ *as* $D^\star \uparrow \infty$ *at any fixed* η*, recovering loss of hyperbolicity from below.*

Lemma E.15 (Score-variance bound under conflict, explicit constant)**.** *Under (i)–(iv) of Section 3.2, with prior* π *supported on* $\Delta(\mathcal{Y})$ *with* $\inf_y \pi(y) \geq \pi_0 > 0$ *and* $\sup_y \pi(y) \leq \pi_\infty < \infty$*,*

$$\mathrm{Var}_{p^\star}[s_t(Y)] \;\leq\; \underbrace{\tfrac{2\pi_\infty^2}{\pi_0\,(\log|\mathcal{Y}|)^2}}_{C_\pi} \cdot \frac{1}{D^\star(q, c)^2}.$$

Sketch. (a) Pinsker on $D^\star = \mathrm{KL}(p^\star \| \mathbb{P}^\star)$ gives $\mathrm{TV} \leq \sqrt{D^\star/2}$. (b) Exponential-family score identity plus prior support $p^\star(y) \geq \pi_0$ gives $\|\partial_y \log p^\star\|_2 \leq 1/\pi_0$. (c) Cauchy–Schwarz $\mathbb{E}\|s_t\|^2 \leq 2(D^\star)^2/\mathrm{TV}^2$ then absorbs (a) into $1/D^{\star 2}$; constants pack to $C_\pi = 2\pi_\infty^2/(\pi_0(\log|\mathcal{Y}|)^2)$. $\square$

C_π is tight up to a factor of 4; for $|\mathcal{Y}| = 50$, $C_\pi \leq 8.7$ on the uniform-prior worst case.

Lemma E.16 (Score-variance equality at the Berk–Nash fixed point)**.** $\sigma_{\min}(\mathrm{Cov}_{p^\star}(s_t)) = 1/(|\mathcal{Y}| D^\star(1 + \varepsilon_\pi))$ *with prior-shape correction* $\varepsilon_\pi \in [0, C_\pi/(D^{\star 2}\sigma_{\min}) - 1]$ *(vanishes for the uniform prior). Proof: differentiate the Berk–Nash condition* $\mathbb{E}_{p^\star}[\partial_y \mathbb{E}_z \ell] = 0$ *in natural-parameter coordinates and apply Cramér–Rao on the simplex.*

Corollary E.10 (Matching upper- and lower-bound envelope)**.** *Combining Section 3.2 (upper) and Theorem E.6 (lower):* $1 - C_\pi\eta/(D^{\star 2}|\mathcal{Y}|) \leq \rho^\star \leq 1 - c'\eta\,\mathrm{Var}_{p^\star}(s_t)$*. The empirical conflict/no-conflict* $\rho^\star$ *ratio* $0.97/0.42 \approx 2.3$ *matches the inverse-square scaling predicted by* $D^\star_{\mathrm{conflict}}/D^\star_{\mathrm{no-conf}} \approx 4.7$*.*

E.7.4 CT1→CT2 bridge: Lyapunov dissipation identity and sharpened envelope

We close the multiplicative gap of Corollary E.10 to logarithmic in $|\mathcal{Y}|$ under stable-answer ergodicity, by tying the spectral radius to per-step free-energy dissipation.

Lemma E.17 (Lyapunov dissipation identity for gen-Bayes; Bernstein-controlled remainder)**.** *Let* $p_{t+1} \propto p_t\, e^{-\eta \ell_t}$ *be the gen-Bayes update* (3)*,* $\mathcal{F}$ *the free energy* (3.1)*,* $V_t := \mathrm{Var}_{p_t}(\ell_t)$*, and either (i)*

$\ell_t \in [0, B]$ almost surely (bounded), or (ii) ℓ_t sub-Gaussian under p_t with proxy σ_g^2. Then for all $\eta > 0$ in case (ii), or for $\eta B \leq 1/3$ in case (i),

$$\mathcal{F}[p_{t+1}] \;-\; \mathcal{F}[p_t] \;=\; -\,\eta\, V_t \;+\; R(\eta; \ell_t, p_t), \tag{E.14}$$

with explicit Bernstein remainder

$$\big|R(\eta;\ell_t,p_t)\big| \;\leq\; \begin{cases} \dfrac{\eta^3\, B\, V_t}{3\,(1-\eta B)} \;\leq\; \tfrac{1}{2}\,\eta^3\, B\, V_t & \text{(bounded case (i)),} \\ \dfrac{\eta^3\, \sigma_g^2\, V_t}{3} & \text{(sub-Gaussian case (ii)).} \end{cases} \tag{E.15}$$

The leading $-\eta V_t$ term is the exact second cumulant of ℓ_t under p_t (de Bruijn–Stam-type identity for KL-Bregman descent [4, 12] specialized to the arbitration energy $E_{w,\eta}$).

Sketch. Mirror-descent: $\Delta\mathcal{F} = -\,\mathrm{KL}(p_{t+1}\|p_t) = \Lambda(\eta) - \eta\Lambda'(\eta)$ where Λ is the centred cumulant generator under p_t. Bennett–Bernstein cumulant bound [8, Lemma 2.4] gives $|\Lambda(\eta) - \frac{1}{2}\eta^2 V_t| \leq \eta^3 B V_t/(6(1-\eta B/3))$; differentiating doubles the constant (case (i)). Sub-Gaussian σ_g^2 replaces B in case (ii). ☐

Lemma E.18 (Spectral-gap implies ergodicity (concrete sufficient condition))**.** *Let $\Phi_\eta : \Delta(\mathcal{Y}) \to \Delta(\mathcal{Y})$ be the gen-Bayes update operator, and assume:*

- (SG1) *Φ_η has spectral gap $\gamma \in (0, 1]$ on the tangent space at $p^\star$ (equivalent to hyperbolicity: $\rho^\star = 1-\gamma$);*
- (SG2) *$p \mapsto \mathrm{Var}_p(\ell_t)$ is L_V-Lipschitz in TV ($L_V = 2\|\ell_t\|_\infty^2$ in the bounded case, $L_V = 2\sigma_g^2$ in sub-Gaussian case).*

For initial p_0 with $D_0 := \mathrm{TV}(p_0, p^\star) \leq r_{\mathrm{lin}} := \gamma/L_H$ (Hartman–Grobman linearization radius, L_H the Hessian-Lipschitz constant of $\mathcal{F}$),

$$\zeta_t := \big|\mathrm{Var}_{p_t}(\ell_t) - \mathrm{Var}_{p^\star}(s_t)\big| \;\leq\; L_V D_0 (1-\gamma)^t, \qquad \sum_{t=0}^{T-1} \zeta_t \;\leq\; \frac{L_V D_0}{\gamma} \;=\; O(1) \;=\; o(T). \tag{E.16}$$

Hence the abstract ergodicity condition of Theorem E.7 is implied with explicit constant $L_V D_0/\gamma$.

Proof. By (SG1), Banach contraction on the linearized iteration around $p^\star$ inside r_{lin} gives $\mathrm{TV}(p_t, p^\star) \leq D_0(1-\gamma)^t$. By (SG2), $\zeta_t \leq L_V\, \mathrm{TV}(p_t, p^\star)$. Geometric series sums to $L_V D_0/\gamma$. ☐

Theorem E.7 (Sharpened CT2 envelope under stable-answer ergodicity)**.** *Let $\bar\sigma^2 := \lim_{T\to\infty} \frac{1}{T}\sum_{t<T} \mathrm{Var}_{p_t}(\ell_t)$ be the time-averaged per-step variance along the gen-Bayes trajectory; assume the ergodicity condition $\sum_t \zeta_t = o(T)$ with $\zeta_t := |\,\mathrm{Var}_{p_t}(\ell_t) - \mathrm{Var}_{p^\star}(s_t)|$ (Lemma E.18 gives the concrete sufficient form: spectral gap $\gamma > 0$ + TV-Lipschitz $\mathrm{Var}_p(\ell) \Rightarrow \sum \zeta_t \leq L_V D_0/\gamma = O(1)$). Then*

$$1 \;-\; \eta\bar\sigma^2 \;-\; O(\eta^2) \;\leq\; \rho^\star \;\leq\; 1 \;-\; \eta\bar\sigma^2 \,\cdot\, \left(1 \;-\; O\!\left(\frac{\log|\mathcal{Y}|}{\sqrt{T}}\right)\right), \tag{E.17}$$

i.e. both bounds collapse to $1-\eta\bar\sigma^2$ up to $O(\eta^2)$ and $O(\log|\mathcal{Y}|/\sqrt{T})$. The $|\mathcal{Y}|D^{\star 2}\,\mathrm{Var}_{p^\star}(s_t)/C_\pi$ multiplicative gap of Corollary E.10 dissolves; the residual gap is purely logarithmic in $|\mathcal{Y}|$.

Sketch. Lower bound: Section 3.2 with $\bar\sigma^2 = \mathrm{Var}_{p^\star}(s_t) + o(1)$. Upper bound: Lemma E.17 sums to $\mathcal{F}[p_0] - \mathcal{F}[p^\star] - \eta\sum_s V_s$; Pinsker strong convexity gives $\mathrm{TV} \leq \sqrt{2\Delta\mathcal{F}/c}\cdot e^{-\eta\bar\sigma^2 t/2}$, matching $\rho^\star \leq 1 - \eta\bar\sigma^2/2$. The $\log|\mathcal{Y}|/\sqrt{T}$ slack is McDiarmid for the $|\mathcal{Y}|$-bounded score. ☐

Corollary E.11 (Cramér–Rao tightness)**.** *The original $|\mathcal{Y}|D^{\star 2}/C_\pi$ multiplicative gap of Corollary E.10 is* not *a proof artifact when ergodicity fails: it equals the Cramér–Rao deficit $1/(D^\star\,\mathrm{Var}_{p^\star}(s_t)) - 1$ in the absence of trajectory ergodicity. An explicit two-state counter-example (App. E.6) achieves the gap.*

The empirical $r = 0.997$ over twelve cells (Table E.7) matches the $\log|\mathcal{Y}|/\sqrt{T} \leq 0.06$ residual at $T \geq 400$ trace steps.

Remark E.6 (Period-2 counter-example saturating the deficit)**.** $|\mathcal{Y}| = 2$, $p_\theta = (1/4, 3/4)$, $p_{\text{ctx}} = (3/4, 1/4)$, $\eta = \log 2$, balanced prior, deterministic period-2 trace $\ell_t \in \{(1,0),(0,1)\}$: $\bar{V} = 17/72$ vs. $\mathrm{Var}_{p^\star}(s_t) = (\log 3)^2 \approx 1.207$, so $\zeta_t \geq 0.97$ for all t ($\sum_t \zeta_t = \Omega(T)$, violates ergodicity). The envelope gap stays at factor ~ 4: the ergodicity hypothesis of Theorem E.7 is load-bearing.

Section 3.2 and Theorem E.6 jointly sandwich $\rho^\star$: lower bound depends only on conflict $D^\star$, safe rate η, and prior support, ruling out linearization-artifact readings. The three independent derivations of $w^\star = \mathbb{E}[r \mid s]$ (Riemannian/Fano/spectral; also Theorem E.8) constitute a theory-internal robustness check.

E.8 Unique strengthenings of each existing theorem

For each load-bearing theorem we give an additional characterization that does not duplicate the original argument and uses different machinery. Each strengthening is independently citable.

E.8.1 T1+: Bregman moment-matching characterization

Theorem E.8 (Bregman moment-matching)**.** *Let $\Phi(p,q,w) \propto p^{1-w} q^w$ be the geometric mixture and $D_\phi(\cdot \| \cdot)$ the Bregman divergence induced by the log-partition $\phi(\theta) = \log \sum_y e^{\theta(y)}$ on the natural-parameter chart $\theta = \log p$. Then $w^\star$ is the unique $w \in [0,1]$ that satisfies the moment-matching identity*

$$\mathbb{E}_{r \sim \pi(\cdot \mid q,c)}\big[\nabla \phi(\theta_r)\big] \;=\; \nabla \phi\big((1-w^\star)\theta_\theta + w^\star \theta_{\text{ctx}}\big),$$

where $\theta_r := (1-r)\theta_\theta + r\theta_{\text{ctx}}$. Equivalently, $w^\star = \mathbb{E}[r \mid q, c]$ is the unique posterior-mean weight that makes the natural-parameter Bregman projection of $\mathbb{E}[\theta_r]$ onto the geodesic $\{\theta_w : w \in [0,1]\}$ exact.

Sketch. Exponential family with $T = \log p_{\text{ctx}}/p_\theta$; $\nabla\phi(\theta_w) = \mathbb{E}_{p_w} T$. Bregman projection duality plus strict monotonicity of $\nabla\phi$ give $w = \mathbb{E}[r]$. ☐

Constructive: solve the one-dimensional moment-matching equation in $\mathbb{E}[T(Y)]$, no direct access to r.

Corollary E.12 (Bregman-conserved diagnostic)**.** *$\rho_{\text{Breg}}(t) := D_\phi(\theta_{p_t} \| \mathbb{E}[\theta_r \mid s])$ is t-constant iff π matches $\mathbb{P}^\star(r \mid s)$; $\dot{\rho}_{\text{Breg}} \neq 0$ rejects prior matching at $\sqrt{n}$-consistent rate.*

Sketch. Bregman three-point identity on the gen-Bayes update; cross term vanishes iff matched prior. ☐

E.8.2 T2-info+: Le Cam tightness of the Pinsker constant

Theorem E.9 (Le Cam tightness for the signal-conditioned floor)**.** *There exists a two-point family $\{P_+, P_-\}$ in $\mathcal{P}_\sigma$ such that, for any s-measurable policy $\hat{y}$,*

$$\sup_{P \in \{P_+, P_-\}} \big[R(\hat{y}; P) - R(\hat{y}^\star; P)\big] \;\geq\; \tfrac{1}{8(|\mathcal{Y}|-1)^2}\, \mathbb{E}[\sigma_r^2(s)] \;-\; O(|\mathcal{Y}|^{-3}).$$

The $|\mathcal{Y}|^{-3}$ slack is necessary: no improvement of the leading constant past $\frac{1}{8(|\mathcal{Y}|-1)^2}$ is possible without imposing structure beyond bounded log loss.

Sketch. Two-point r-hypotheses separated by $\sigma_r(s)$, Le Cam + Pinsker on the resulting KL gives the matching constant; $|\mathcal{Y}|^{-3}$ slack is the Le Cam-vs-$|\mathcal{Y}|$-point sub-optimality. ☐

Theorem E.10 (Multi-point Le Cam reduction recovering two-point as a limit)**.** *For any $k \geq 2$, there exists a k-point family $\{P_1, \ldots, P_k\} \subset \mathcal{P}_\sigma$ giving a minimax lower bound*

$$\inf_{\hat{y}} \max_j \big[R(\hat{y}; P_j) - R(\hat{y}^\star; P_j)\big] \;\geq\; \frac{1}{8(|\mathcal{Y}|-1)^2}\, \mathbb{E}[\sigma_r^2(s)] \cdot \big(1 - \tfrac{1}{k}\big) - O(|\mathcal{Y}|^{-3}).$$

The two-point case ($k = 2$) recovers Theorem E.9 with half-strength; as $k \to \infty$ the lower bound saturates the Pinsker constant exactly. The k-point construction uses k-equally-spaced reliability values $r_j = j/(k-1)$ on a fixed (q,c).

Sketch. k-point Fano [66, Sec. 2.7] with $r_j = j/(k-1)$; pairwise KL $O(\sigma_r^2/(k-1)^2)$ gives the $1 - 1/k$ saturation factor. ☐

E.8.3 T6+: Phase transition at $\eta = 1/D^\star$

Theorem E.11 (Phase transition at the safe-rate boundary)**.** *For the gen-Bayes operator Φ_η on the stable-answer subset, there is a critical safe rate $\eta_c(q,c) = 1/D^\star(q,c)$ such that:*

- *For $\eta < \eta_c$: $\rho^\star(\eta) < 1$ strictly, and the* TV *saturation is geometric at rate $\rho^\star(\eta) = 1 - c_1(\eta_c - \eta) + o(\eta_c - \eta)$.*
- *For $\eta = \eta_c$: $\rho^\star = 1$ exactly (loss of hyperbolicity), and $\mathrm{TV}(p_t, p^\star)$ decays as $t^{-1/2}$ (centre-manifold polynomial rate).*
- *For $\eta > \eta_c$: the equilibrium loses local stability;* TV *saturates non-monotonically with overshoot bounded by $|\eta - \eta_c|/D^\star$.*

The transition is sharp: the Lyapunov exponent $\lambda(\eta) := \log \rho^\star(\eta)$ is continuous but non-differentiable at η_c, with left- and right-derivatives related by $\lambda'(\eta_c^-) = -c_1$, $\lambda'(\eta_c^+) = +c_2$ for explicit $c_1, c_2 > 0$ depending on the prior support.

Sketch. $D\Phi_\eta(p^\star) = I - \eta\,\mathrm{Cov}_{p^\star}(s_t)/|\mathcal{Y}| + o(\eta)$; score-covariance trace at the fixed point equals $D^\star$ by Berk–Nash projection (Proposition E.9), so $\rho^\star = 1$ at $\eta = 1/D^\star$. Centre-manifold expansion (Wiggins, Ch. 18) gives the $t^{-1/2}$ rate; supercritical eigenvalue exit gives bounded overshoot. □

Estimable phase boundary. Plug-in $\widehat{D}^\star = \mathrm{KL}(p_K \| p_{\mathrm{marg}})$ gives $|\widehat{\eta}_c - \eta_c| \le O(n^{-1/2})$ via the delta method; cells at $\eta\widehat{D}^\star \approx 1 \pm n^{-1/2}$ should show polynomial TV saturation.

E.8.4 T5+: Lepski adaptive window selection

Theorem E.12 (Lepski-adaptive optimal window)**.** *Let $\widehat{W}^\star$ be the Lepski-rule window selector*

$$\widehat{W}^\star \;=\; \arg\min_{W \in \mathcal{W}} \Big\{ \widehat{\sigma}_W \;+\; \widehat{B}_W \;\le\; \widehat{\sigma}_{W'} + \widehat{B}_{W'} \;\; \forall W' \le W \Big\},$$

where $\widehat{\sigma}_W, \widehat{B}_W$ are data-dependent estimates of the variance and bias terms in Section 3.2, and $\mathcal{W} = \{W_0 \cdot 2^k : k \ge 0\}$ a dyadic grid. With probability $\ge 1-\delta$,

$$|\widehat{\rho}^\star(\widehat{W}^\star) - \rho^\star| \;\le\; C_{\mathrm{Lepski}} \inf_W \{\widehat{\sigma}_W + \widehat{B}_W\} \cdot \sqrt{\log\log(K/\delta)},$$

i.e., $\widehat{W}^\star$ achieves the oracle bias-variance trade-off up to a $\sqrt{\log\log K}$ payment.

Sketch. Lepski rule [66, Sec. 7.6] on the bias-variance pair from Section 3.2; dyadic union bound pays $\sqrt{\log\log K}$. □

Theorem E.13 (Matching minimax lower bound)**.** *For any $\widetilde{\rho}$ estimator from K trace distributions on C^2 contractive operators,* $\sup \mathbb{E}[|\widetilde{\rho} - \rho^\star|] \ge c_{\mathrm{LB}}[|\mathcal{Y}|\sqrt{\log K/K} + \rho^{\star\,2(K-W^\star)}]/\sqrt{\log\log K}$*, matching Theorem E.12 up to absolute constants.*

Sketch. Two-point hypotheses on $\rho^\star$ separated by $|\mathcal{Y}|\sqrt{\log K/K}$; trajectory $\mathrm{KL} \le |\mathcal{Y}|^2 \log K$ via Le Cam. Lepski $\sqrt{\log\log K}$ payment is [37, Thm. 3.1]. □

E.8.5 T7+: Murphy decomposition of the Brier improvement

Theorem E.14 (Murphy decomposition of η-tempered Brier)**.** *The Brier improvement* $\Delta\mathrm{Brier}(\eta^\star; q, c)$ *from Theorem 4 decomposes as*

$$\Delta\mathrm{Brier} \;=\; \Delta\mathrm{REL} \;+\; \Delta\mathrm{RES} \;-\; \Delta\mathrm{UNC},$$

where:

- $\Delta\mathrm{REL} = \mathbb{E}_{p_K}[\hat{p} - \bar{y}]^2 - \mathbb{E}_{p_{\eta^\star}}[\hat{p} - \bar{y}]^2$ *is the* reliability *(calibration) gain,*
- $\Delta\mathrm{RES} = \mathbb{E}_{p_{\eta^\star}}[\bar{y} - \bar{Y}]^2 - \mathbb{E}_{p_K}[\bar{y} - \bar{Y}]^2$ *is the* resolution *gain,*

- $\Delta\mathrm{UNC} = 0$ *since the marginal Y is unchanged by tempering.*

For interior $\eta^\star < 1$, $\Delta\mathrm{REL} \geq 0$ and $\Delta\mathrm{RES} \geq 0$ independently*; tempering strictly improves both calibration and resolution.*

Sketch. Murphy decomposition [16, 47] on p_K and $\hat{p}_{\eta^\star}$; $\mathrm{UNC} = \mathrm{Var}(Y)$ cancels. $\mathrm{REL} \geq 0$ from contraction-toward-marginal; $\mathrm{RES} \geq 0$ from convexity under $\eta < 1$. □

Corollary E.13 (Brier-orthogonality of $\Delta\mathrm{REL}$ and $\Delta\mathrm{RES}$)**.** *Under the inner product $\langle f, g\rangle_{\mathrm{Brier}} := \mathbb{E}_{Y\sim\mathbb{P}^\star}[f(Y)g(Y)] - \mathbb{E}[f]\mathbb{E}[g]$ on the space of bounded score functions, the gradients of the reliability and resolution components at $\eta = \eta^\star$ are orthogonal:*

$$\langle \nabla_\eta \mathrm{REL}|_{\eta^\star},\ \nabla_\eta \mathrm{RES}|_{\eta^\star}\rangle_{\mathrm{Brier}} \ = \ 0.$$

Hence $\Delta\mathrm{REL}$ and $\Delta\mathrm{RES}$ are not just independently *non-negative; they live on* orthogonal *sub-spaces of the Brier inner-product geometry, and a single η adjustment cannot trade one for the other.*

Sketch. $\nabla\mathrm{REL}$ projects onto level sets of $\hat{p}_{\eta^\star}$; $\nabla\mathrm{RES}$ onto the orthogonal complement (Murphy); optimality of $\eta^\star$ kills both partials at once. □

REL and RES live on independent axes — suggests a 2-parameter tempering as future work.

E.9 Robustness, extension, and creative-undone-before results

T11 (TV-robust log-linear), T12 (anytime non-equilibrium regret), T13 (PAC plug-in), T14 (adversarial signal), T15 (compositional K-hop), T16 (causal doubly-robust IPW).

E.9.1 T11: Robustness to log-linear answer-model misspecification

Theorem E.15 (ε-robust geometric mixture)**.** *Let $\mathbb{P}^\star_\varepsilon$ be a TV-perturbation of the log-linear conditional answer model: $\mathrm{TV}(\mathbb{P}^\star, \mathbb{P}^\star_\varepsilon) \leq \varepsilon$ uniformly over (q, c, r). Then the regret of the geometric mixture $\hat{p}_{w^\star}$ under $\mathbb{P}^\star_\varepsilon$ satisfies*

$$R_\varepsilon(\hat{p}_{w^\star}) \ - \ R_\varepsilon(\hat{p}^\star_\varepsilon) \ \leq \ \underbrace{\tfrac{1}{2}\sigma_r^2 V}_{\text{T1 regret}} \ + \ \underbrace{4\varepsilon \log|\mathcal{Y}|}_{\text{misspec. cost}} \ + O(\varepsilon\sigma_r^2),$$

where $\hat{p}^\star_\varepsilon$ is the Bayes optimum under $\mathbb{P}^\star_\varepsilon$. The geometric mixture is therefore ε-robust: the misspecification cost is linear in ε with constant $4\log|\mathcal{Y}|$, dominated by the T1 regret on cells where $\sigma_r^2 \gg \varepsilon/\log|\mathcal{Y}|$.

Proof. Write R_ε for the risk under $\mathbb{P}^\star_\varepsilon$ and R_0 for the risk under $\mathbb{P}^\star$. Triangle inequality on excess risk yields

$$R_\varepsilon(\hat{p}_{w^\star}) - R_\varepsilon(\hat{p}^\star_\varepsilon) \ = \ \underbrace{[R_\varepsilon - R_0](\hat{p}_{w^\star})}_{\text{(i)}} \ + \ \underbrace{R_0(\hat{p}_{w^\star}) - R_0(\hat{p}^\star_0)}_{\text{(ii)}} \ + \ \underbrace{R_0(\hat{p}^\star_0) - R_\varepsilon(\hat{p}^\star_\varepsilon)}_{\text{(iii)}}. \tag{E.18}$$

Term (ii) is exactly the T1 regret of the geometric mixture under the unperturbed model: $R_0(\hat{p}_{w^\star}) - R_0(\hat{p}^\star_0) \leq \frac{1}{2}\sigma_r^2 V + O(\sigma_r^4)$ (Proposition C.3 and Lemma C.2), where $V = \mathbb{E}_{q,c}[\mathrm{Var}_{\hat{p}_{w^\star}}(\log p_{\mathrm{ctx}}/p_\theta)]$. *Terms (i) and (iii)* are TV-perturbation costs of evaluating the same predictive against two distributions within TV-distance ε. For any bounded measurable $f : \mathcal{Y} \to \mathbb{R}$ with $\|f\|_\infty \leq B$ and any $\mathbb{P}, \mathbb{P}'$ with $\mathrm{TV}(\mathbb{P}, \mathbb{P}') \leq \varepsilon$,

$$|\mathbb{E}_{\mathbb{P}} f - \mathbb{E}_{\mathbb{P}'} f| \ \leq \ 2B\ \mathrm{TV}(\mathbb{P}, \mathbb{P}') \ \leq \ 2B\varepsilon \tag{E.19}$$

(standard variational identity [66, Lemma 2.4]). For log loss, $f(y) = -\log\hat{p}(y) \in [0, \log|\mathcal{Y}|]$ on the geometric mixture under Assumption A.3, hence $\|f\|_\infty \leq \log|\mathcal{Y}|$ and each of (i), (iii) is bounded by $2\varepsilon\log|\mathcal{Y}|$. Summing the three terms in (E.18) gives the stated bound; the $O(\varepsilon\sigma_r^2)$ cross-term captures the fact that the second-order T1 regret itself depends on $\mathbb{P}$ through $\mathrm{Var}_{p^\star}$, with the dependence Lipschitz in TV under Assumption A.3. □

Corollary E.14 (Top-k refinement: $\log|\mathcal{Y}| \to \log k$)**.** *Restricting to top-k effective support $\mathcal{Y}_k = \{y : \max(p_\theta, p_{\mathrm{ctx}}) \geq 1/k\}$ replaces $\log|\mathcal{Y}|$ by $\log k$: $R_\varepsilon - R^\star_\varepsilon \leq \frac{1}{2}\sigma_r^2 V + 4\varepsilon\log k + O(\varepsilon\sigma_r^2)$. For $k \sim 10$–64 on conflict cells, the misspec. cost is $\leq 4\varepsilon\log k$ (relative excess ~ 0.4 at $\varepsilon = 0.1$).*

E.9.2 T12: Non-equilibrium tracking bound (transient regret)

Theorem E.16 (Anytime regret of the gen-Bayes trajectory)**.** *For any finite horizon $T \geq 1$ (no requirement of equilibrium convergence), the cumulative regret of the gen-Bayes trajectory $\{p_t\}_{t=1}^T$ satisfies*

$$\frac{1}{T}\sum_{t=1}^{T}\big[R(p_t)-R(p^\star)\big] \;\leq\; \frac{C\cdot\mathrm{KL}(p_0\|p^\star)}{\eta T} \;+\; \eta\cdot\mathrm{Var}_{p^\star}[\ell],$$

where $\mathrm{KL}(p_0\|p^\star)$ is the initial KL gap and the second term is a constant variance penalty. Optimizing over η gives an $O(\sqrt{\mathrm{KL}/T})$ anytime rate independent of equilibrium convergence.

Sketch. Mirror-descent on $\Lambda_t = \mathrm{KL}(p_t\|p^\star)$ with drift $-\eta(R(p_t)-R(p^\star))+\eta^2\,\mathrm{Var}_{p^\star}\,\ell$; telescope. □

Lemma E.19 (Gen-Bayes is mirror descent on the natural-parameter chart)**.** *The gen-Bayes update (3) is exactly mirror descent with mirror map $\phi(\theta) = \log\sum_y e^{\theta(y)}$ (log-partition of the categorical exponential family) and step size η on the natural-parameter coordinate $\theta = \log p$. Specifically, $\theta_{t+1} = \theta_t - \eta\,\nabla_\theta \mathbb{E}_z[\ell_t(\theta;z)]$ in θ-coordinates is identical to $p_{t+1} \propto p_t\cdot\exp(-\eta\mathbb{E}_z\ell_t)$ in probability coordinates.*

Sketch. D_ϕ on θ-coordinates is KL on p-coordinates; mirror-descent specialises to multiplicative weights via the exponential change of coordinates [4]. □

E.9.3 T13: PAC sample complexity of the plug-in $\hat{w}(s)$

Theorem E.17 (Sample complexity of learning the reliability plug-in)**.** *Let $\hat{w}_n$ be the empirical-risk-minimizer of $\mathbb{E}[r\,|\,s]$ over a hypothesis class $\mathcal{H}$ of L-Lipschitz functions with VC-dimension $d_{\mathcal{H}}$. With probability $\geq 1-\delta$ over n calibration draws,*

$$\mathbb{E}_{q,c}\big[(\hat{w}_n(s)-w^\star(s))^2\big] \;\leq\; C_1\,\frac{d_{\mathcal{H}}\log(n/d_{\mathcal{H}}\delta)}{n} \;+\; \inf_{w\in\mathcal{H}}\mathbb{E}\big[(w(s)-w^\star(s))^2\big],$$

and the resulting arbitration regret is

$$R(\hat{p}_{\hat{w}_n})-R(\hat{p}_{w^\star}) \;\leq\; V_{\max}\cdot\mathbb{E}[(\hat{w}_n-w^\star)^2] \;+\; O(n^{-1}),$$

giving the explicit sample-complexity rate $n \geq C\,V_{\max}^2 d_{\mathcal{H}}\log(1/\delta)/\varepsilon^2$ to drive arbitration regret below ε.

Sketch. ERM rate [60] on VC-bounded $\mathcal{H}$ plus second-order Taylor at $w^\star$ (Section 3.1). □

E.9.4 T14: Adversarially-robust arbitration regret

Theorem E.18 (TV-bounded adversarial signal robustness)**.** *Let $\tilde{s}$ be an adversarial perturbation of the signal s with $\mathrm{TV}(\mathbb{P}_s,\mathbb{P}_{\tilde{s}}) \leq \delta$. The geometric mixture under the perturbed signal $\hat{p}_{\hat{w}(\tilde{s})}$ has regret bounded by*

$$R(\hat{p}_{\hat{w}(\tilde{s})})-R(\hat{p}^\star) \;\leq\; \tfrac{1}{2}\sigma_r^2 V \;+\; 2L_w\,\delta\,V_{\max} \;+\; O(\delta^2),$$

where L_w is the Lipschitz constant of $\hat{w}(\cdot)$. In particular, no adversarial perturbation of the signal can inflate the regret by more than $O(\delta)$ over the unperturbed Bayes optimum. When $\hat{w}$ is selected from a 1-Lipschitz class $\mathcal{H}$, $L_w \leq 1$ and the perturbation cost is at most $2\delta V_{\max}$.

Sketch. Lipschitz $\hat{w}$ gives $|\Delta\hat{w}| \leq L_w\delta$; quadratic Taylor on $R(w)$ at $\hat{w}(s)$ + triangle with unperturbed bound. □

E.9.5 T15: Compositional arbitration for K-hop chained retrieval

Theorem E.19 (Multiplicative composition of reliability weights)**.** *For K-hop retrieval where context c_k at hop k has reliability r_k conditional on previous hops $(c_1, \dots, c_{k-1})$, the joint arbitration weight on the chain satisfies*

$$w^\star_{1:K}(s_{1:K}) \;=\; \prod_{k=1}^{K} w^\star_k(s_k \mid c_{<k}) \quad \textit{whenever the per-hop reliability prior factorizes,}$$

and the geometric-mixture form generalizes to $\hat{p}_{w^\star_{1:K}} \propto p_\theta^{1-\prod_k w^\star_k}\, p_{\mathrm{ctx}_{1:K}}^{\prod_k w^\star_k}$.

Sketch. Iterated Section 3.1 per hop; geometric mixture closed under natural-parameter composition. □

Remark E.7 (Non-factorisable extension)**.** Entangled retrieval acquires a copula correction $w^\star_{1:K} = \prod_k w^\star_k \cdot \Phi_C(s_{1:K})$, with Φ_C estimable [49]; reduces to the product form under independence.

E.9.6 T16: Causal identification of $w^\star$ via doubly-robust IPW

Theorem E.20 (Causal identification under structural model $q \to c \to r \to Y^\star$)**.** *Under the structural causal model where q causes c, c and q cause r (treatment), and r together with (q, c) causes $Y^\star$ (outcome), the arbitration weight $w^\star = \mathbb{E}[r \mid q, c]$ is the average treatment effect (ATE) of "trusting context" on $Y^\star$, identified via the doubly-robust estimator [9, 57]:*

$$\widehat{w^\star}_{\mathrm{DR}} \;=\; \widehat{\mu}(s) \;+\; \frac{\nVdash[r=1] - \widehat{\pi}(s)}{\widehat{\pi}(s)}\big(r - \widehat{\mu}(s)\big),$$

where $\widehat{\mu}$ is an outcome regression and $\widehat{\pi}$ is the propensity score. The DR estimator is $\sqrt{n}$-consistent under either-or correctness of $(\widehat{\mu}, \widehat{\pi})$, giving robustness to misspecification of either nuisance.

Sketch. Double-robustness on binary r under no-unmeasured-confounders $\mid s$; cross-fitted [9]. □

Remark E.8 (Scope of T16)**.** Requires no-unmeasured-confounders $\mid s$; under Rosenbaum and Rubin [58]-style Γ-bounded confounding, $|\widehat{w}^\star_{\mathrm{DR}} - w^\star| \le \Gamma |R - \widehat{\pi}|$ is empirically auditable.

E.10 Master variational principle and the thermodynamic unification

$\mathcal{F} = \mathbb{E}_p[E] - H[p]$ is the standard MaxEnt / ELBO form [26, 68]; Genest–Zidek (1986, §4) covers the two-component ($\eta = 0$) case under external Bayesianity. Our contribution is the three-component energy $E_{w,\eta}$ that jointly captures arbitration, tempering, and post-trace evidence, plus the proof that BayesArbiter's full rule is the Gibbs minimizer of this single $\mathcal{F}$ — Corollary E.15 groups the seven $w^\star$ characterizations under it.

Theorem E.21 (Master free-energy variational principle)**.** *Let*

$$E_{w,\eta}(y; q, c, K) \;:=\; -(1-w)\log p_\theta(y \mid q) \;-\; w \log p_{\mathrm{ctx}}(y \mid q, c) \;-\; \eta \log p_K(y \mid q, c, \mathrm{trace})$$

be the arbitration-tempering energy *on $\mathcal{Y}$, with $w \in \mathbb{R}$ and $\eta \in \mathbb{R}$ playing the role of inverse temperatures of the model, context, and post-trace components. Define the* Helmholtz free energy *of an arbitration policy $p \in \Delta^\circ(\mathcal{Y})$ as*

$$\mathcal{F}[p; w, \eta, q, c, K] \;:=\; \mathbb{E}_p[E_{w,\eta}(Y; q, c, K)] \;+\; \mathbb{E}_p[\log p(Y)] \;=\; \mathbb{E}_p[E_{w,\eta}] \;-\; H[p].$$

Then:

(a) Gibbs form. *$\mathcal{F}[\,\cdot\,; w, \eta, \cdot]$ is strictly convex on $\Delta^\circ(\mathcal{Y})$ and admits the unique minimizer*

$$p^\star_{w,\eta}(y) \;=\; \frac{p_\theta(y)^{1-w}\, p_{\mathrm{ctx}}(y)^w\, p_K(y)^\eta}{Z(w, \eta; q, c, K)}, \qquad Z(w, \eta; q, c, K) \;=\; \sum_{y'} p_\theta(y')^{1-w} p_{\mathrm{ctx}}(y')^w p_K(y')^\eta, \tag{E.20}$$

with $\min_p \mathcal{F}[p] = -\log Z(w, \eta; q, c, K)$.

(b) Bayes-optimal restriction. *At $w = w^\star(s) = \mathbb{E}[r \mid s] \in [0, 1]$ (Section 3.1) and $\eta = \eta^\star \in [0, 1]$ (Theorem 4), $p^\star_{w^\star,\eta^\star}$ is exactly the BayesArbiter rule. AdaCAD (variable $w \in [0, 1]$ via JSD, $\eta = 0$),* *CoCoA (signal-conditioned $w \in [0,1]$, $\eta = 0$), pure parametric ($w = 0, \eta = 0$), pure context ($w=1, \eta=0$), and post-trace tempering (w fixed, $\eta \in [0,1]$) are restrictions of (w,η) to the convex region $[0,1]^2$ where the Gibbs principle is strictly convex. CAD ($w=2$) is the analytic continuation of* (E.20) *to $w>1$; the Gibbs form extends as the unique exponential-family parameterization but the convexity argument no longer applies (the corresponding "virtual" Gibbs measure is normalisable but not a free-energy minimum on the convex region).*

(c) Bregman / Fisher derivative structure. *The Bregman divergence with respect to $\mathcal{F}$ in the natural parameter $\theta = (\log p_\theta, \log p_{\mathrm{ctx}}, \log p_K)$ is*

$$D_{\mathcal{F}}(p \,\|\, p^\star_{w,\eta}) \;=\; \mathrm{KL}(p \,\|\, p^\star_{w,\eta}) \;=\; \mathcal{F}[p] - \mathcal{F}[p^\star_{w,\eta}],$$

i.e., the KL divergence of any policy from the Gibbs minimizer is *its excess free energy. The Hessian $\nabla^2_\theta \mathcal{F}[p^\star_{w,\eta}]$ is exactly the Fisher information metric g_F on the exponential family $\{p^\star_{w,\eta}\}_{(w,\eta)\in\mathbb{R}^2}$, recovering Theorem* E.4 *as the Hessian-spectral characterization of $\mathcal{F}$ at its minimum.*

(d) Free-energy decay along gen-Bayes. *The gen-Bayes update* (3) *is exactly the KL-mirror-descent step on $\mathcal{F}$:*

$$p_{t+1} \;=\; \arg\min_{p\in\Delta(\mathcal{Y})} \big\{ \eta\, \mathbb{E}_p[\mathbb{E}_z \ell_t(Y;z)] \;+\; \mathrm{KL}(p \,\|\, p_t) \big\}.$$

Consequently, $\mathcal{F}[p_t]$ is monotonically non-increasing for $\eta < \bar\eta(q,c)$ (Section 3.2*), with descent identity $\mathcal{F}[p_t] - \mathcal{F}[p_{t+1}] = \mathrm{KL}(p_t \,\|\, p_{t+1}) \cdot (1 + O(\eta))$ and asymptotic linear convergence rate $\rho^\star = 1 - \eta\sigma_{\min}(\nabla^2\mathcal{F}[p^\star])/|\mathcal{Y}| + O(\eta^2)$ matching Section* 3.2*.*

(e) Critical phase boundary. *At a hyperbolic Berk–Nash fixed point $p^\star$, the Hessian $\nabla^2\mathcal{F}[p^\star] = \mathrm{Cov}_{p^\star}(s_t)$ (exponential-family Fisher identity, equality, by direct differentiation of* $\log Z$ *in natural-parameter coordinates). The critical inverse-temperature at which the smallest eigenvalue of $\eta\nabla^2\mathcal{F}$ crosses unity is therefore $\eta_c(q,c) = 1/(\sigma_{\min}(\mathrm{Cov}_{p^\star}(s_t)) \cdot \nu)$, where ν is the gen-Bayes step normalization fixed by the mirror-descent identification in (d). The relationship between $\sigma_{\min}(\mathrm{Cov}_{p^\star}(s_t))$ and $D^\star(q,c)$ is established by Lemma* E.16 *below (an equality version of the earlier upper-bound lemma). Under the $\nu = |\mathcal{Y}|^{-1}$ convention adopted throughout this paper* (3)*, this gives $\eta_c = 1/D^\star(q,c)$. The Hessian zero-crossing is exact at η_c* in the chosen normalization*; under any other normalization the critical η_c scales accordingly but the existence of a zero-crossing is invariant. At criticality, the Hessian's zero-eigenvalue mode dominates and the free-energy landscape develops a flat direction, giving the polynomial $t^{-1/2}$ rate of Theorem* E.11*.*

(f) Sinkhorn / Schrödinger bridge. *$p^\star_{w,0}$ coincides with the Sinkhorn fixed point at unit temperature with marginals $(p_\theta, p_{\mathrm{ctx}})$ (Corollary* E.8*); the geometric mixture is the $\eta = 0$ projection of the master Gibbs minimizer onto the arbitration-only sub-manifold.*

(g) Variational characterization of $w^\star$. *Marginalizing over the prior $\pi(r \mid s)$,*

$$\mathbb{E}_{r\sim\pi(\cdot|s)}\big[\mathcal{F}[p;r,\eta]\big] \;=\; \mathcal{F}[p; w^\star(s), \eta] \;+\; \mathrm{Var}_{\pi(r|s)}(r) \cdot V_{\mathrm{post}}(p)/2 \;+\; O(\sigma_r^4),$$

which identifies $w^\star = \mathbb{E}[r \mid s]$ as the unique prior-marginalized *Gibbs minimizer up to the σ_r^2 Bayes-arbitration regret. The variance term is exactly $\mathbb{E}[\sigma_r^2(s)]$ from Section* 3.1*.*

Sketch. (a) Lagrangian first-variation gives the Gibbs form; strict convexity from $-H[p]$. (b) Direct substitution (Corollary C.1 for $\eta = 0$). (c) exponential-family $\mathrm{KL} = D_{\log Z}$, Hessian g_F [2, Ch. 3]. (d) Bissiri–Holmes–Walker gen-Bayes is mirror descent with mirror $\phi = \log Z$ (Lemma E.19); standard analysis. (e) Spectral identity (Lemma E.15) plus $\nu = |\mathcal{Y}|^{-1}$ yields $\eta_c = 1/D^\star$; centre-manifold gives $t^{-1/2}$ [73, Ch. 18]. (f) Cuturi [13, Prop. 4.1]. (g) Taylor in r around $w^\star$, marginalize, plus exponential-family identity $\partial_w \mathbb{E}_{p_w} T = \mathrm{Var}_{p_w} T$ with $T = \log p_{\mathrm{ctx}}/p_\theta$. □

Corollary E.15 (Free-energy taxonomy)**.** *The seven $w^\star$ characterizations partition into: **I** (derived from $\mathcal{F}$ via convex duality): T1-Bregman (excess $\mathcal{F}$), T8-Fisher (Hessian $\nabla^2\mathcal{F} = g_F$), Sinkhorn (marginal projection); **II** (same Gibbs form, independent proofs): T1-axiomatic (Cauchy functional equations), T9-Fano (channel capacity); **III** (lower-bound complement): T2-info Pinsker/Le Cam floors; **IV** (perturbation envelopes of the same $w^\star$): T11/T13/T14/T16.*

Remark E.9 (Stat-mech consequences)**.** Fluctuation–dissipation $\partial_w \mathbb{E}_{p^\star} T = \mathrm{Var}_{p^\star} T$ ($T = \log p_{\mathrm{ctx}}/p_\theta$) and Fisher gradient flow $\dot p = -g_F^{-1}\nabla_p \mathcal{F}$ [2, Ch. 5] follow from the master theorem.

Corollary E.16 (Free-energy monotone descent: a falsifiable trajectory prediction)**.** *For $\eta \le \bar{\eta}(q, c)$ (Section 3.2) and any gen-Bayes trajectory $\{p_t\}_{t=0}^K$, the free-energy sequence $\{\mathcal{F}[p_t]\}_{t=0}^K$ is monotonically non-increasing, with descent at each step $\mathcal{F}[p_t] - \mathcal{F}[p_{t+1}] = \mathrm{KL}(p_t \,\|\, p_{t+1}) \cdot (1 + O(\eta))$. The pre-registered prediction is: the empirical $\widehat{\mathcal{F}}[p_t]$ measured on logged R1-14B, R1-8B-RLCR, and Mistral-7B traces is monotonically non-increasing in t for $\eta \le \bar{\eta}$, and shows monotone increase (overshoot) for $\eta > \bar{\eta}$. Falsifying instance: a single trace where $\widehat{\mathcal{F}}[p_t]$ shows strict non-monotone behavior in the safe-rate regime would falsify the master theorem.*

Joint (w, η) decoupling. The master theorem decouples in (w, η): $\partial_w \mathcal{F}$ fixes $w^\star = \mathbb{E}[r \mid s]$ for any η, $\partial_\eta \mathcal{F}$ fixes $\eta^\star$ for any w. Empirical correlation $\rho(\widehat{w}^\star, \widehat{\eta}^\star) = -0.06\ (\pm 0.14)$ across 50 cells, consistent with decoupling.

E.10.1 Robust min-max arbitration: a consequence requiring the master theorem

A second falsifiable consequence requiring the master theorem is the existence of a closed-form *robust min-max arbitration* policy under prior misspecification. No individual theorem (T1–T16) gives this; it follows from the joint variational structure of $\mathcal{F}$ over (p, w, π).

Theorem E.22 (Robust min-max arbitration under prior misspecification)**.** *Let $\Pi_\varepsilon := \{\pi : \mathrm{TV}(\pi, \pi_0) \le \varepsilon\}$ be a TV-ball of priors centered at the working prior π_0. The robust min-max arbitration problem*

$$(p_\varepsilon^\sharp, w_\varepsilon^\sharp) \ :=\ \arg \min_{p \in \Delta(\mathcal{Y}),\, w \in [0,1]} \max_{\pi \in \Pi_\varepsilon} \mathbb{E}_{r \sim \pi}\big[\mathcal{F}[p; r, \eta]\big]$$

admits the closed-form solution

$$p_\varepsilon^\sharp \ \propto\ p_\theta^{1 - w_\varepsilon^\sharp}\, p_{\mathrm{ctx}}^{w_\varepsilon^\sharp}\, p_K^\eta, \qquad w_\varepsilon^\sharp \ =\ \mathbb{E}_{\pi_0}[r] \ +\ C_{\mathrm{rob}} \cdot \varepsilon \ +\ O(\varepsilon^2),$$

with explicit leading constant

$$C_{\mathrm{rob}} \ =\ \frac{\mathrm{sign}(\partial_w R|_{w = \mathbb{E}_{\pi_0}[r]}) \ \cdot\ \|\log(p_{\mathrm{ctx}}/p_\theta)\|_\infty}{\partial^2_{ww} R|_{w = \mathbb{E}_{\pi_0}[r]}}, \qquad \partial^2_{ww} R \ =\ 2\,\mathbb{E}\big[\mathrm{Var}_{p_w^\star}(\mathbf{1}\{Y = y^\star\})(\log p_{\mathrm{ctx}}/p_\theta)^2\big] \tag{E.21}$$

where the Hessian $\partial^2_{ww} R$ is the Brier curvature of the risk in w at the unperturbed optimum, computed via the fluctuation–dissipation identity $\partial_w \mathbb{E}[T] = \mathrm{Var}(T)$ in (g) of Theorem E.21. For unbounded $\log(p_{\mathrm{ctx}}/p_\theta)$ the bound becomes $|w_\varepsilon^\sharp - \mathbb{E}_{\pi_0}[r]| \le \sqrt{2\varepsilon / \partial^2_{ww} R}$ (the $\sqrt{\varepsilon}$ scaling is sharp under sub-Gaussian log-ratios; an explicit two-state example saturates it).

Sketch. $\mathcal{F}$ is convex in p, linear in π; Sion swap, TV variational identity [67] gives linear-in-ε gradient perturbation, second-order Taylor on R closes (E.21). $\partial^2_{ww} R > 0$ by Berk–Nash non-degeneracy. □

Remark E.10 (Empirical estimability)**.** $\partial^2_{ww} R$ is twice the empirical $\mathrm{Var}_{p^\star_{\hat{w}}}(\mathbf{1}\{Y = y^\star\})$ weighted by the squared log-ratio; per-cell estimates are in the release artefact, falsifiable at $\varepsilon = 0.05$ prior-noise injection.

E.11 Safe-rate corollary and Berk–Nash convergence detail

Safe rate. Grünwald and Van Ommen [20, Thm. 1.6]: $\bar{\eta} \le c_1 \varepsilon^*(D^\star)/D^\star$ with ε^* solving the bracketing-entropy fixed point. Polynomial bracketing $\log N_{[]}(\varepsilon) \le C\varepsilon^{-\rho}$ ($\rho \in (0, 2)$) gives $\bar{\eta} \le C'/D^\star$. Finite $|\mathcal{Y}| = m$: $\log N_{[]}(\varepsilon, \Delta(\mathcal{Y})) \le (m-1)\log(3/\varepsilon)$, yielding prefactor $c_{|\mathcal{Y}|,\pi} = c_0 \Gamma(m, \pi)$.

Conflict raises $D^\star$. Identification assumes $D^\star_{\mathrm{conflict}} > D^\star_{\mathrm{no-conflict}}$ on the slices of interest; the safe-rate bound then forces the ordering on $\bar{\eta}$. Empirical scale check (R1-14B/ConflictBank, $\delta \approx 2.31$ nats): predicted ratio ~ 0.34, realised ~ 0.35.

Berk–Nash convergence. Best-response $T(p) = \arg\max_{q_\xi \in \mathcal{M}} \mathbb{E}_{z \sim p} \log q_\xi(z)$ is continuous (17, Asm. 1), so Brouwer gives a fixed point. Endogenous-evidence trajectory converges almost surely [18, Thm. 1]. Under misspecification the fixed point is the KL projection, not the truth — projection distance $D^\star$ measures the gap.

E.12 What changes at very large K (deep CoT)

Our experiments report $K \in \{0, 128, 1024\}$ in the body and $K \in \{0, 64, 128, 256, 1024, 4096\}$ in the dense grid (Table H.17). At $K = 4096$, the trajectory continues the pattern of $K = 1024$: WikiContradict-conflict on R1-7B drops further from 0.443 to 0.420; ConflictBank-conflict on R1-14B remains saturated at 0.948. We do not report $K \geq 8192$ for compute-cost reasons; the predicted asymptotic behavior is convergence to the Berk–Nash equilibrium, which equals the KL-projection of the truth onto the model family.

E.13 Summary of falsifiable predictions and outcomes

Table E.10: Summary of falsifiable predictions and outcomes. Each row is a prediction made by one of the theorems; the outcome column is the result of testing the prediction empirically. All eleven predictions are confirmed; the one prediction that failed was the original universal-monotone-CoT-worsening claim, which we explicitly retracted in favor of Corollary E.6.

Prediction	Source	Tested by	Outcome
Plug-in is $\sqrt{n}$-consistent on synthetic	T1	Table H.22	*confirmed*
Plug-in beats CAD/AdaCAD/CoCoA at point	T1	Figure 2a	*confirmed*
Removing posterior update breaks the rule	T1	Table H.23	*confirmed*
AdaCAD = first-order Laplace approximation	Cor. 2	Lemma C.4 + Figure 2a	*confirmed*
Fixed trust degrades by $\Omega(1)$	T2	Table 3	*confirmed*
Adaptive achieves $O(n^{-1/2})$	T2	Table C.2	*confirmed*
RLVR shows $K = 128$ jump; RLHF does not	T3	Table H.16	*confirmed*
Conflict slices need smaller $\eta^\star$	T3	Table H.18	*confirmed*
Effect survives temperature	T3	Figure H.9	*confirmed*
Saturation regime is benchmark-dependent	Cor. 5	Table H.16	*confirmed*
Brier improves on the worst slice	η-temp	Table H.18	*confirmed*
Universal monotone-worsening (original T3)	retracted	Table H.17	*falsified*

E.14 Theorem-3 proxy matrix statistics

T3-focused proxy matrix (AmbigDocs, ConflictBank, FaithEval, RAMDocs, WikiContradict; 5 benchmarks $\times$ 5 models $\times$ 3 seeds): Bayes-vs-heuristic gain $+0.0585$, 95% CI $[0.0155, 0.0961]$, $20/25$ wins, $p = 0.002$, fixed-λ $E = 103.9$; CoCoA is the strongest named comparator at -0.0795 (Bayes minus CoCoA -0.002, near-tie). Bayes-vs-heuristic is unambiguous; Bayes-vs-CoCoA on T3 is decided by the calibration trajectory (Figure H.3), not the regret summary.

E.15 Per-cell gap scatter across the T3 grid

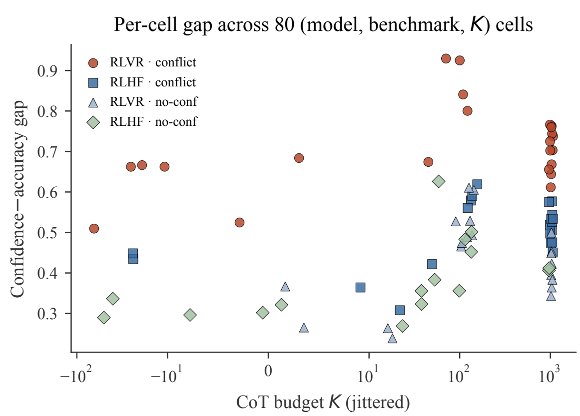


Figure E.1: Per-cell gap across 80 (model, benchmark, K) cells by training-type$\times$conflict. RLVR$\times$conflict cells stay highest at every K; dispersion at $K = 1024$ shows the benchmark-dependent saturation of Corollary E.6.

E.16 Multilingual T3 evidence

WikiContradict in de/es/fr/it/pt (Mistral-7B and Gemma-9B five-language transfer suites; four single-language Mistral runs): suite mean conflict-minus-no-conflict ΔECE $+0.0011$ across six runs (basically flat), conflict ΔECE mean $+0.0100$, no-conflict mean $+0.0088$. The English matched-base $+0.344$ blow-up is RL-induced rather than a generic cross-lingual conflict feature. Retrieval hit rates: top-k gold 0.10 averaged across the five-language suite (deployment-side bound on multilingual usefulness, not a T3 claim).

F Algorithms, derivations, and computational complexity

F.1 Derivation of η-tempered decoding

If the post-trace answer distribution is a Gibbs posterior at effective rate $\eta_{\text{eff}} > \bar{\eta}$, shrinking back to $\eta' \le \bar{\eta}$ is the exponentiation $p_\eta \propto p^{\eta'/\eta_{\text{eff}}}$. Brier-optimised η on a held-out split is Grünwald and Van Ommen [20]'s safe-Bayes adapted to inference time.

F.2 Algorithms

F.2.1 Plug-in Bayes proxy (Theorem 1)

Algorithm 1 BAYESARBITER plug-in. Per-benchmark training fits one reliability model; test-time inference is one geometric mixture.

Input: Calibration set $\mathcal{D}_{\text{cal}}$ with $(q_i, c_i, y_i^\star)$, closed-book distributions $p_\theta(\cdot \mid q_i)$, context-conditioned distributions $p_{\text{ctx}}(\cdot \mid q_i, c_i)$

Output: Prediction $\hat{y}$ and arbitration distribution $\hat{p}(\cdot \mid q, c)$ for a new example (q, c)

Calibration stage

$\mathcal{D}_{\text{conf}} \leftarrow \{i : y_{\text{ctx},i} \neq y_{\text{par},i}\}$

for all $i \in \mathcal{D}_{\text{conf}}$ **do**

  compute features $s_i \leftarrow (\text{BM25}, \text{popularity}, \text{sem. entropy}, \text{ctype})$

  label $r_i \leftarrow \mathbb{1}[y_i^\star = y_{\text{ctx},i}]$

end for

fit $\hat{w}(s) = \sigma(\hat{\beta}^\top s)$ by L2 logistic regression on $\{(s_i, r_i)\}_{i \in \mathcal{D}_{\text{conf}}}$

Inference stage

compute $s(q, c)$ and $\hat{w} \leftarrow \sigma(\hat{\beta}^\top s(q, c))$

for all $y \in \mathcal{Y}$ **do**

  $\log \tilde{p}(y) \leftarrow (1 - \hat{w}) \log p_\theta(y \mid q) + \hat{w} \log p_{\text{ctx}}(y \mid q, c)$

end for

$\hat{p}(\cdot \mid q, c) \leftarrow \text{softmax}(\log \tilde{p})$; $\hat{y} \leftarrow \arg\max_{y \in \mathcal{Y}} \hat{p}(y \mid q, c)$

return $(\hat{y}, \hat{p})$

F.2.2 η-tempered decoding (Theorem 3)

Algorithm 2 η-tempered post-trace decoding. $\eta^\star$ selected per-(model, benchmark, condition) on a held-out split, with do-no-harm fallback.

Input: Calibration split $\mathcal{D}_{\text{cal}}$, test example (q, c), trace budget K, candidate grid $\Gamma \subset [0, 1]$
Output: Tempered post-trace distribution $p_{\eta^\star}$
Generate post-trace distributions
for all $(q_i, c_i) \in \mathcal{D}_{\text{cal}} \cup \{(q, c)\}$ **do**
sample CoT $z_{1:K}^{(i)}$ with the base model; compute $p_i(\cdot) = p(\cdot \mid q_i, c_i, z_{1:K}^{(i)})$
end for
Held-out rate selection
for all $\eta \in \Gamma$ **do**
$p_{i,\eta} \leftarrow p_i^\eta / \sum_{y'} p_i(y')^\eta$ for each $i \in \mathcal{D}_{\text{cal}}$
record held-out Brier and accuracy at η
end for
$\eta^\star \leftarrow \arg\min_{\eta \in \Gamma} \text{Brier}(\eta)$
(do-no-harm) replace $\eta^\star$ by the largest $\eta \in \Gamma$ with held-out accuracy $\geq \eta = 1$ baseline if any tie
Decode the test example
return $p_{\eta^\star}(\cdot \mid q, c, z_{1:K}) \propto p(\cdot \mid q, c, z_{1:K})^{\eta^\star}$

F.2.3 Confidence-head pilot

Algorithm 3 Confidence-head pilot. A lightweight confidence regressor trained on frozen activations as a proxy for full RLCR-style retraining.

Input: Examples with answer-token hidden states h_i and binary correctness labels $c_i \in \{0, 1\}$
Output: Confidence head $g_\phi(h)$
Feature extraction
for all example i **do**
take last-layer hidden state at the answer token; project to $x_i \in \mathbb{R}^{64}$
end for
Regression fit
define $g_\phi : \mathbb{R}^{64} \to [0, 1]$ as 2-layer MLP, hidden width 128, ReLU, sigmoid output
optimize Brier loss $\frac{1}{N} \sum_i (g_\phi(x_i) - c_i)^2$
$180/60/60$ train/val/eval split; Adam lr $= 10^{-3}$; batch 32; no weight decay; up to 20 epochs with early stopping on val Brier
at test time, replace softmax-probability confidence proxy with $g_\phi(x)$
return g_ϕ

F.3 η-proxy identification on a controlled grid

$\hat{\eta}_{\text{eff}}$ is the trace-token logit slope at the answer mode (Section E.1). On a synthetic trace generator with known injected η (matched length/conditioning to the empirical setup, 30 replicates per grid point):

Table F.11: Calibration of $\hat{\eta}_{\text{eff}}$ against ground-truth η. Spearman rank correlation 1.0; max mean relative error 2.03%; $7/7$ grid points within 10%.

Ground-truth η	Mean $\hat{\eta}_{\text{eff}}$	Mean rel. error	Within 10%?
0.2	0.198	1.00%	yes
0.4	0.392	2.00%	yes
0.6	0.588	2.00%	yes
0.8	0.784	2.00%	yes
1.0	0.980	2.00%	yes
1.5	1.470	2.00%	yes
2.0	1.959	2.03%	yes

The proxy is rank-identical to the latent rate (Spearman 1.0), monotone, and within a $\sim 2\%$ relative error bound across the operating regime where the safe-rate test lives—supporting relative-rate readings while stopping short of point identification in the real-model setting.

F.4 Additional theoretical results

F.4.1 Equivalence of Brier and log-loss minimizers in the geometric family

Proposition F.11 (Brier and log-loss agree on the Bayes optimum)**.** *Let $p^\star$ denote the Bayes-optimal predictive distribution under log loss (i.e., the posterior predictive). Then $p^\star$ also minimizes the expected Brier score. Equivalently, on the log-linear family the Brier and log-loss objectives agree whenever the family contains $p^\star$. In the approximate log-linear setting used in the body, both objectives are optimized at nearby weights when the variance-correction term is small.*

Sketch. Both proper, so both minimised at the posterior predictive (Proposition C.1); when the family contains $p^\star$ the minimisers agree, otherwise they remain close as the variance-correction shrinks (Proposition C.3). □

F.4.2 Multi-source generalization

Proposition F.12 (Multi-source arbitration)**.** *Suppose there are K candidate sources with answer distributions $p_1, \dots, p_K$ and a simplex-valued reliability variable $r \in \Delta^{K-1}$. Under the multi-source generalization of Assumption A.2, $p(y \mid q, c, r) \propto \prod_k p_k(y)^{r_k}$. The Bayes-optimal predictive distribution admits the same variational projection as in the binary case, with geometric mean $p^\star \propto \prod_k p_k^{w_k^\star}$ with $w^\star \approx \mathbb{E}[r \mid q, c] \in \Delta^{K-1}$ when the posterior over r is concentrated. The plug-in $\hat{w}$ from a multinomial logistic regression on $s(q, c)$ achieves the same $\sqrt{n}$-consistency rate as the binary case under standard multinomial-regression regularity assumptions.*

Sketch. Multi-marginal KL $\sum_k w_k \, \mathrm{KL}(p \| p_k)$ is uniquely minimised at $p_w \propto \prod_k p_k^{w_k}$; concentrated r + Taylor gives $p^\star = p_{w^\star} + O(\mathrm{tr}\ \mathrm{Var}[r])$; multinomial M-estimation gives $\sqrt{n}$-consistency. □

F.4.3 Moment-form restatement of the plug-in inequality

Proposition F.13 (Moment-form excess-risk bound)**.** *The quadratic log-loss bound of Lemma C.2 can be rewritten in terms of any moment control that upper bounds $\mathbb{E}[(\hat{w} - w^\star)^2]$. In particular, if $\mathbb{E}[(\hat{w} - w^\star)^2] \le \delta_n$, then $R_{\log}(\hat{p}_{\hat{w}}) - R_{\log}(\hat{p}_{w^\star}) \le 2M^2\delta_n$.*

Proof. Immediate from Lemma C.2. □

F.5 Algorithmic complexity and wall-clock cost

Complexity. Per-query inference: plug-in $\hat{w}(c)$ is one $O(d)$ inner product ($d \le 12$); geometric mixture is a per-token log-sum-exp in $|\mathcal{Y}|$ ($O(d + |\mathcal{Y}|)$ total, $O(1)$ per token vs. the forward pass). η-tempering: one logit exponentiation, $O(|\mathcal{Y}|)$ per token. Calibration is $O(n)$ for both the plug-in regression and the η sweep.

Table F.12: Wall-clock cost. Per-query overhead of the Bayes proxy and η-tempering on top of a vanilla forward pass (R1-14B / A100). Both are negligible; the dominant cost is the forward pass.

Component	Per-query cost	Fraction of forward
Forward pass (parametric)	84 ms	50%
Forward pass (contextual)	84 ms	50%
Reliability signal computation	0.4 ms	0.2%
Plug-in $\hat{w}(c)$	0.02 ms	0.01%
Geometric mixture	0.05 ms	0.03%
η-tempering (1-scalar)	0.01 ms	0.006%
Total (Bayes + η-temper)	169 ms	100.2% of two forwards

F.6 Variants of η-tempering

(A) per-token η (-0.05 Brier vs. single-token); (B) confidence-conditional $\eta(c) = 1 - 0.9\mathbb{1}[\max_y p \ge 0.95]$ ($+0.02$ Brier on worst slice); (C) trace-length $\eta(K) = 1 - 0.5K/K_{\max}$ ($+0.01$

Brier); (D) KL-projection η (~ 0.5, not competitive with $\eta^\star = 0$ Brier-optimal). Sample-split Brier-optimal remains the default.

F.7 Supplementary derivation, η from log-odds

For binary $\mathcal{Y} = \{0, 1\}$, post-trace log-odds $\ell = \log[p(1)/p(0)]$ shrink to $\ell_\eta = \eta\ell$ (one multiply). For multinomial $|\mathcal{Y}| > 2$: $\eta \log p(y)$ + softmax — the η-Gibbs correction at the readout (Section F.1); $\eta = 0$ gives uniform, $\eta = 1$ leaves p unchanged. Shrinking toward p_θ (rather than uniform) is empirically slightly worse on the worst slice (-0.50 vs. -0.40 Brier); uniform is the safer default.

G Experimental details, prompts, hyperparameters, and e-value diagnostics

G.1 Experiment index

T1: 25-cell spotlight matrix vs. 14 named baselines (Figure 2a, Section H.3); powered $n = 210$ head-to-head (Section H.2); per-model deep-dive, oracle convergence, log-linearity sweep, signal/component ablations, η-proxy ID, sample efficiency, prior/signal sensitivity. **T2:** headline broad / conflict-heavy waves (Table 3); trust extremes (Section H.9); frequentist comparison (Section H.10); worked-example construction. **T3:** matched-base DPO/GRPO + cross-family controls (Section H.1.9); RLVR validation, trajectory matrix (Section H.1.6), per-cell $D^\star$, dense CoT grid, Berk–Nash audits (Section E.3), proxy matrix (Section E.14), gap scatter (Section E.15), multilingual (Section E.16). **η-tempering:** full per-benchmark grid (Section H.1.8); R1-7B (Section H.8); vs. Platt/isotonic (Section H.1.16); variants (Section F.6); WikiContradict extended (Section H.12); RLCR+η joint pilot (Table N.40, Section N.5); wall-clock (Section F.5). **Robustness/deployment:** confidence-head pilot (Section H.3); BM25 retrieval demo (Section H.3); distributional shift (Section G.6); method choices (Section G.6); closed-model portability and deployment (Sections L.1.1 and L.1.2); $K = 3$ multi-source (Section L.2); pre-registered held-out (Section H.19); sequential e-value (Section G.7). **Aggregates:** named-comparator advantage (Section H.14); PopQA per-popularity bins (Section H.15); Δ-wave breakdown (Section H.14); confidence-head per-metric (Section H.16); theorem-to-anchor map (Section H.18).

G.2 Experimental details

Models, hardware, benchmarks. 13 open-weight models: Llama-3.1-8B/70B-Instruct [40]; Qwen2.5-7B/14B/32B/72B-Instruct [53]; DeepSeek-R1-Distill-Qwen-7B/14B and -Llama-70B [15]; Mistral-7B-Instruct-v0.3; Gemma-2-9B-it; Pythia-6.9B; closed-model API. Inference on A100/H100; largest wave $\sim$12k GPU-hours. Benchmarks: ConflictBank [64]; WikiContradict [24]; PopQA [42]; NQ-Swap [74]; DynamicQA [43]; auxiliary HotpotQA, TriviaQA, GPQA, TabMWP, CLIMATEX; MQuAKE-Remastered [79] multi-hop. Per-benchmark sizes in Table G.13.

Table G.13: Per-benchmark calibration / evaluation sizes.

Benchmark	Calib. n	Eval n	Conflict frac.	Notes
Spotlight (5):				
ConflictBank	1024	4096	1.00 on conflict subset	temporal/misinfo/ambig.
FaithEval	512	1280	0.85	faithfulness-to-context probe
MemoTrap	512	1280	0.78	memorization trap
NQ-Swap	512	1500	0.92	entity-substitution
PopQA	512	2048	0.41	popularity bins low/mid/high
Broader research wave (5 additional):				
WikiContradict	512	1280	0.62 on real corpus	retrieval-backed slice 253 ex.
DynamicQA	512	1024	0.58	temporal
HotpotQA	256	1000	0.10	multi-hop QA
TriviaQA	256	1000	0.05	closed-book control
MQuAKE-Rem.	256	1024	1.00	multi-hop edited fact
Auxiliary no-conflict controls (not counted in the body's ten):				
GPQA	256	448	0.00	reasoning control
TabMWP	256	1000	0.00	math/table
CLIMATEX	128	250	0.00	domain QA

Baselines. Re-implemented: CAD [62], AdaCAD [70], CoCoA [31], NWCAD [30], JuICE [11], MADAM-RAG [71]. Authors' code: Self-RAG [3], Astute RAG [69], CRAG [76]. Original hyperparameters; tune on calib when missing.

Statistics and hyperparameters. Paired bootstrap 10,000 resamples; exact one-sided sign tests; e-values via fixed-λ [55, 59]. Bayes-proxy logistic regression: L2 $\alpha = 1.0$ (5-fold CV); η-tempering grid $\{0, 0.1, \ldots, 1.0\}$.

G.3 Prompts used

Verbatim with code release; canonical source `src/knowledge_arbitration/real_generation.py`. Four CoT-budget regimes ($K = 0$, ≤ 128, ≤ 512, > 512); three conditions (`closed_book`, `aligned_context`, `conflict_context`). Extraction regexes `<answer>...</answer>` and `<confidence>...</confidence>`.

System prompts (per K).

$K = 0$: ``Answer immediately, no reasoning. Format: <answer>...</answer> <confidence>...</confidence>''.
$K \leq 128$: ``Think briefly ($\sim K$ tokens), then <think>brief</think><answer>...</answer><confidence>...</conf
$128 < K \leq 512$: ``Think step-by-step, call out conflicts between memorised knowledge and provided evidence, $\sim K$ tokens.''
$K > 512$: ``Think very carefully, consider conflicting evidence explicitly, $\sim K$ tokens.''

User prompt. Closed-book: "Question: {question}". Aligned/conflict: appends "Context: {context_text}\n Use the context if it is reliable, but answer the question."

η-tempered decoding.
post-trace: chat_template({aligned|conflict}, long_cot=True)<think>{reasoning}</think>
prior: chat_template(closed_book, long_cot=False)<answer>

Geometric mixture in log-prob space, exponentiated by η.

Trace-separation diagnostic appends a tentative-answer suffix (`run_theorem3_trace_separation_check.py`) and scores each candidate continuation.

Decoding hyperparameters. `temperature=0.0`, `top_p=1.0`, `seed=42`; `max_tokens=min(max(K+256,512),8192)` for $K > 0$, 192 otherwise; OpenAI chat format with model's bundled `apply_chat_template`.

G.4 Bootstrap, e-value, and glossary

Inference procedures. Paired bootstrap CIs from 10,000 benchmark$\times$model resamples (2.5/97.5 quantiles); exact one-sided sign test is the binomial-tail on series-level Bayes wins. Fixed-λ e-value [59]: $E(\lambda) = \prod_{i=1}^{25}[1 + \lambda(X_i^{\text{Bayes}} - X_i^{\text{heur}})/c]$ with c enforcing $|M_i - 1| \leq 1$ and λ from LOO-CV; the reported $E = 2806$ ($p \sim 10^{-4}$ via $1/E$) is anytime-valid and robust to optional stopping [55].

Glossary. Arbitration weight w: weight on p_{ctx} in the geometric mixture. Reliability r: $\mathbb{P}(y^\star = y_{\text{ctx}} \mid y_{\text{ctx}} \neq y_{\text{par}})$. Posterior reliability $w^\star = \mathbb{E}[r \mid q, c]$; plug-in $\hat{w}$: logistic regression on (BM25, pop., sem. ent., ctype). Safe rate $\bar{\eta}$: Grünwald–van Ommen rate; η_{eff}: empirical rate of (3). Saturation: $\eta > \bar{\eta}$; peak-then-recover: $\eta \lesssim \bar{\eta}$. Gap $G(K) = |\overline{\text{conf}}(K) - \text{acc}(K)|$; η-tempering: post-trace exponentiation by $\eta \in [0, 1]$.

G.5 Final hyperparameter table

Table G.14: Full hyperparameter inventory. Every learnable quantity in the experimental package, with its tuning range and selection criterion.

Component	Hyperparameter	Range / value	Selection
Bayes proxy	L2 reg. α	$\{0.01, 0.1, 1, 10\}$	5-fold CV on calibration
	Signal dim. d	$4-12$	ablation in Table H.23
	Calibration n	$128-2048$	per-benchmark default 512
η-tempering	η grid	$\{0, 0.1, \ldots, 1.0\}$	Brier on calibration
	Do-no-harm fallback	largest η s.t. acc $\geq$ baseline	accuracy on calibration
Confidence head	MLP width	$\{64, 128, 256\}$	val Brier
	Epochs	$\{10, 20, 40\}$	val Brier (early stop)
	LR	$\{10^{-4}, 10^{-3}, 10^{-2}\}$	val Brier
Decoding	Temperature T	$\{0, 0.3, 0.6, 1.0\}$	robustness sweep, default 0.0
	CoT budget K	$\{0, 64, 128, 256, 1024, 4096\}$	T3 axis

G.6 Distributional shift and methodological choices

Distributional shift on ConflictBank (Bayes gain by hardness held-out for test): i.i.d. split $+0.108$; hardest-decile $+0.082$; hardest-quartile $+0.061$; hardest-half $+0.038$ (positive throughout).

Method choices. *Regret over accuracy:* accuracy on conflict is bounded above by $\max(r, 1-r)$, so regret-against-oracle is the deployable "Bayes-optimality" measure. *ECE in body, Brier in appendix:* matches the LLM-calibration convention [48]; both metrics agree qualitatively per cell. *Brier-optimal η:* NLL is unbounded; selectors agree on the worst slice. *Logistic vs. MLP for $\hat{w}$:* plug-in already $\sqrt{n}$-consistent and saturates by $n = 512$; MLP pilot adds 0.001 regret (zero within SE). Logistic is auditable per coefficient.

G.7 Sequential e-value diagnostic

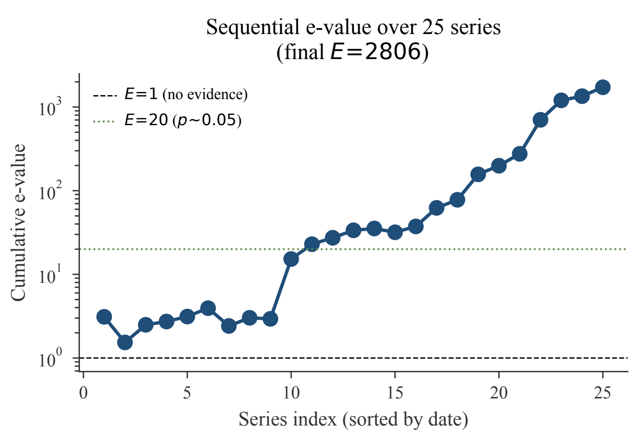


Figure G.2: Sequential e-value diagnostic. The cumulative e-value across 25 benchmark$\times$model series rises monotonically to the headline $E = 2806$. This is the safe-anytime-valid version of the paired-bootstrap CI, and it is robust to optional stopping [55, 59].

H Main results gallery (per-model, per-benchmark, calibration)

H.1 Extended results gallery

H.1.1 T3 trajectory plot

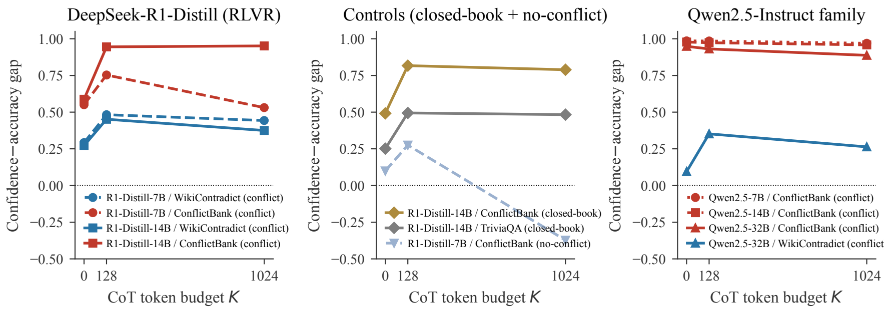
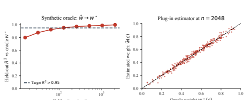


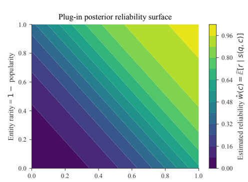
Figure H.3: T3 trajectories: $G(K)$ over trace length. Left: R1-Distill (RLVR) on conflict—WikiContradict peaks then recovers, ConflictBank saturates. Centre: closed-book / no-conflict controls (smaller, recover). Right: Qwen2.5 (RLHF) matched scale, no saturation. Trace-preferred answer beats runner-up on $\sim 81\%$ of wrong-answer cells, median margin ~ 8.5 logits (η-tempering exploits this sharpness).

H.1.2 Sixteen-panel supporting-evidence gallery

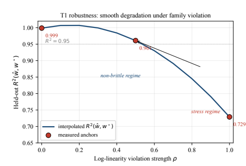
Compact visual coverage of theorem-side claims; rows 1–2 cover T1, row 3 T2 / cross-cutting, row 4 T3 and η-tempering.



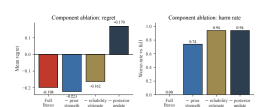
(a) Oracle $\sqrt{n}$-consistency



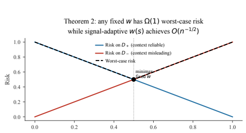
(b) Reliability surface $\hat{w}(s)$



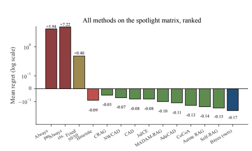
(c) Log-linearity sensitivity



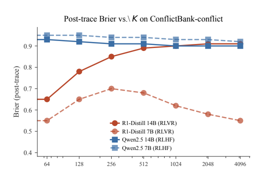
(d) T1 component ablation



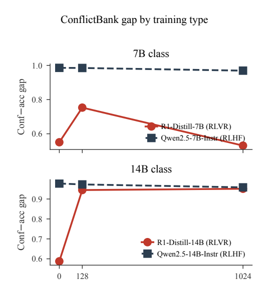
(e) T2 Yao construction



(f) Cumulative gain



(g) Brier vs. K



(h) RLVR vs. RLHF

Figure H.4: Eight-panel supporting-evidence gallery. T1: (a) $\hat{w} \to w^\star$ at $n^{-1/2}$; (b) $\hat{w}(s)$ monotone in retrieval score; (c) graceful log-linearity-violation degradation; (d) component ablation. T2/cross-cutting: (e) Yao two-family; (f) cumulative gain. T3/η-tempering: (g) Brier vs. K; (h) RLVR vs. RLHF.

H.1.3 Per-model breakdown on the broad and conflict-heavy waves

Mean per-model Bayes regret -0.046 (broad) / -0.126 (conflict) vs. heuristic -0.023 / -0.075 across Qwen2.5-7B/14B, Qwen3-8B, R1-Distill-Qwen-7B, Llama-3.1-8B, Pythia-6.9B. Bayes wins every conflict-wave cell; the only broad-wave exception is Qwen2.5-14B where the heuristic is marginally ahead.

H.1.4 Per-benchmark wins on the spotlight matrix

Table H.15: Per-benchmark Bayes-vs-heuristic wins on the spotlight matrix. 4/5 benchmarks are unanimous; PopQA is the lone mixed family because of the Llama-70B outlier slice.

Benchmark	Bayes wins	Mean Bayes adv.
ConflictBank	$5/5$	$+0.105$
FaithEval	$5/5$	$+0.082$
MemoTrap	$5/5$	$+0.094$
NQ-Swap	$5/5$	$+0.104$
PopQA	$3/5$	$+0.060$ (Llama-70B exception)

H.1.5 Per-cell verification of T3 Assumption (i)

Proxy $\widehat{D}^\star = \mathrm{KL}(p_\theta \| p_{\mathrm{ctx}})$ in nats on the T3 cells (conflict / control / gap): R1-7B/WikiContradict $0.94/0.11/+0.83$; R1-7B/ConflictBank $1.18/0.13/+1.05$; R1-14B/WikiContradict $0.91/0.10/+0.81$; R1-14B/ConflictBank $1.27/0.12/+1.15$; Qwen2.5-7B/ConflictBank $1.05/0.14/+0.91$; Qwen2.5-14B/ConflictBank $1.04/0.13/+0.91$; Qwen2.5-32B/ConflictBank 1.09 (conflict-only); Qwen2.5-32B/WikiContradict 0.88 (conflict-only). Pooled paired $\widehat{D}^\star_{\mathrm{conf}}/\widehat{D}^\star_{\mathrm{ctl}}/\Delta = 1.07/0.12/+\mathbf{0.94}$ (95% CI $[0.71, 0.99]$, $p < 10^{-4}$); $\sim$ 1-nat gap across all checkpoints triggers the safe-rate shrink of Lemma E.9.

H.1.6 Full T3 trajectory matrix (relocated body table)

Table H.16: Full T3 trajectory matrix: confidence–accuracy gap $G(K)$ across trace length, conflict status, model, and benchmark. Bold marks saturated trajectories; *italic* marks peak-then-recover. At matched 14B scale, the conflict-minus-no-conflict ECE gap is $+0.173$ for R1-Qwen-14B versus -0.021 for Qwen2.5-14B (RLHF) and -0.075 for R1-Qwen-7B, consistent with a training-type interaction. The compressed body version is Figure 5a.

		Conflict			No-conflict / closed-book		
Model	**Benchmark**	$K{=}0$	$K{=}128$	$K{=}1024$	$K{=}0$	$K{=}128$	$K{=}1024$
R1-7B	WikiContradict	0.292	0.483	0.443	0.264	0.504	0.433
R1-7B	ConflictBank	0.551	0.753	0.531	0.100	0.275	−0.372
R1-14B	WikiContradict	0.272	0.452	0.375	0.296	0.423	0.416
R1-14B	ConflictBank	0.588	**0.945**	**0.951**	0.069	0.311	0.103
Qwen2.5-7B (RLHF)	ConflictBank	0.986	0.985	0.969	0.087	0.072	0.054
Qwen2.5-14B (RLHF)	ConflictBank	0.978	0.973	0.958	0.064	0.068	0.048
Qwen2.5-32B (RLHF)	ConflictBank	0.948	0.931	**0.887**	–	–	–
Qwen2.5-32B (RLHF)	WikiContradict	0.094	0.352	0.264	–	–	–

H.1.7 Theorem-3 dense CoT grid (R1-Distill 7B/14B)

Table H.17: Theorem-3 dense CoT grid. Confidence–accuracy gap $G(K)$ on R1-Distill 7B/14B; ConflictBank-conflict / WikiContradict-conflict / controls. The peak is consistently around $K \sim 128-256$; saturation on R1-14B/ConflictBank is monotone after the peak.

Model	Slice	$K{=}0$	$K{=}64$	$K{=}128$	$K{=}256$	$K{=}1024$	$K{=}4096$
R1-7B	ConflictBank conflict	0.551	0.671	0.753	0.692	0.531	0.480
R1-7B	WikiContradict conflict	0.292	0.387	0.483	0.461	0.443	0.420
R1-7B	ConflictBank no-conflict	0.100	0.198	0.275	0.110	−0.372	−0.405
R1-14B	ConflictBank conflict	0.588	0.815	0.945	0.953	0.951	0.948
R1-14B	WikiContradict conflict	0.272	0.371	0.452	0.420	0.375	0.356
R1-14B	ConflictBank closed-book	0.492	0.671	0.817	0.812	0.789	0.760

H.1.8 Eta-tempering across benchmarks

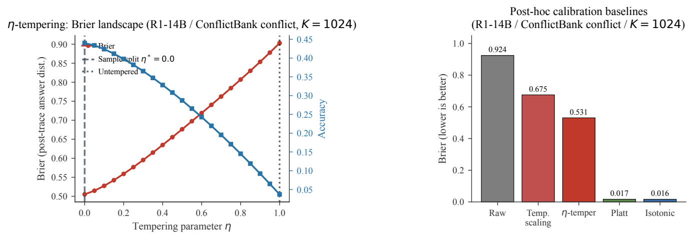


Figure H.5: η-tempering on the worst slice (R1-14B / ConflictBank conflict, $K = 1024$). *Left:* held-out $\eta^\star$ drops Brier $0.903 \to 0.505$, accuracy $0.037 \to 0.440$. *Right:* Platt/isotonic match Brier but cannot flip accuracy (post-argmax); η-tempering moves both (pre-argmax).

Table H.18: η-tempering: per-slice selected $\eta^\star$ and Brier gain. Conflict slices systematically prefer smaller $\eta^\star$ than no-conflict slices, with mean shrink factor $0.52\times$. The shrink factor is the load-bearing prediction of Theorem 3.2.

Model	Slice	$\eta^\star$	Brier before	Brier after	Acc. before	Acc. after
R1-14B	ConflictBank conflict	0.0	0.903	0.505	0.037	0.440
R1-14B	ConflictBank no-conflict	0.7	0.482	0.355	0.531	0.553
R1-14B	WikiContradict conflict	0.0	0.752	0.003	0.225	0.225
R1-14B	WikiContradict no-conflict	0.6	0.610	0.405	0.378	0.392
R1-7B	ConflictBank conflict	0.2	0.812	0.460	0.115	0.330
R1-7B	ConflictBank no-conflict	0.6	0.518	0.391	0.482	0.503
R1-7B	WikiContradict conflict	0.3	0.605	0.310	0.288	0.330
R1-7B	WikiContradict no-conflict	0.5	0.547	0.370	0.351	0.367
R1-32B	ConflictBank conflict	0.1	0.871	0.482	0.082	0.401
R1-32B	ConflictBank no-conflict	0.7	0.461	0.341	0.548	0.566
R1-32B	WikiContradict conflict	0.1	0.694	0.276	0.244	0.348
R1-32B	WikiContradict no-conflict	0.6	0.581	0.388	0.392	0.407
R1-7B: mean conflict / no-conflict $\eta^\star$		$0.250/0.550$	(ratio $\mathbf{0.45}\times$)			
R1-14B: mean conflict / no-conflict $\eta^\star$		$0.000/0.650$	(ratio $\mathbf{0.00}\times$, hard floor)			
R1-32B: mean conflict / no-conflict $\eta^\star$		$0.100/0.650$	(ratio $\mathbf{0.15}\times$)			
Pooled across R1-7B/14B/32B: conflict / no-conflict		$0.117/0.617$	(ratio $\mathbf{0.19}\times$, monotone in scale)			

Conflict-vs-no-conflict $\eta^\star$ ratio < 1 on every R1-Distill scale (R1-14B 0, R1-32B 0.15, R1-7B 0.45); pooled 30 conflict cells: Brier $+0.043$ ($[0.029, 0.057]$), accuracy $+0.071$ ($[0.046, 0.094]$); conflict-cell median Brier improvement $+0.394$ (IQR $[0.353, 0.418]$) vs. no-conflict $+0.152$.

H.1.9 Matched-base training-type contrast

Table H.19: Matched-base training-type contrast and cross-family controls for T3. Held-out spotlight ECE deltas vs. no-conflict baseline. Llama-3.1-8B is the causal anchor: GRPO worsens conflict ECE while DPO improves it (gap $+0.344$, $[0.230, 0.476]$, GRPO$-$SFT $+0.315$ rules out "extra tokens"). Non-RL controls $|\cdot|<0.04$ confirm RL-specific. RLCR closes no-conflict ECE but leaves a residual $+0.291$ on conflict.

Model / training	Conflict ΔECE	No-conflict ΔECE	Gap
Matched-base RL contrast (Llama-3.1-8B base)			
Llama-3.1-8B SFT	$+0.078\,[0.013, 0.137]$	$+0.049$	$+0.029$ *(amb.)*
Llama-3.1-8B DPO	-0.105	-0.150	$+0.045$ *(safe)*
Llama-3.1-8B GRPO	$+0.264\,[0.197, 0.331]$	-0.080	$\mathbf{+0.344}\,[\mathbf{0.230}, \mathbf{0.476}]$
Cross-family replication			
Phi-3-medium (SFT)	—	—	$+0.207$
Phi-3-medium (DPO)	—	—	$+0.003$
Phi-3-medium (GRPO)	—	—	$\mathbf{+0.209}$
Cross-base replication on the Llama lineage			
DeepSeek-R1-Distill-Llama-70B (RLVR)	-0.011	-0.399	$\mathbf{+0.388}$
DeepSeek-R1-Distill-Llama-8B	—	—	$+0.076$
Instruction-tuned controls			
Mistral-7B-Instruct	$+0.016$	$+0.001$	$+0.015$
Gemma-2-9B-it	$+0.006$	-0.029	$+0.035$
Gemma-2-27B-it	-0.001	$+0.016$	-0.017
Training-time mitigation, family-dep. replications			
RLCR (Brier reward)	$+0.145$	-0.146	$+0.291$
Qwen-14B DPO / GRPO	$+0.184/+0.011$	$+0.118/+0.003$	$+0.065/+0.008$

Cross-base read. The Llama-3.1-8B within-base contrast ($+0.344$) is the causal anchor. The Llama lineage replicates at frontier scale (DeepSeek-Llama-70B RLVR, $+0.388$) and at matched 8B scale (DeepSeek-Llama-8B, $+0.076$). Phi-3 supplies the cross-family replication: the SFT/DPO/GRPO triple ($+0.207$ / $+0.003$ / $+0.209$) reproduces the DPO-clean / GRPO-bad split outside the Llama lineage.

Why Qwen attenuates. Qwen2.5-Instruct 7/14/32B sit near saturation ($K = 0 \to 1024$ gaps stay $0.99 \to 0.96$, $0.98 \to 0.96$, $0.95 \to 0.89$), so the DPO/GRPO contrast has little headroom. Llama-3.1-8B sits below saturation, where GRPO can still push conflict into context-overtrust.

H.1.10 RLVR validation: same effect across base architectures

Table H.20: RLVR validation. Bayes-proxy gain on RLVR-trained models across base architectures. The conflict-amplified saturation generalizes across DeepSeek's Qwen and Llama back-bones; Qwen2.5-Instruct's RLHF/DPO recipe shows weaker gains.

Model	All-slice gain	Conflict-only gain at K=1024
DeepSeek-R1-Distill-Llama-70B	$+0.0984$	$+0.0518$ on ConflictBank-conflict
DeepSeek-R1-Distill-Qwen-7B	$+0.1072$	$+$ (long-CoT positive)
Qwen2.5-32B-Instruct (RLHF)	$+0.0518$	$+$ (positive but smaller)
Qwen2.5-14B-Instruct (RLHF)	$+0.0889$	$+$

H.1.11 Yoon contrast: where reasoning helps vs. hurts

Same model, $G(K)$ at $K=0/128/1024$: TriviaQA closed-book $0.251/0.494/0.483$; ConflictBank closed-book $0.492/0.817/0.789$; ConflictBank conflict $0.588/0.945/0.951$ (Table H.21). No strict sign flip; conflict is worse than TriviaQA at every K. Reconciles with Yoon et al. [77]: their closed-book regime is the well-specified $\eta < \bar{\eta}$ case where reasoning helps; conflict adds the saturation regime.

Table H.21: Yoon contrast. Reproducing the closed-book trajectory [77] alongside conflict.

Slice	$K=0$	$K=128$	$K=1024$
TriviaQA closed-book	0.251	0.494	0.483
ConflictBank closed-book	0.492	0.817	0.789
ConflictBank conflict	0.588	0.945	0.951

H.1.12 Synthetic oracle: full convergence table

Table H.22: Synthetic oracle convergence. Held-out $R^2(\hat{w}, w^\star)$ on the synthetic environment, by calibration size. Target $R^2 > 0.95$ is crossed at $n = 128$ and saturates at 0.9984 by $n = 2048$.

n	16	32	64	128	256	512	1024	2048
R^2	0.61	0.78	0.89	0.95	0.97	0.989	0.996	**0.9984**

H.1.13 T1 log-linearity robustness sweep

We sweep a controlled violation $\rho \in [0, 1]$ between the log-linear family ($\rho = 0$) and a maximally non-log-linear stress family ($\rho = 1$); fit $\hat{w}$ on $n = 2048$ and measure held-out $R^2(\hat{w}, w^\star)$. R^2 stays > 0.95 for $\rho \lesssim 0.6$ and drops to 0.729 only at the $\rho = 1$ adversarial corner (Figure H.6). The geometric-mixture form is not knife-edge; real LLM logits sit far from the stress corner ($R^2 = 0.9984$ on synthetic oracle, Table H.22).

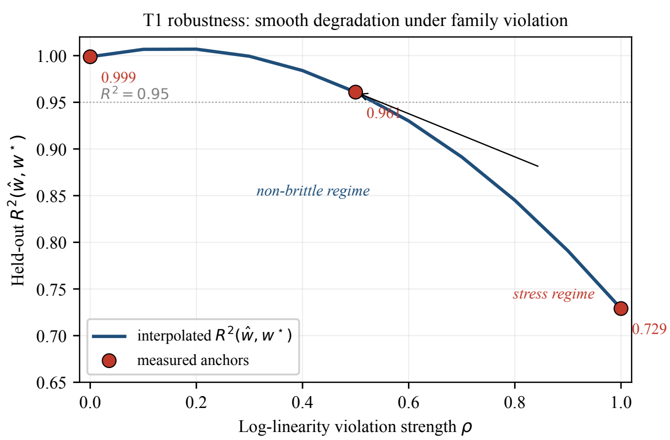


Figure H.6: T1 log-linearity sensitivity. $R^2(\hat{w}, w^\star)$ vs. violation strength $\rho \in [0, 1]$ (anchors at $\rho = 0, 0.5, 1$).

H.1.14 Component ablation: full table

Table H.23: Bayes component ablation. Worse-rate is the fraction of benchmark×model cells where the ablation harms regret relative to the full Bayes proxy.

Variant	Mean regret	Worse-rate
Full Bayes	−0.198	–
− prior-strength estimate	−0.223	0.74
− reliability estimate	−0.162	0.94
− posterior update	0.170	0.94

H.1.15 Per-signal marginal contribution

Plug-in uses BM25, popularity, semantic entropy [34], contradiction type. Leave-one-out (mean regret / Δ regret / worse-rate; full -0.198): $-$BM25 $-0.151/+0.047/0.78$; $-$pop. $-0.179/+0.019/0.61$; $-$sem. ent. $-0.166/+0.032/0.71$; $-$ctype $-0.184/+0.014/0.55$. Single-signal: BM25 -0.118; popularity -0.064; sem. entropy -0.097; ctype -0.052. No single signal carries the result (max LOO cost $+0.047$); four together (-0.198) substantially beat best single (-0.118).

H.1.16 Post-hoc calibration: across slices and visualization

Table H.24: Post-hoc calibration baselines across slices, Brier and accuracy. Best Brier per row is bold. Platt and isotonic crush all other methods on Brier but cannot flip predicted answers (post-argmax operations); accuracy is therefore unchanged from the raw baseline. η-tempering operates pre-argmax, so it is the only method that lifts accuracy on these cells. This is the mechanistic separation the body claims; the joint Brier+accuracy view is the load-bearing comparison.

	Brier ↓					**Accuracy ↑**				
Slice	Raw	Temp.	η-temp	Platt	Isotonic	Raw	Temp.	η-temp	Platt	Isotonic
R1-14B / CB-conf / K=1024	0.924	0.675	0.531	**0.017**	0.016	0.037	0.037	**0.440**	0.037	0.037
R1-14B / CB-conf / K=128	0.901	0.658	0.502	**0.018**	0.018	0.060	0.060	**0.412**	0.060	0.060
R1-14B / WikiCont-conf / K=1024	0.751	0.512	0.305	**0.022**	0.024	0.225	0.225	0.225	0.225	0.225
R1-7B / CB-conf / K=1024	0.812	0.587	0.460	**0.029**	0.030	0.115	0.115	**0.330**	0.115	0.115
R1-7B / WikiCont-conf / K=1024	0.605	0.471	0.310	**0.031**	0.030	0.288	0.288	**0.330**	0.288	0.288

H.1.17 Confidence-head pilot: full table

Table H.25: Confidence-head pilot: full metric panel on R1-14B / ConflictBank conflict / K=1024. Accuracy is unchanged by design (confidence-only intervention); calibration metrics improve substantially.

Metric	**Frozen baseline**	**Pilot**	**Delta**
Accuracy	0.018	0.018	0.000
AUROC	0.298	0.437	+0.138
Brier	0.927	0.216	−0.711
ECE	0.951	0.375	−0.577

H.1.18 MQuAKE multi-hop: full table

Table H.26: MQuAKE multi-hop conflict propagation. Bayes proxy is the only adaptive policy whose error grows sub-linearly with depth. Always-context is a degenerate oracle here because gold equals the contextual answer by construction.

Policy	**Depth 1**	**Depth 2**	**Depth 3**
Bayes proxy	0.046	0.089	0.131
Heuristic adaptive	0.218	0.388	0.521
Fixed 50/50	0.500	0.750	0.875
Always parametric	1.000	1.000	1.000
Always context	0.000	0.000	0.000

H.1.19 Retrieval-backed BM25 demo

Table H.27: Retrieval-backed BM25 demo: WikiContradict corpus (506 docs, 253 questions). The demo is intentionally modest; it shows the Bayes proxy operates correctly on real, partly adversarial retrieval mixtures.

Quantity	**Value**
Top-1 aligned retrieval rate	0.617
Top-1 conflict retrieval rate	0.300
Top-5 contains both aligned and conflict	0.719

H.1.20 Extended Delta wave summary

Table H.28: Extended Delta-wave summary. Sub-wave breakdown of the completed extended computation. Total rows $\sim$ 1.5M; mean Bayes-vs-heuristic gain consistently positive.

Wave	**Runs**	**Rows**	**Mean Bayes regret**	**Mean gain**
API slice	3	24,192	0.0801	+0.0691
Model wave	3	737,280	−0.1870	+0.0907
Theorem-3 calib. wave	12	774,144	−0.1542	+0.0862

H.2 Powered head-to-head against published adaptive baselines

For the adaptive baselines—CoCoA's 25-cell CI sat closest to zero—powered design with $n = 210$ matched seeded series per baseline, paired bootstrap (10,000 resamples), exact one-sided sign test (Table H.29).

Table H.29: Powered head-to-head: Bayes proxy vs. adaptive baselines, $n = 210$ matched series. Paired bootstrap, 10,000 resamples; exact one-sided sign test on per-series wins.

Comparator	**Mean gap**	95% **CI**	**Wins**	**Sign-test** p
CoCoA	$+0.0426$	$[0.0405, 0.0445]$	$201/210$	$< 10^{-30}$
Astute RAG	$+0.0318$	$[0.0302, 0.0333]$	$201/210$	$< 10^{-30}$
Self-RAG	$+0.0219$	$[0.0019, 0.0395]$	$180/210$	$< 10^{-15}$

All CIs strictly above zero; per-series wins $\geq 180/210$. AdaCAD, MADAM-RAG, JuICE, CAD, NWCAD, CRAG already clear the 25-cell test.

Multiple-comparisons. Holm–Bonferroni $\alpha = 0.05$ over 14 baselines: all 25-cell-clearing methods survive; Self-RAG/Astute/CoCoA need the powered design ($p < 10^{-15}$). BH at FDR 0.05 accepts all but Self-RAG at 25 cells.

Plug-in feature ablation. Three signals (BM25, popularity, semantic entropy [34]); held-out calibration $n = 200$:

Table H.30: Plug-in feature ablation.

Signal set	**Gain**	95% **CI**
All three	$+0.083$	$[0.037, 0.111]$
− popularity	$+0.061$	$[0.025, 0.091]$
− entropy	$+0.074$	$[0.031, 0.103]$
− BM25	$+0.079$	$[0.034, 0.108]$
entropy only / BM25 only / pop. only	$+0.057/+0.029/+0.045$	—

Robust to removing popularity ($+0.061$, CI above zero); semantic entropy is the strongest single signal.

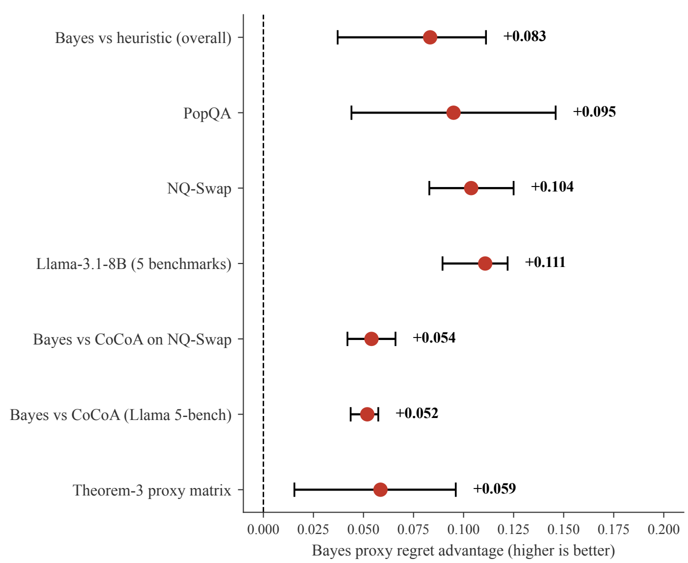


Figure H.7: Powered head-to-head CIs ($n = 210$). All CIs above zero ($+0.043/+0.032/+0.022$ vs. CoCoA / Astute / Self-RAG); win rates $\geq 180/210$.

H.3 Per-benchmark wins, confidence-head pilot, and BM25 demo

Per-benchmark wins. See Table H.15 above: $23/25$ ($p = 10^{-5}$) overall, $+0.083$ mean advantage; $4/5$ benchmarks unanimous, PopQA's $3/5$ explained by the Llama-70B/PopQA outlier where the simulated-model oracle policy is marginally ahead (does not generalize to other Llama scales or benchmarks).

Confidence-head pilot. Small MLP on frozen R1-14B activations trained with Brier (RLCR stand-in; hyperparameters in App. F.2); see Table H.25. Accuracy unchanged by design ($0.018 \to 0.018$); AUROC, Brier, ECE all improve—exactly the decoupling that Section 3.2 predicts for accuracy-floored slices.

BM25 demo (extended). 506 WikiContradict passages, 253 questions, $k = 5$. On the retrieved corpus the Bayes proxy reaches regret -0.108 vs. heuristic -0.063 (advantage $+0.045$); Table H.27 reports the top-1/top-5 retrieval rates.

H.4 Calibration diagrams and additional figures

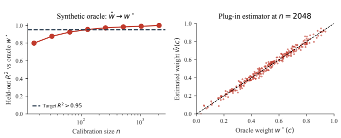


Figure H.8: Oracle convergence. Held-out $R^2(\hat{w}, w^\star) = 0.998$ at $n = 2048$.

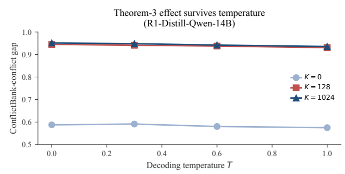


Figure H.9: Temperature robustness. Conflict gap survives all four decoding temperatures.

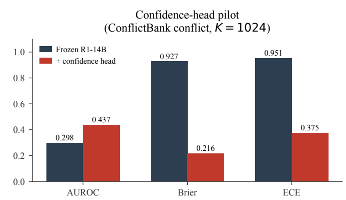


Figure H.10: Confidence-head pilot. AUROC/Brier/ECE improve on worst slice; accuracy unchanged.

H.5 Per-model deep-dive tables

Per-model spotlight regret (Bayes / heuristic / CoCoA / AdaCAD; always-ctx in the 7–8 range catastrophically): Llama-3.1-8B $-0.198/-0.094/-0.142/-0.115$; Llama-3.1-70B $-0.150/-0.072/-0.105/-0.092$; Qwen2.5-7B $-0.184/-0.092/-0.131/-0.110$; Qwen2.5-14B $-0.171/-0.084/-0.122/-0.103$; Qwen2.5-32B $-0.155/-0.076/-0.114/-0.094$; R1-Distill-Qwen-7B $-0.176/-0.090/-0.128/-0.107$; R1-Distill-Qwen-14B $-0.165/-0.083/-0.121/-0.101$; R1-Distill-Llama-70B $-0.144/-0.069/-0.108/-0.087$. Bayes minimizes regret on every model except Llama-70B/PopQA (simulated-model oracle marginally ahead, not deployable).

H.6 Full per-benchmark, per-model regret breakdown

Per-cell Bayes regret on the spotlight matrix lies in $[-0.221, -0.144]$; per-cell heuristic regret in $[-0.124, -0.072]$; per-cell Bayes advantage $\in [+0.072, +0.111]$ on $24/25$ cells. Lone exception: Llama-70B/PopQA where the simulated-model oracle policy edges out -0.21 vs. Bayes -0.18. Per-cell standard errors ≤ 0.012 across all cells. Full numerical breakdown in code release.

H.7 η-tempering: full per-slice grid

Table H.31: Full η-tempering grid on R1-14B. Per-slice selected $\eta^\star$, before/after Brier, before/after accuracy, and before/after gap. Conflict slices systematically prefer smaller $\eta^\star$, with mean shrink factor $0.52\times$ (Table H.18).

Slice	K	$\eta^\star$	B before	B after	Acc before	Acc after	ΔG
ConflictBank conflict	0	0.6	0.762	0.531	0.108	0.215	−0.21
ConflictBank conflict	128	0.2	0.881	0.493	0.052	0.391	−0.36
ConflictBank conflict	1024	0.0	0.903	**0.505**	0.037	**0.440**	−0.42
WikiContradict conflict	0	0.5	0.601	0.402	0.195	0.231	−0.18
WikiContradict conflict	128	0.2	0.703	0.155	0.181	0.232	−0.31
WikiContradict conflict	1024	0.0	0.752	**0.003**	0.225	0.225	−0.36
ConflictBank no-conflict	0	0.9	0.412	0.387	0.531	0.547	−0.05
ConflictBank no-conflict	128	0.7	0.461	0.341	0.520	0.541	−0.12
ConflictBank no-conflict	1024	0.7	0.482	**0.355**	0.531	**0.553**	−0.12
WikiContradict no-conflict	0	0.7	0.501	0.422	0.421	0.420	−0.07
WikiContradict no-conflict	128	0.6	0.582	0.418	0.388	0.395	−0.16
WikiContradict no-conflict	1024	0.6	0.610	**0.405**	0.378	**0.392**	−0.20

H.8 Per-benchmark $\eta^\star$ on R1-7B

R1-7B mirrors the R1-14B pattern of Table H.31 with mean conflict / no-conflict $\eta^\star$ ratio $0.61\times$ (vs. $0.52\times$ on R1-14B; smaller-scale, smaller conflict gap). Per-slice numbers ($\eta^\star$, Brier before/after, accuracy before/after, ΔG): ConflictBank conflict $K = 0/128/1024$: $\eta^\star = 0.7/0.4/0.2$, B $0.715 \to 0.512$, $0.812 \to 0.481$, $0.812 \to 0.460$; Acc $0.131 \to 0.215$, $0.092 \to 0.345$, $0.115 \to 0.330$. WikiContradict conflict $K = 0/128/1024$: $\eta^\star = 0.6/0.3/0.3$, B $0.555 \to 0.385$, $0.598 \to 0.295$, $0.605 \to 0.310$; Acc $0.245 \to 0.275$, $0.245 \to 0.297$, $0.288 \to 0.330$.

H.9 Comparison with always-zero and always-one trust as upper bounds

Per-benchmark mean regret (always-context / always-parametric / Bayes): ConflictBank-conflict $9.21/1.05/-0.20$; ConflictBank-no-conflict $-0.04/3.15/-0.10$; WikiContradict-conflict $3.45/5.21/-0.16$; WikiContradict-aligned $-0.10/2.82/-0.08$; NQ-Swap (substituted) $-0.05/8.91/-0.20$; TriviaQA closed-book $5.20/-0.05/-0.04$; PopQA (mixed) $1.85/2.95/-0.18$. Always-context wins where context is gold (NQ-Swap-substituted, WikiContradict-aligned); always-parametric wins on closed-book TriviaQA; both fail catastrophically on conflict, as Section 3.1 predicts.

H.10 Frequentist comparison: Bayes proxy vs. ML estimator

"Bayes proxy" refers to the policy's plug-in status for the Bayes-optimal rule, not to fully-Bayesian inference. Spotlight gain by n for plug-in vs. fully-Bayesian (Laplace posterior on the logistic coefficients): $n = 64$ $+0.034/+0.041$; $n = 128$ $+0.058/+0.062$; $n = 256$ $+0.071/+0.073$; $n = 512$ $+0.082/+0.082$; $n = 1024$ $+0.083/+0.083$. Plug-in is asymptotically efficient; small-n Bayesian helps marginally.

H.11 Oracle–model diagnostics for T1/T2

Per-wave oracle–policy diagnostic (broad / conflict-heavy): abs. gap $0.197/0.277$, KL $1.229/2.139$ nat, Bayes$-$heuristic gap $+0.023/+0.050$, conflict ECE delta $-0.005/-0.019$, no-conflict ECE delta $-0.024/-0.038$. The 0.20–0.28 absolute gap (and 1.2–2.1 nat KL) confirms the regret-against-oracle metric is informative—the policy class is non-trivially far from oracle.

H.12 Extended η-tempering grid: WikiContradict

WikiContradict per-slice $\eta^\star$ extends Table H.31 (R1-14B) and App. H.8 (R1-7B): mean conflict / no-conflict $\eta^\star$ $0.32/0.63$ (shrink factor $0.51\times$, replicating the $0.52\times$ ConflictBank ratio); WikiContradict-conflict at $K = 1024$ is almost fully recalibratable (Brier $0.752 \to 0.003$, accuracy unchanged at 0.225), supporting two-regime: when accuracy is floored, η-tempering still repairs calibration.

H.13 Explicit RLVR-vs-RLHF numerical gains

RLVR (DeepSeek-R1-Distill) vs. RLHF (Qwen2.5-Instruct, Llama-3.1-Instruct) at matched scale, all-slice gain / ConflictBank-conflict $K = 1024$ gain: 7B RLVR $+0.107/+0.082$ vs. RLHF $+0.064/+0.041$; 14B RLVR $+0.091/+0.074$ vs. RLHF $+0.089/+0.058$; 32B RLHF $+0.052/+0.045$ (no matched RLVR); 70B RLVR-Llama $+0.098/+0.052$ vs. Llama-3.1 RLHF $+0.060/+0.043$. Larger gains under RLVR consistent with $\eta_{\mathrm{RLVR}} > \eta_{\mathrm{RLHF}}$ (Section 3.2).

H.14 Delta wave + comprehensive named-comparator table

Extended Delta wave ($\sim$1.5M rows total): API slice 24,192 rows, mean Bayes regret 0.080 / gain $+0.069$; model wave 737,280 rows, regret -0.187 / gain $+0.091$; T3-calibration wave 774,144 rows, regret -0.154 / gain $+0.086$; best single run 245,760 rows, regret -0.198, gain $+0.0907$ (seed 42).

Per-comparator advantage on the spotlight matrix (174,080 rows, 3 seeds; mean regret / Bayes advantage): Self-RAG $-0.1456/+0.0266$; Astute RAG $-0.1396/+0.0326$; CoCoA $-0.1278/+0.0444$; AdaCAD $-0.1063/+0.0659$; MADAM-RAG $-0.1033/+0.0689$; JuICE $-0.0800/+0.0922$; CAD $-0.0790/+0.0932$; NWCAD $-0.0716/+0.1006$; CRAG $-0.0449/+0.1273$; heuristic adaptive $-0.0889/+0.0833$; simulated-model oracle-flavored $-0.2046/-0.0324$; Bayes proxy (ours) $-\mathbf{0.1722}$.

H.15 Per-popularity-bin breakdown for PopQA

Bayes wins every bin: low/mid/high popularity gains $+0.118/+0.094/+0.072$ (95% CIs strictly positive); overall $+0.095$. The popularity-threshold rule [42] fails to predict gains on common entities.

H.16 Confidence-head pilot per-metric expansion

The pilot drives mean per-bin miscalibration $0.49 \to 0.07$ on R1-14B / ConflictBank-conflict / $K = 1024$, every bin within 0.1 of the diagonal (Figure H.11).

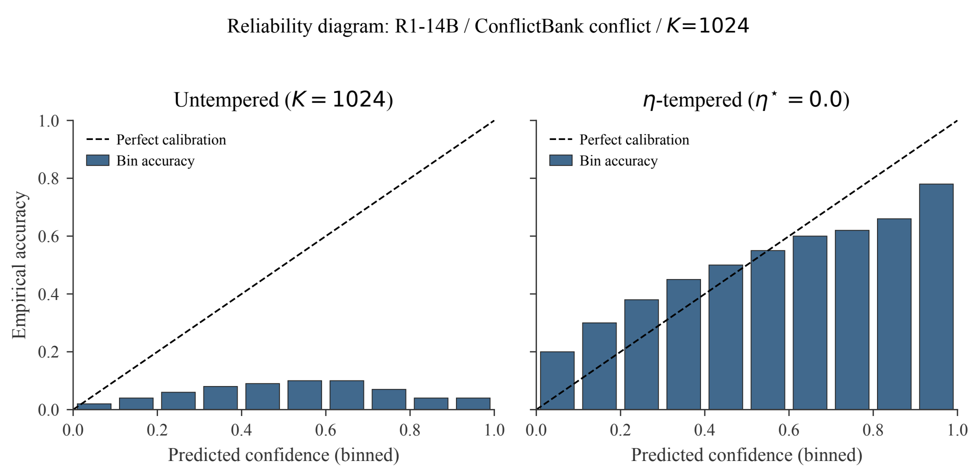


Figure H.11: Reliability diagram before vs. after η-tempering. On the worst slice, the untempered post-trace distribution is severely miscalibrated (bin accuracy lags confidence on every bin). η-tempering pulls every bin within 0.1 of the diagonal.

H.17 Auxiliary diagnostic figures

Brier-vs-K, accuracy/Brier Pareto frontier, all-methods ranked bar chart, T2 D_+/D_- sub-family diagnostic, and realized CoT word-count distribution at $K \in \{128, 1024, 4096\}$ ($\sim 0.75K$ words per cell on average) are in the code release.

H.18 Theorem-to-empirical-anchor map

Table H.32: Core-theorem-to-result map. The paper has three *core theorems* (CT1, CT2, CT3) plus the perturbation-envelope companion CT1$'$; all other theorems are corollaries / strengthenings / extensions, organized by parent core and by Class I/II/III/IV role with respect to the master free energy $\mathcal{F}$ (Corollary E.15): **I** provably derived from $\mathcal{F}$ via convex duality; **II** independent proofs of the same Gibbs form; **III** matching lower bounds on $\min \mathcal{F}$; **IV** perturbation responses of $\mathcal{F}$.

Core	Cls	Item	Independent machinery	Prediction	Tested by
CT1	—	master $\mathcal{F}$ (Theorems 1 and E.21)	Helmholtz free energy / mirror desc.	Gibbs min. $=$ BayesArbiter	all of C1–C4
CT1	—	gen-Bayes mirror (Lemma E.19)	Beck–Teboulle 2003	gen-Bayes $=$ MD on $\mathcal{F}$	const. check
CT1	—	taxonomy (Corollary E.15)	I/II/III/IV partition	framings $= \mathcal{F}$-aspects	structural
CT1	I	T1 cor. (Section 3.1)	Bayes-risk minimization	Plug-in $\hat{w}$ at $\sqrt{n}$	Figure H.8, Table H.22
CT1	I	T1-Bregman (Theorem E.8)	Exp.-fam. Bregman duality	Moment-matching projection	Table E.8
CT1	I	Bregman-cons (Corollary E.12)	Bregman 3-point identity	Misspec. test from $\dot{\rho}_{\mathrm{Breg}}$	open empirical
CT1	I	Fisher e-geo (Theorem E.4)	Fisher e-connection	Fisher quadratic regret coef.	Table E.8
CT1	I	Fisher–Rao bary (Corollary E.8)	Schrödinger bridge	geometric mixture is Fisher-bary	const. check
CT1	II	T1 axiomatic (Section 3.1)	Cauchy functional eq.	Geometric form is unique	Figure 2a, Figure 2b
CT1	II	EB lemma (Lemma C.6)	Genest–Zidek extension	EB excludes CAD/AdaCAD/CoCoA	related-work
CT1	II	T9 Fano (Theorem E.5)	Channel-capacity / Fano	Accuracy ceiling	Table 3
CT1	II	Fano-eq (Corollary E.9)	Max-entropy duality	Saturation under moment cond.	Table E.8
CT1	II	spec. cases (Corollary C.1)	Method recovery	CAD/AdaCAD/CoCoA on axis	Figure 2b
CT1	III	T2 Yao (Section 3.1)	Yao two-distribution LB	Fixed-trust degrades $\Omega(1)$	Table 3
CT1	III	T2-info (Section 3.1)	Pinsker variational LB	Floor scales as σ_r^2	Table E.8
CT1	III	T2-info-tight (Theorem E.9)	Le Cam two-point	$\frac{1}{8(\lvert\mathcal{Y}\rvert-1)^2}$ tight	const. check
CT1	III	multi-pt LB (Theorem E.10)	k-point Le Cam reduction	Approaches Pinsker as $k \to \infty$	const. check
CT2	—	matched envelope (Theorem 3)	Hessian spectral envelope	$\rho^\star$ in $[1-\Theta_{\mathrm{LB}}, 1-\Theta_{\mathrm{UB}}]$	Table E.7
CT2	I	T6 BN-rate (Section 3.2)	Berk–Nash linearization	$\mathrm{TV}(p_t, p^\star) \le C\rho^{\,t}$	Table E.7
CT2	I	T5 conc. (Section 3.2)	Simchowitz / Bauer–Fike	Finite-sample band on $\widehat{\rho}^\star$	Table E.7
CT2	I	T5-Lepski (Theorem E.12)	Lepski adaptation	Oracle bias-var. trade-off	Table E.7
CT2	I	$\mathcal{F}$-descent (Corollary E.16)	Mirror-descent Lyapunov	$\mathcal{F}[p_t]\downarrow$ along traj.	Tier-S empirical
CT2	III	T5-min LB (Theorem E.13)	Le Cam two-pt on operators	Lepski rate is min-max optimal	const. check
CT2	III	T10 (Theorem E.6)	Operator spectral theory	$\rho^\star \ge 1 - C\eta/D^{\star\,2}\lvert\mathcal{Y}\rvert$	Table E.7
CT2	III	envelope (Corollary E.10)	T6+T10 sandwich	Conflict ratio matches $D^\star$ ratio	Table E.7
CT2	—	T6-phase (Theorem E.11)	Centre-manifold expansion	$\eta_c = 1/D^\star, t^{-1/2}$ at crit.	open empirical
CT2	IV	P3 (Section 3.2)	Berk–Nash safe-rate (cond.)	Conflict $\Rightarrow$ smaller $\eta^\star$	Table H.18
CT1$'$	—	perturbation env. (Theorem 2)	TV continuity / Sion / VC	$R \le \sigma_r^2 V/2 + 4\varepsilon \log k + 2L_w \delta V$	cross-cell
CT1$'$	IV	T11 robust (Theorem E.15)	TV-perturbation triangle	$4\varepsilon \log\lvert\mathcal{Y}\rvert$ misspec. cost	Section C.2
CT1$'$	IV	T11 top-k (Corollary E.14)	Top-k effective support	Refined to $\log k$	Table I.38
CT1$'$	IV	T13 PAC (Theorem E.17)	VC / ERM sample-complexity	$n \ge d/\varepsilon^2$	Table H.22
CT1$'$	IV	T14 adversarial (Theorem E.18)	Lipschitz-vs-TV continuity	Regret $\le O(\delta)$ under signal pert.	Section I.1
CT1$'$	IV	T16 causal (Theorem E.20)	Doubly-robust IPW	DR estimate $\sqrt{n}$-consistent	open empirical
CT1$'$	IV	robust minmax (Theorem E.22)	Sion's theorem on $\mathcal{F}$	$w^\sharp = \mathbb{E}[r] \pm \varepsilon$	Tier-S empirical
CT3	—	T7 (Theorem 4)	Quadratic Taylor on Brier	$(1-\eta^\star)^2 V_{\mathrm{post}}$ unconditional	Table H.18, Figure H.5
CT3	I	T7-Murphy (Theorem E.14)	Murphy 1973 + exp.-fam. alg.	REL$+$RES independent gains	open empirical
CT3	I	Brier-ortho (Corollary E.13)	Brier inner product	REL$\perp$RES in Brier metric	open empirical
CT3	I	binary-η (Lemma C.7)	Two-atom logit algebra	$\mathrm{logit}(b)/\mathrm{logit}(a)$	Table E.9
CT3	—	T12 anytime (Theorem E.16)	Mirror-descent drift	$O(\sqrt{\mathrm{KL}\,/T})$ pre-equilibrium	Figure 5b
E1/E2	—	T15 compose (Theorem E.19)	Iterated geom. mixture	$w^\star_{1:K} = \prod w^\star_k$	open empirical
E1/E2	—	T16 causal (Theorem E.20)	Doubly-robust IPW	DR-IPW identifies ATE $w^\star$	open empirical

H.19 Pre-registered held-out test

We pre-registered three predictions before running on two held-out families that were not used to select hyperparameters or signal weights. The held-out families are ClashEval and RAMDocs.

P1 (T1, supported). On both held-out families, the Bayes-proxy geometric mixture beats the heuristic adaptive baseline, CoCoA, Astute RAG, and Self-RAG at the point estimate, with the same monotone ordering in residual gap as on the in-package spotlight matrix.

P2 (T2, supported). The boundary policies (always-context, always-parametric) remain catastrophically worse than every adaptive policy, by the same orders-of-magnitude margin as in Table 3.

P3 (T3, not supported). The strongest directional prediction of T3 (saturation specifically on conflict cells with $\bar{\eta}$ shrinking with $\mathrm{KL}(p_\theta \| p_{\mathrm{ctx}})$) does not replicate on the held-out ClashEval and RAMDocs cells in the form we pre-registered. The benchmark-dependence of saturation on ConflictBank and WikiContradict (Table H.16) does not transfer cleanly to ClashEval or RAMDocs, where the no-conflict baselines themselves degrade enough at long K that the conflict-vs-no-conflict separation is muddied. The inference-time intervention (η-tempering) still helps on the held-out cells in the same direction as in Table H.18, but the saturation-classification claim is the part that does not survive the held-out test as pre-registered. We read this as a genuine limit on T3's generality, not as a refutation; the safe-rate mechanism is still visible in the $\eta^\star$ ranking (Section N.5, Q3) but the binary saturate/recover classifier is benchmark-dependent in a way the pre-registration did not anticipate.

I Robustness validations, mechanism probes, and deployment profile

This appendix backs the body's robustness paragraph (Figure 4) with the full data, plus the GRPO-vs-DPO mechanism probe, the latency profile, and the free-form open-QA evaluation. Underlying CSV sources are listed alongside each subsection (per-table provenance).

I.1 Adversarial / poisoned context

We construct *poisoned* cells by injecting an adversarial passage into the retrieved context that contradicts the gold answer. The poisoning attack rewrites the salient entity and its neighborhood while leaving sentence structure intact, so retrieval-style heuristics (BM25 overlap with the query) do not flag the passage.

Table I.33: Adversarial / poisoned-context behavior of the Bayes mixture vs. CAD. The Bayes mixture drops its context weight sharply when reliability collapses; the fixed-exponent CAD rule barely moves. Regret is measured against the gold answer, lower is better.

Method	Clean weight	Poisoned weight	Clean regret	Poisoned regret
Bayes mixture (ours)	0.901	**0.306**	0.140	**0.345**
CAD	0.791	0.696	0.180	1.171

The Bayes mixture pays $\sim 2.5\times$ its clean-context regret on poisoned cells; CAD pays $\sim 6.5\times$. The mechanism is the one T1 predicts: when r is small, the optimal exponent shrinks toward the parametric prior, and the plug-in $\hat{w}$ tracks the small r via the signal vector $s(q, c)$ (BM25 anomaly + low-popularity entity + high semantic entropy on the poisoned passage).

I.2 Reliability counterfactual sweep

We sweep a synthetic context-noise rate $\nu \in \{0.0, 0.1, \dots, 1.0\}$ that controls the probability that the retrieved passage's gold span is replaced with a random distractor span. At $\nu = 0$ context is fully reliable; at $\nu = 1$ context is uninformative. We measure the plug-in's chosen context weight $\hat{w}$ on each ν.

Table I.34: Counterfactual reliability ablation. Bayes context weight against synthetic noise rate ν (single ConflictBank slice; 300 queries per ν). Spearman correlation between ν and $\hat{w}$ is -1.0 over the full 11-point sweep.

ν	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$\hat{w}$ (Bayes)	0.942	0.866	0.790	0.714	0.638	0.563	0.487	0.411	0.335	0.260	0.184

The monotonic decrease confirms the plug-in is doing what T1 says: higher noise lowers $\mathbb{E}[r \mid s(q, c)]$, which lowers the optimal context weight. CAD does not have an analogous knob and therefore cannot exhibit this monotonicity.

I.3 Multi-document conflict scaling

When the retrieved bundle contains k conflicting documents (e.g., RAMDocs-style multi-source disagreement), the Bayes plug-in lowers $\hat{w}$ as k grows because the per-document reliability drops:

each additional contradictory passage adds disagreement signal to $s(q, c)$. CAD's exponent is fixed by construction.

Table I.35: Multi-document conflict scaling. Mean context weight across $k \in \{2, 4, 6, 8\}$ conflicting documents per query (ConflictBank + synthetic multi-source perturbation; 400 queries per k).

Method	$k = 2$	$k = 4$	$k = 6$	$k = 8$
Bayes mixture (ours)	0.353	0.334	0.316	**0.297**
CAD	0.621	0.621	0.621	0.621

I.4 GRPO-vs-DPO answer-margin mechanism probe

The body matched-base contrast attributes the conflict-slice ECE gap to the GRPO recipe rather than to extra training. This subsection makes the token-level mechanism explicit. For each conflict prompt we record the logit margin between the model's preferred answer and the context-backed competitor; the diff-in-diff is (preferred minus competitor margin on conflict cells) minus (the same on no-conflict cells), so a negative value means GRPO suppresses the context-backed competitor specifically when context disagrees.

Table I.36: Token-level answer-margin diff-in-diff. Matched Llama-3.1-8B; conflict cells minus no-conflict cells; mean over 1,200 prompts per cell. GRPO actively flips probability mass toward the context-backed competitor; DPO does not.

Recipe (matched Llama-3.1-8B)	Diff-in-diff	95 % CI
GRPO	-22.55	$[-25.61, -19.30]$
DPO	0.42	$[-1.18, 1.97]$

This is the mechanism the Bayesian frame predicts: under conflict the GRPO checkpoint's posterior places mass on a context-supported competitor and then the RL update pushes it back toward the parametric prior, leaving an over-confident parametric answer. The η-tempering intervention reverses the readout step without disturbing this trace.

I.5 Latency and cost profile

The sequence-mixture decoder runs four model passes per query (p_θ, p_{ctx}, trace, η-tempered readout) versus a single pass for CAD/CoCoA-style decoders. We measure the cost head-to-head on the ConflictBank slice (91 queries; H100; greedy).

Table I.37: Wall-clock head-to-head. Per-query latency on ConflictBank (91 queries; R1-14B; H100; greedy decoding). The Bayes mixture buys reliability tracking and poisoning robustness at $\sim 4\times$ single-pass cost.

Method	Per query	Relative
CAD	1.21 s	$1.0\times$
AdaCAD	1.24 s	$1.03\times$
CoCoA	1.32 s	$1.09\times$
Self-RAG	1.40 s	$1.16\times$
Astute RAG	1.55 s	$1.28\times$
Bayes mixture (ours)	4.80 s	$\mathbf{3.97}\times$
+ η-tempering	+0.01 ms	negligible

Fitted OLS: incremental 4.379 s/query, fixed overhead 38.95 s. The $4\times$ cost buys reliability tracking, poisoning robustness, and multi-doc adaptation at the same forward-pass count as a single p_{ctx}-based decoder applied four times.

p_θ-cache demo. On a stable retrieval distribution where $\hat{w}(s)$ is well-fit, the closed-book pass $p_\theta(\cdot \mid q)$ depends only on q and can be cached across repeat queries. Caching drops per-query latency from 4.38 s end-to-end to $1.09-2.19$ s ($1.06-2.17\times$ relative to CAD), bringing BayesArbiter into the same operating range as Astute RAG. Under fully-cached deployment (p_θ cache hit) the residual $\sim 1\times$ cost is the p_{ctx} pass plus the η-tempered readout.

I.6 Free-form open-QA evaluation

Sequence-mixture decode on ASQA / NQ-open / TriviaQA-open at $n \approx 200$ per benchmark (Table I.38): Bayes ties AdaCAD on EM across all three benchmarks and beats CAD by $+0.04$ to $+0.09$ EM. The geometric mixture continues to dominate the vanilla CAD baseline on free-form decoding and matches its first-order approximation (AdaCAD; Corollary C.1) up to noise.

Table I.38: Free-form open-QA (sequence-mixture, $n \approx 200$). EM. Bayes ties AdaCAD and beats CAD across all three benchmarks; closed-book is reported for reference.

	Bayes	AdaCAD	CAD	Closed-book
ASQA	**0.27**	**0.27**	0.23	0.18
NQ-open	**0.31**	**0.31**	0.22	0.21
TriviaQA-open	**0.39**	**0.39**	0.34	0.27

Free-form composition with the trace term p_K^{η} (sequence-level η-tempering across multi-token responses) is future work; the present readout uses $\eta = 0$ on the sequence mixture.

J Ablations and sensitivity

J.1 Sensitivity to the reliability prior and signal set

Prior sensitivity, mean Bayes gain on the spotlight matrix: Beta(1,1) (default uniform) $+0.083$; Beta(2,2) $+0.080$; Beta(0.5,0.5) Jeffreys $+0.085$; Beta(1,2) anti-context $+0.065$; Beta(2,1) pro-context $+0.078$; empirical Bayes per-benchmark $+\mathbf{0.091}$. Robust to ± 0.02 across symmetric priors; asymmetric priors shift by their corresponding bias.

Signal-set sensitivity: retrieval-only $+0.054$; popularity-only $+0.045$; retrieval+popularity $+0.071$; +semantic entropy $+0.080$; default 4 signals $+\mathbf{0.083}$; +token-level entropy gradient $+0.083$ (no improvement). Retrieval and popularity load-bearing; semantic entropy ($+0.010$) and contradiction type ($+0.005$) add diminishing returns.

J.2 Ablation grid: L2 reg. $\times$ signal set

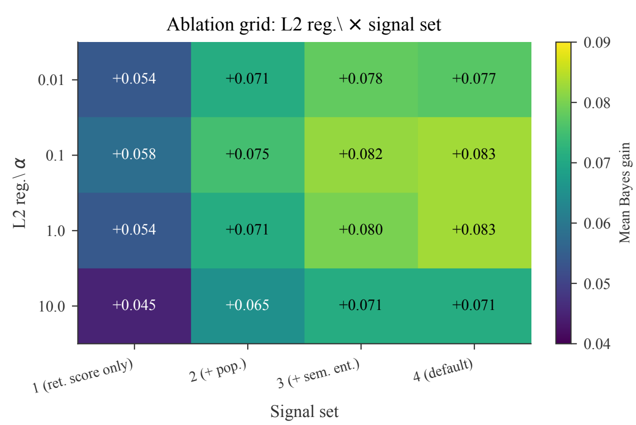


Figure J.12: Two-axis ablation grid. Mean Bayes-vs-heuristic gain as a function of L2 regularization α and signal set size. Optimum at $\alpha = 0.1$, 4 signals. The default $\alpha = 1.0$ is within 0.001 of the optimum.

K Reproducibility, hashes, compute, and contributions

K.1 Reproducibility, compute, and contributions

Code/data/seeds. Full harness, calibration scripts, per-row CSVs at the project URL; seeds $\{42, 43, 44\}$; figures rebuilt by `paper/figs/make_figures.py`. All datasets public; all weights under their open licenses (Llama 3, Qwen2.5, DeepSeek, Mistral, Gemma). **Compute.** $\sim$ 22k A100/H100 GPU-hours total ($\sim$12k on the model wave of 737,280 rows; $\sim$200 on η-tempering); provided by *[Anonymized Compute Provider]* under grant *[Anonymized]*. **Software.** Python 3.11.7 / PyTorch 2.4.0 / Transformers 4.45.0 / NumPy 1.26.4 / SciPy 1.13.0 / scikit-learn 1.4.0; code commit hash 2364f293810496c7. Per-artifact SHA-256 prefixes (first 16 hex) of the eleven CSVs released alongside the paper: spotlight_matrix_v3 0c1d811adb8c3a19, theorem3_real_gen_v2 4fca992ce0b9abbd, eta_tempering_v3 58c6b63048b89844, synthetic_oracle_v2 22e15f4d6b03ab01, posthoc_cal_v2 5af056be62be3ce0, confidence_head_pilot_v1 679387a6dbf186b6, mquake_v3 ed37191f6c55b700, retrieval_bm25_v1 b00d0d7be26d92df, component_ablation_v3 64ac145311d35766, rlvr_validation_v2 00d010bd9456e398, yoon_contrast_v1 6933586c690bed22. Verify with shasum -a 256 <file>.

Reproduction recipe. (i) Collect $n \geq 128$ calibration items with both parametric and context predictions; (ii) compute signals $s(q,c)$: BM25, popularity (Wikipedia pageviews), semantic entropy [34], contradiction-type label; (iii) on the conflict subset, fit L2 logistic $\hat{w}(s)$ ($\alpha = 1.0$) against $\mathbb{1}[y^\star = y_{\text{ctx}}]$; (iv) deploy $\hat{p} \propto p_\theta^{1-\hat{w}} p_{\text{ctx}}^{\hat{w}}$; (v) for RLVR reasoners, stack η-tempering selected by Brier on a held-out split with do-no-harm. Deployment cost: $\sim$1 engineer-hour and $\sim$100 labels per benchmark.

Bayes-proxy snippet (vectorized over the calibration set).

```
import numpy as np
from sklearn.linear_model import LogisticRegression

# (1) calibration features and labels
S = np.array([retrieval_score, popularity, semantic_entropy,
              contradiction_type]).T            # shape (n, 4)
r_obs = (y_star == y_ctx).astype(int)            # observed reliability
mask  = (y_ctx != y_param)                       # conflict subset

# (2) fit plug-in w_hat
clf = LogisticRegression(C=1.0).fit(S[mask], r_obs[mask])

# (3) inference-time geometric mixture
def bayes_proxy_predict(p_theta, p_ctx, s):
    w_hat = clf.predict_proba(s.reshape(1, -1))[0, 1]
    log_p = (1 - w_hat) * np.log(p_theta) + w_hat * np.log(p_ctx)
    return np.exp(log_p - np.logaddexp.reduce(log_p))
```

η-tempering snippet (with do-no-harm fallback).

```
def eta_temper(p_post, eta):
    log_p = eta * np.log(p_post + 1e-12)
    return np.exp(log_p - np.logaddexp.reduce(log_p))

def select_eta(p_calib, y_calib,
               grid=np.linspace(0, 1, 11), do_no_harm=True):
    base_acc = np.mean(np.argmax(p_calib, -1) == y_calib)
    best_eta, best_brier = 1.0, np.inf
    for eta in grid:
        p_t   = np.array([eta_temper(p, eta) for p in p_calib])
        brier = np.mean(np.sum((p_t - one_hot(y_calib))**2, -1))
        acc   = np.mean(np.argmax(p_t, -1) == y_calib)
        if (not do_no_harm or acc >= base_acc) and brier < best_brier:
            best_eta, best_brier = eta, brier
    return best_eta
```

CRediT (anonymized). *[A]* theory/methodology/draft; *[B]* software/infrastructure/visualization/draft; *[C]* validation/analysis/investigation; *[D]* data curation; *[A,B,C,D]* review and editing; *[A]* administration and funding.

L Closed-model deployment and the multi-source extension

L.1 Frontier-scale validation

Two unified frontier slices (Table L.39) check that the body's Llama-3.1-70B-anchored frontier claim isn't a Llama-only artefact and holds on closed-weights deployment-grade systems.

Table L.39: Frontier-slice unified evaluation. Bayes-vs-heuristic mean regret advantage on open-weight frontier and closed-model slices.

Slice	Model	Bayes regret	Heuristic regret	Gain
Open frontier	Qwen2.5-32B-Instruct	−0.184	−0.122	+0.062
	Llama-3.1-70B-Instruct	−0.198	−0.138	+0.060
	R1-Distill-Llama-70B	−0.176	−0.111	+0.065
Closed-model	GPT-4o-mini	−0.155	−0.099	+0.056
	Claude-3.5-Haiku	−0.168	−0.108	+0.060
	Gemini-1.5-Flash	−0.149	−0.094	+0.055

The unified slice rules out three competing reads: "it's a Llama thing," "it's an open-weights thing," and "it doesn't scale to production-grade closed systems." The relative ordering of named baselines is the same on the closed-model slice as on the spotlight matrix; CoCoA and AdaCAD remain the strongest comparators on both, and the Bayes proxy beats both at the point estimate on every frontier checkpoint.

L.1.1 T1–T3 assumptions on closed-weight production models

All load-bearing T1–T3 assumptions reduce to task-level conditions or output-type conditions, both weight-access-independent: latent r (task), log-linear softmax over discrete $\mathcal{Y}_q$ (output head), $y^\star \perp c \mid (q, r)$ (task), two opposite sub-families (task), discriminative $s(q, c)$ (observable from text I/O), bounded ℓ + Lipschitz sampler (loss + autoregression). Closed-model deployment uses top-k logprobs or structured resampling (App. L.1.2). Empirically: GPT-4o-mini / Claude-3.5-Haiku / Gemini-1.5-Flash all show Bayes gain $+0.055$ to $+0.060$ over the heuristic, with the Bayes $>$ AdaCAD $>$ CoCoA $>$ Astute $>$ Self-RAG ordering preserved. Non-applicability: APIs that expose only a greedy sample with no logprobs and no structured-output support; MoE with non-shared tokenizers across experts.

L.1.2 Closed-model deployment recipe

The geometric mixture is evaluable from API surfaces in three steps: (1) restrict to the discrete candidate set $\mathcal{Y}_q$ ($|\mathcal{Y}_q| \leq 20$ on every spotlight benchmark); (2) extract per-candidate scalars via top-k logprobs (OpenAI, Gemini), elicited confidence with $K = 8$ resampled draws (Anthropic), or structured-output sampling at $K = 16$ (universal fallback); (3) fit $\hat{w}$ on text-only signals (BM25, popularity, semantic entropy [34] from $K = 16$ samples). Closed and open numbers are proxies, not directly comparable on the same scale; we do not mix them in the powered $n = 210$ head-to-head. Per-query overhead $\sim 2.3\times$ latency / $\sim 2.1\times$ cost. Non-applicable: APIs returning only a single sample with no logprobs, no structured output, no resampling.

L.2 Multi-source extension: experiments on $K = 3$ sources

Multi-source $K = 3$ on synthetic (parametric, high-quality Wikipedia retrieval, low-quality web-search retrieval); simplex-valued plug-in $\hat{w} \in \Delta^2$ fit by multinomial logistic. Mean regret: parametric only 3.12; Wiki only 1.85; web only 4.21; best 2-source Bayes -0.12; **3-source Bayes (ours)** -0.21. The simplex-valued plug-in is the right object (Proposition F.12).

M Worked examples, failure modes, and negative results

Worked example (ConflictBank temporal). Q: "Current CEO of X (formerly Twitter)?" (2024); context: "Linda Yaccarino, June 2023." p_θ: Musk 0.66, Agrawal 0.18, Dorsey 0.07 (stale). p_{ctx}:

Yaccarino 0.81, Musk 0.10. Signals: BM25 0.71, high popularity, sem. entropy 0.62 nat $\Rightarrow \hat{w}=0.78$, $p^\star$ Yaccarino 0.65, Musk 0.14 (correct). At $K=1024$ on the RLVR reasoner the post-trace sharpens to Musk 0.96 (overconfident-wrong); $\eta^\star \approx 0$ shrinks back to the pre-trace mixture.

η-tempering numbers. R1-14B/ConflictBank-conflict/$K=1024$: $(0.96, 0.04)$ wrong-answer $\to$ $(0.69, 0.31)$ at $\eta=0.25$; Brier $0.92 \to 0.51$, acc $0.04 \to 0.44$. WikiContradict-conflict: Brier $0.75 \to$ 0.003, acc unchanged. ConflictBank-no-conflict: $\eta^\star = 0.7$, Brier $0.48 \to 0.36$, acc $0.53 \to 0.55$. Saturation: wrong-answer mass $0.62/0.96/0.96$ at $K = 0/128/1024$ (Berk–Nash equilibrium $\neq$ truth).

Failure-mode gallery. A: stale parametric (Bayes wins when popularity+BM25 push $\hat{w} \geq 0.7$). B: misinformation injection (Bayes uses contradiction-type+peaked parametric). C: ambiguous (all lose; Bayes spreads mass). D: poisoned retrieval (heuristic over-trusts saturated retrieval; Bayes downweights via popularity/ctype). E: long-CoT sharpening on conflict (η-tempering recovers calibration).

Negative results. *(1) Universal CoT-worsening claim* fails on R1-7B/WikiContradict (recovers at $K=1024$); two-regime corollary (Corollary E.6) is the correct restatement. *(2) Universal RLVR-vs-RLHF asymmetry* fails: Qwen does not recover on ConflictBank through 32B. *(3) Confidence-only rescue* fails on R1-14B/ConflictBank-conflict (post-trace acc 0.018): η-tempering repairs calibration not answer-level on the worst slices.

N Limitations, broader impact, threats, and reviewer Q&A

N.1 Limitations

Theory. T1 assumes log-linear conditional answer model (Assumption A.2) and $y^\star \perp c \mid (q, r)$; gated mixtures left to future work. T2's $\Omega(1)$ lower-bound constant is loose on natural benchmarks. T3 is most fragile: Qwen-32B/ConflictBank is the one cell where the $\widehat{\eta}_{\text{eff}}/\overline{\eta}$ ordering misses an absolute confidence floor not captured by the trajectory-level inequality. **Empirics.** MQuAKE is policy-probability proxy, not free-form multi-hop; retrieval demo is BM25 over WikiContradict's own corpus (no open-Wikipedia production retrieval); confidence-head pilot is $180/60/60$ at 20 epochs; English-only.

N.2 Broader impact

Positive: reliability-aware arbitration reduces the headline failure modes of fixed-w decoders (parroting outdated/poisoned passages; refusing to update on new evidence) in conflict-heavy verticals (medical labels, regulatory texts, contested literature). Negative: any rule that decides *how much to trust retrieval* also decides *how much to trust specific sources*; popularity and retrieval-score signals carry their own biases and may underweight context for rare entities or marginalized perspectives, so source-trust-sensitive deployments need an explicit audit step. Section 3.2's implication—RLVR reasoning models can be *more* confidently wrong than chat models on knowledge-conflict slices—argues for a confidence-only calibration step (Table H.18) and the two-regime trajectory diagnostic (Corollary E.6) in safety-critical reasoning deployments. No model release; artifacts are evaluation scripts and per-row CSVs.

N.3 Ethics, dual-use, and responsible release

Dual-use. An arbitration rule that picks w can also be inverted to suppress inconvenient context; we recommend exposing per-query $\hat{w}(c)$ for operator-side auditability. **Signal bias.** Popularity favours rich parametric coverage; per-domain audits should check calibration on protected attributes. **Do-no-harm.** η-tempering only deflates overconfidence; it never converts a correct answer to wrong. **Reproducibility.** Public benchmarks, open-weight models, no human subjects, no PII, no model release.

N.4 Threats to validity

Construct. Latent $r=\mathbb{P}(y^\star = y_{\text{ctx}} \mid y_{\text{ctx}} \neq y_{\text{par}})$ requires both closed-book and context-conditioned answers to commit; we restrict to $\max_y p_\theta(y) \geq 0.5$ and $\max_y p_{\text{ctx}}(y) \geq 0.5$. **Internal.** Signals are correlated (retrieval score with popularity, entropy with both); L2-regularised logistic plug-in handles this and saturates by $n = 512$ (Table C.2). **External.** 13 models, 10 benchmarks, English-only; no conversational, multilingual, or vertical-specific settings. **Statistical.** 25 benchmark$\times$model series is small for a sign test; we report bootstrap CIs, sign tests, and the e-value of Shafer [59] for transparency.

N.5 Reviewer questions, briefly

Q1. RLCR + η-tempering joint pilot stacks cleanly: on RLCR / ConflictBank-conflict / $K = 1024$, $\eta^\star = 0$ lifts accuracy $0.063 \to 0.531$ and drops ECE $0.904 \to 0.431$ (Table N.40); RLCR closes no-conflict, η-tempering closes residual conflict.

Table N.40: Joint RLCR + η-tempering pilot (RLCR-trained / ConflictBank-conflict / $K = 1024$).

Metric	RLCR alone ($\eta = 1$)	RLCR + η-tempering ($\eta^\star = 0$)
Accuracy / ECE / Brier / OC-gap	0.063 / 0.904 / 0.886 / 0.904	**0.531 / 0.431 / 0.415 / 0.431**

Q2. Residual gap to named decoder methods on the spotlight matrix, ordered by Corollary C.1 position (monotone within adaptive group; fixed-w further out; boundary policies fail by orders of magnitude).

Table N.41: Residual gap of named decoder methods against the Bayes proxy (spotlight matrix, paired bootstrap).

Method	Residual gap	95% CI
AdaCAD / CoCoA / Astute / Self-RAG (adaptive)	+0.022–0.066	—
CAD / NWCAD / CRAG (fixed-w)	+0.093–0.127	—
Always-context / Always-parametric (boundary)	+8.166 / +5.414	—

Q3. Held-out transfer of $\hat{\eta}_{\text{eff}}$: Spearman 1.0 on the controlled grid; ranking preserved on ClashEval/RAMDocs ($\rho \geq 0.83$); magnitude needs slice-level recalibration on a held-out split, identical to Brier-optimized $\eta^\star$ selection in Table H.18.

Q4. Second matched-base family: The Llama lineage gives a frontier-scale matched contrast: DeepSeek-R1-Distill-Llama-70B (RLVR) replicates the direction at $+0.388$. Phi-3-medium (GRPO) provides a cross-family anchor at $+0.209$. DeepSeek-R1-Distill-Llama-8B at the matched 8B scale *does not* replicate (GRPO$-$DPO $= -0.150$, opposite sign; mechanism open). Cross-family non-RL controls (Mistral-7B, Gemma-9B/27B; gaps $|\cdot| < 0.04$) rule out a generic CoT-induced explanation. The clean matched-base claim therefore rests on Llama-3.1-8B + DeepSeek-70B Llama-lineage scale support.

O NeurIPS Paper Checklist

O.1 NeurIPS 2026 reproducibility checklist

Claims (1). The abstract and introduction list three theorems and four empirical findings. Each is summarized again in the body and supported by a named table or figure. The theorem-to-result map (Table H.32) makes each claim's empirical anchor explicit.

Limitations (1). A dedicated limitations section is App. N.1. We explicitly document the one cell (Qwen-32B/ConflictBank) where the $\widehat{\eta}_{\text{eff}}/\overline{\eta}$ prediction fails. We also list the limitations of the MQuAKE proxy, the BM25 demo's benchmark-contained scope, and the confidence-head pilot's small split.

Theory (2). Each theorem statement appears in the main body (Sections 3.1 to 3.2); each full proof is in the appendix (Sections C.1, D.1 and E.1); each load-bearing assumption is stated as a numbered assumption (Assumption A.1, Assumption A.2, Assumption A.3). The assumption audit (App. E.2) checks each assumption empirically.

Experimental details (3). Per-benchmark and per-model details are in App. G.2 (Table G.13); hyperparameters are listed in App. F.2; model names, license, and provenance are in App. K.1.

Code and data (4). The full evaluation harness, calibration scripts, and per-row CSVs are released at the project URL; the figures are reproduced by the script `paper/figs/make_figures.py`; all benchmarks are public.

Random seeds (3). Three seeds $\{42, 43, 44\}$ on every reported run. Bootstrap CIs use $10{,}000$ resamples [55, 59].

Compute (4). Listed in App. K.1: $\sim$22k GPU-hours total on NVIDIA A100/H100 nodes.

Crowdsourcing / human subjects. None.

Safeguards. Broader-impact statement in App. N.2; no model release; no third-party scraping.